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Systematic Benchmarking of Biological Information Sources for Viral Mutation Hotspot Detection

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Abstract

This dissertation presents **MutBench** (currently scoped to the RNA-virus single-position regime; 11 RNA viruses, surface glycoproteins, point substitutions only), a region-level, single-position hotspot-detection benchmark that systematically compares 10 biological information types (20 scoring formulas) $\times$ 14 detection families (39 variants) $\times$ 11 RNA viruses = 8,580 evaluations. The headline result: within the 20-type single-position scoring regime evaluated on the 11-pathogen RNA-virus panel, the optimal information source is pathogen-dependent, with the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$, 11-pathogen cluster-bootstrap 95% interval [0.195, 0.333]) as the largest modeled variance component, and HIV-1 vaccine-escape enrichment of $7.19\times$ (Bonferroni $p = 2.5 \times 10^{-16}$) as the primary external anchor (retrospective enrichment; no prospective time-forward validation).

Detecting mutation hotspots in viral genomes is essential for variant surveillance and vaccine design, yet existing approaches have relied almost exclusively on mutation frequency and Shannon entropy, leaving comparatively unexplored a rich landscape of complementary information sources (phylogenetic selection signals, protein language models, structural features). Concurrent fitness-regression benchmarks (ViroGym, EVEREST v3, ProteinGym v1.3) target per-variant prediction; to our knowledge, MutBench is the broadest *region-level, single-position hotspot-detection* benchmark to date for RNA viruses, with 8,580 (scoring$\times$detector$\times$pathogen) evaluation cells. MutBench comprises three core components: (1) a three-layer ground truth system integrating literature-curated positive markers

(Layer A: a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, with immune-escape positions dominating the curated set), conserved structural regions under purifying selection as negative controls (Layer B), and Deep Mutational Scanning (DMS) experimental fitness data as functional validation (Layer C); (2) a composite evaluation metric defined as hotspot-score $= (\text{recall}+\text{precision}+\text{stability})/3$; and (3) a two-stage main experimental design plus a Stage 3 information-integration layer, in which Stage 1 conducts in-depth analysis on SARS-CoV-2, Stage 2 performs the large-scale cross-pathogen comparison, and Stage 3 evaluates multi-source feature integration.

Stage 1 provides a controlled SARS-CoV-2 sanity check for the framework (MutClust-Hybrid hotspot-score 0.778). Stage 2 establishes that, within the 20-type single-position scoring regime, the optimal information source varies systematically across the 11-pathogen RNA-virus panel: $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$ (11-pathogen cluster-bootstrap 95% interval [0.195, 0.333]; 6-category aggregation gives $\omega^2 = 0.103$ as a collinearity-robust lower bound; within-cell residual $\omega^2 \approx 0.39$). Friedman test shows no universally best combination ($\chi^2 = 7.69$, $p = 0.990$), and LOPO yields 0/11 exact matches with a 0.265 oracle-vs-generalized MCC gap (LOPO 0/11 alone is null-consistent under permutation; the conclusion rests on the interaction effect plus the vaccine-escape audit, not on LOPO in isolation). Contribution 3, practical search-space reduction validated on HIV-1 vaccine-escape: the optimal scoring–detection combination narrows the candidate list from $\sim$1,000 to 20–80 positions per protein with 7–9$\times$ enrichment for escape sites, transforming exhaustive experimental screening into guided investigation. Under a three-tier evidence rule (Bonferroni survival $\times$ Layer A overlap $< 33\%$), HIV-1 is the primary external anchor: full-set Fisher enrichment $7.19\times$ ($p_{\text{adj}} = 2.5 \times 10^{-16}$, 82% Layer-A-disjoint), with a direct novel-only enrichment of 7.16–$8.24\times$ on the 37 Layer-A-disjoint positions (worst-case $p = 5 \times 10^{-12}$; Bonferroni $p_{\text{adj}} \approx 3.9 \times 10^{-9}$ across 780 combinations). H3N2 ($9.36\times$, 69% overlap) is a self-consistency check and SARS-CoV-2 ($7.63\times$) is exploratory after Bonferroni. Validation is restricted to 3/11 pathogens by curated antibody-escape annotation availability, not by panel design. A multi-source feature-ablation robustness check (nested-LOPO on the 12-pathogen Stage 3 panel: 4-feature core paired delta $+0.011$ MCC, 95% CI [-0.018, $+0.042$], permutation $p = 0.61$) confirms the compact core (homoplasy, pLDDT, entropy, frequency) is statistically indistinguishable from the full 10-feature EqualWeight ensemble; EqualWeight integration also achieves H3N2 escape enrichment $4.012\times$ ($p = 0.0023$).

A systematic analysis of the 20 scoring types spanning six information categories (frequency, MSA-derived, phylogenetic, structural, AI-based, and composite) reveals 9 distinct optimal information types across 11 pathogens (Tranception/ESM-2 channel collinearity is not separated at headline granularity; see Chapter 2): FUBAR phylogenetic selection scores achieve the highest MCC for H3N2 (0.534), protein language model scores dominate for SARS-CoV-2 and RSV, and homoplasy scores excel for Norovirus. The external practical anchor for downstream search-space reduction remains HIV-1, where the Layer-A-disjoint escape subset supports 7.16–$8.24\times$ enrichment under the novel-only audit.

Keywords: mutation hotspot detection, MutBench, biological information types, viral genomics, multi-source information analysis, pathogen-dependent optimality, SARS-CoV-2, Deep Mutational Scanning

Chapter 1

Introduction

1.1 Research Background

When a new viral pandemic emerges, identifying which positions in the viral genome are mutating in functionally important ways is critical for public health response. These mutation "hotspots" help researchers prioritize which variants to monitor, which vaccine targets to update, and which drug resistance mutations to anticipate. SARS-CoV-2, which emerged in late 2019, caused more than 700 million confirmed cases worldwide, reaffirming the critical importance of viral genome analysis [8]. Viruses continuously accumulate mutations during replication, and the phenomenon whereby mutations concentrate at specific genomic positions is termed a hotspot. Because hotspots are directly associated with major phenotypic changes in viruses—including increased transmissibility, immune evasion (the ability to escape pre-existing immune responses), and drug resistance—accurately detecting them is a core task in vaccine design, antiviral drug development, and real-time genomic surveillance [8, 9]. Table 1.1 defines the key biological terms used throughout this dissertation.

Existing computational approaches for hotspot detection have relied predominantly on two types of information: mutation frequency and Shannon entropy. While these capture the most direct signals of mutational enrichment, they fail to distinguish biologically

Table 1.1: Key biological terms used in this dissertation

Term	Definition
Mutation	A change in the nucleotide or amino acid sequence during viral replication
Hotspot	A genomic position where mutations accumulate more frequently than expected by chance
Convergent evolution	Independent emergence of the same mutation in unrelated lineages, indicating positive selection
Purifying selection	Evolutionary removal of harmful mutations that disrupt protein structure or function
DMS	Laboratory technique measuring the functional impact of every possible amino acid substitution
MSA	Computational alignment of multiple protein sequences to identify per-position variation

meaningful hotspots from founder-effect artifacts and cannot leverage the growing wealth of complementary information sources—including phylogenetic selection signals, protein language model predictions, structural features, and deep mutational scanning data. Concurrent benchmarks evaluate per-variant fitness regression on overlapping virus panels (notably ViroGym [10], 13 viruses / 79 DMS / 7 phenotypes; EVEREST v3 [11], 45 DMS / 31 forecasting clades / 40 WHO priority viruses; ProteinGym [12, 13], 217 DMS / 90+ baselines; EVEscape [14], an immune-escape predictor combining fitness, accessibility, and antibody dissimilarity; and the variant-effect benchmark of Livesey & Marsh [15]), but none has performed a region-level hotspot-detection comparison that systematically decomposes which biological information types are most effective for which pathogens, leaving researchers to default to frequency-based approaches regardless of the virus under study.

Formally, given a multiple sequence alignment (MSA) of viral protein sequences, the detection task identifies a subset of genomic positions with statistically elevated mutation rates; yet no standardized ground truth, composite evaluation metric, or cross-pathogen validation methodology has been established for the region-level detection task itself (concurrent benchmarks above target per-variant fitness regression rather than region selection), despite partial benchmark studies in cancer genomics [16]. The author previously developed a systematic benchmark framework for multi-omics integration (MOSD [17]) and an adaptive hotspot detection algorithm (MutClust [18]); the experience of developing MutClust without a quantitative evaluation framework directly motivated this research.

Current viral genomics platforms such as Nextstrain [19] and PANGO lineage classification [20] monitor variant emergence but do not provide systematic evaluation of which biological information sources are most effective for hotspot detection. The absence of standardized hotspot-detection benchmarks (as opposed to fitness-regression benchmarks) means that the rich landscape of available information types—from phylogenetic signals to protein language models—remains comparatively unexplored at the region level, and researchers cannot make informed choices about which information source to prioritize for a given pathogen.

In practice, mutation hotspot detection follows a three-step pipeline (sequence collection $\rightarrow$ computational analysis $\rightarrow$ experimental validation); to avoid lexical collision with the Stage 1/2/3 experimental design introduced in Sections 1.4–1.5, the present chapter uses the term "step" for the bench-side workflow. In the first step (*sequence collection*), viral genomes from infected individuals are sequenced over weeks to months at research institutions worldwide and deposited in public databases such as GISAID and NCBI GenBank; for SARS-CoV-2, millions of sequences have been accumulated since 2020, while for influenza, tens of thousands of sequences span several decades. These sequences are then aligned to a reference genome to produce the MSA that enables position-by-position comparison of mutation patterns. In the second step (*computational analysis*), statistical and algorithmic methods are applied to the MSA to identify positions where mutations are significantly concentrated—the hotspot candidates. This step, which encompasses frequency analysis, entropy computation, density-based clustering, wavelet transforms, and other approaches, is the focus of this dissertation. In the third step (*experimental validation*), the computationally identified hotspot candidates are individually verified through laboratory experiments such as Deep Mutational Scanning (DMS), neutralization assays, and cell infection assays. Because each experimental validation requires weeks to months per position, it is prohibitively resource-intensive to test all positions; computational analysis therefore serves as a critical filter that prioritizes positions for experimental investigation. The effectiveness of this filtering—how well computational methods concentrate biologically meaningful positions in their top-ranked candidates—is precisely what MutBench evaluates (within the RNA-virus single-position scoping clarified in Chapter 2; extension to DNA viruses, eukaryotic pathogens, and prokaryotic systems is left to future work in Chapter 4, Section 4.7). In practice, this means that for a virus with 1,000 amino acid positions, computational hotspot detection narrows the candidate list from 1,000 to approximately 50–100 positions (top 5–10%), saving months of laboratory effort and enabling rapid response during emerging outbreaks.

1.2 Related Research

This section reviews prior work in viral mutation hotspot detection and identifies the benchmarking gap that motivated MutBench.

1.2.1 Mutation Hotspot Detection Methods

Mutation hotspots are regions within a genome where mutations are concentrated in a non-uniform manner, likely reflecting functionally important sites under structural or immunological selective pressure. The concept originates from the observation that mutations are not distributed uniformly across viral genomes: certain positions or regions accumulate mutations at rates far exceeding the background, driven by positive selection for immune evasion, receptor binding optimization, or adaptation to new host environments. For RNA viruses, which exhibit mutation rates of 10^{-3} to 10^{-5} substitutions per nucleotide per replication cycle [21, 22], the interplay between high mutation supply and strong selective pressure creates pronounced mutational clustering in functionally critical regions. Identifying hotspots in viral genomes is critical for three practical applications: (1) predicting variant emergence by identifying positions predisposed to future mutations, (2) selecting vaccine targets by avoiding highly mutable epitopes or targeting conserved adjacent regions, and (3) developing broadly effective therapeutics that target structurally constrained positions unlikely to mutate without fitness cost. Existing hotspot detection approaches can be categorized into frequency-based, entropy-based, density-based, and selection-based methods.

Frequency-based approaches calculate mutation frequency at the amino acid or nucleotide level and designate positions with high frequency as hotspots. While intuitive and easy to implement, they have two fundamental limitations. First, **lineage bias**: when a specific lineage is overrepresented in the database, its defining mutations receive disproportionately high frequency scores. Mutations such as D614G, observed in nearly all modern SARS-CoV-2 sequences, reflect genuine fitness advantages, but the rate of near-fixation is amplified by founder effects during the early pandemic, making it difficult to distinguish the relative contributions of selection and genetic drift from frequency data alone. Second, **neglect of rare variants**: mutations that occur at low frequency but exhibit diverse forms—potentially indicating immune evasion or viral adaptation—are systematically overlooked.

Entropy-based approaches use Shannon entropy to measure nucleotide diversity at a given position. Mullick et al. [23] combined Shannon entropy with K-means clustering to detect hotspots in the SARS-CoV-2 Spike protein. However, conventional nucleotide entropy reflects the distribution of all nucleotides regardless of whether mutation has occurred, failing to capture effective diversity conditioned on the occurrence of mutation. Furthermore, K-means clustering requires the number of clusters to be specified in advance and does not account for density structure. Large-scale frequency-based analyses have been applied to SARS-CoV-2 [24, 25], but these studies used single information types (mutation frequency or fitness proxies) without systematic comparison against alternative scoring approaches.

Cancer genomics methods. In cancer genomics, a rich ecosystem of methods has been developed to detect somatic mutation hotspots. MutSigCV [26] identifies significantly mutated genes by modeling background mutation rates that account for gene expression, replication timing, and chromatin state. OncodriveCLUST [27] detects genes with significant spatial clustering of mutations in protein sequence space, under the assumption that driver mutations cluster in functional domains. HotMAPS [28] maps mutations onto protein three-dimensional structures to identify spatial hotspots in 3D, and MutSpot [29] models position-specific background mutation rates using epigenomic features. While these tools represent sophisticated approaches to hotspot detection, they target somatic mutations in tumor genomes, which are more than 1,000 times larger than viral genomes and involve fundamentally different mutation dynamics and selective pressures (neutral somatic mutation followed by clonal selection vs. population-level viral evolution), making direct transfer infeasible.

An important limitation of frequency-based mutation analysis is **phylogenetic non-independence**: mutations observed in closely related sequences are not independent events [30]. Phylogeny-aware methods such as TreeTime [31], BEAST [32], and HyPhy [33] can address this but require high-quality phylogenetic trees and face scaling and placement challenges in densely sampled datasets, motivating ultra-fast tree-placement frameworks such as UShER [34]. Several tools detect homoplasic (independently recurrent) mutations on phylogenies. HomoplasyFinder [35] identifies positions where the same mutation arises independently on multiple branches of a phylogenetic tree, which provides evidence for convergent evolution driven by positive selection rather than shared ancestry. ConDor [36] extends this approach to detect convergent amino acid substitutions across large protein alignments, using statistical tests to distinguish genuine convergence from chance recurrence. Homoplasy-based approaches are valuable because they explicitly account for phylogenetic non-independence, but they identify individual recurrent sites rather than performing the spatial clustering and multi-information-type comparison that MutBench provides. In MutBench, homoplasy is incorporated as one of the 10 underlying biological information families, which are expanded into 20 scoring channels (with normalization or compositing variants) and evaluated across pathogens (Section 3.5).

Simpler approaches to hotspot identification include sliding window methods, which scan fixed-length windows across the genome and flag those exceeding a mutation count threshold. While computationally straightforward, sliding window methods suffer from two inherent limitations: the window size must be specified in advance (too small misses extended hotspots; too large merges distinct hotspots), and they impose a fixed spatial resolution that cannot adapt to the variable density of mutations across different genomic regions. Signal processing techniques such as wavelet transforms offer a multi-scale alternative, decomposing mutation frequency profiles to identify localized peaks at multiple spatial scales simultaneously. However, these methods treat the genome as a one-dimensional signal and do not incorporate biological information about mutation importance or selective pressure.

Selection-based methods represent a fundamentally different approach. Rather than detecting spatial clusters of mutations, codon-based methods estimate the ratio of nonsynonymous to synonymous substitution rates ($\omega = dN/dS$) at each codon position. The HyPhy phylogenetic platform [33] packages a family of such tests: MEME (Mixed Effects Model of Evolution) [37] detects episodic positive selection at sites where $\omega > 1$ on a subset of phylogenetic branches, while FUBAR (Fast Unconstrained Bayesian AppRoximation) [38] provides a Bayesian estimate of per-site selection pressure pooled across the whole tree. Closely related tests in the same family include FEL (Fixed Effects Likelihood) and SLAC (Single Likelihood Ancestor Counting) [39], which provide site-level ω estimates under a maximum-likelihood and a counting framework respectively, and BUSTED (Branch-Site Unrestricted Statistical Test for Episodic Diversification) [40], which tests for episodic diversifying selection along a designated set of branches. These methods provide a principled evolutionary framework for identifying positions under positive selection, but they detect individual selected sites rather than spatial clusters of hotspot regions, and they require reliable phylogenetic trees and codon-aligned sequences—prerequisites that are not always available for all pathogens. MutBench incorporates four phylogenetic selection-pressure scoring channels—three FUBAR-derived posterior summaries (FUBAR posterior probability, FUBAR positive-selection indicator, FUBAR Bayes factor) plus a Nei–Gojobori dN/dS proxy (Chapter 2, Section 2.2.4)—enabling direct comparison of selection-based features against frequency-based and PLM-based alternatives within a unified framework. FEL, SLAC, and BUSTED are reviewed here for completeness of the HyPhy family but are not benchmarked as scoring channels in the present panel; coverage of the broader selection-test family is left to future work (Chapter 4, Section 4.7).

1.2.2 Density-Based Clustering

DBSCAN [41] is a representative density-based clustering algorithm that defines high-density regions as clusters and classifies low-density regions as noise. The algorithm operates on two fundamental concepts: the ε-*neighborhood* of a point p, defined as all points within distance ε from p, and *core points*, defined as points whose ε-neighborhood contains at least MinPts points. Points reachable from core points through chains of ε-neighborhoods form clusters, while points not reachable from any core point are classified as *noise*. This formulation enables discovery of clusters with arbitrary shapes without requiring the number of clusters to be specified in advance—a key advantage over partitional methods such as K-means.

However, DBSCAN uses a **fixed global epsilon**, making it unable to form appropriate clusters in data with substantial density variation. In viral genomics, mutation density varies dramatically across the genome: immunodominant surface proteins (e.g., the Spike receptor-binding domain) accumulate mutations at far higher density than conserved polymerase regions. A single epsilon value cannot simultaneously capture tight clusters in high-density regions and sparser clusters in lower-density functional domains. This problem manifests in extreme form: direct application to the SARS-CoV-2 genome with default parameters results in 99.76% of the data being merged into a single massive cluster; even with manually tuned parameters ($\varepsilon = 0.15$, MinPts $= 5$), only 4 undifferentiated clusters are obtained, rendering biologically meaningful hotspot distinction impossible [18].

HDBSCAN [42] addresses the fundamental limitation of DBSCAN's fixed epsilon by constructing a hierarchy of density-based clusterings. Rather than fixing a single epsilon value, HDBSCAN computes a *mutual reachability distance* between all pairs of points and builds a minimum spanning tree, which is then condensed into a cluster hierarchy. Clusters are extracted by selecting the most persistent components across density levels, effectively adapting the neighborhood radius to local data density. It operates with a single parameter, min_cluster_size, without requiring explicit specification of epsilon. It can detect clusters of arbitrary shape, automatically identifies noise points, and provides probabilistic interpretation of cluster membership through cluster stability scores. For viral hotspot detection, HDBSCAN's adaptive density handling is conceptually appealing, but it was designed for general-purpose spatial data and does not incorporate domain-specific biological information such as mutation importance scores.

OPTICS [43] takes a complementary approach by producing a *reachability plot*—an ordering of data points where the reachability distance of each point reflects the density required to connect it to the preceding point. Valleys in this plot correspond to clusters, and their depth indicates cluster density relative to surrounding regions. This visualization reveals clustering structure at all density levels simultaneously, but requires post-processing (e.g., the ξ-extraction method) to convert the reachability plot into discrete clusters. The multi-scale perspective of OPTICS is valuable for exploratory analysis, but the additional extraction step introduces subjectivity in cluster boundary determination.

The limitations of standard density-based methods in the viral genomics context motivated the development of adaptive approaches that integrate biological scoring into the clustering framework. MutClust [18] addressed this by replacing DBSCAN's fixed epsilon with a position-specific radius proportional to the H-score (a mutation importance metric combining frequency and entropy), thereby assigning wider neighborhoods to biologically significant positions and narrower neighborhoods elsewhere.

1.2.3 Deep Mutational Scanning as Ground Truth

Deep Mutational Scanning (DMS) is an experimental technique that systematically measures the functional effects of all possible single amino acid substitutions within a protein [44]. In a typical DMS experiment, a library of mutant variants—each carrying a single amino acid substitution at one position—is generated through saturation mutagenesis. These variants are subjected to a selection assay (e.g., receptor binding, viral entry, or replication), and the relative fitness effect of each mutation is quantified by comparing variant frequencies before and after selection using deep sequencing. The result is a comprehensive *fitness landscape*: a matrix of fitness scores for every possible substitution at every position in the protein.

DMS is the gold standard for characterizing mutational effects because it is **unbiased** (every position and substitution is tested, not just those observed in nature), **comprehensive** (thousands of mutations are measured simultaneously in a single experiment), and **experimental** (fitness effects are directly measured rather than computationally predicted). Unlike natural sequence variation, which reflects the combined effects of selection, drift, and sampling bias, DMS directly measures the intrinsic functional consequence of each mutation.

Several landmark DMS studies have shaped our understanding of viral protein fitness landscapes. Starr et al. [45] measured ACE2 binding affinity and protein expression for all single amino acid substitutions in the SARS-CoV-2 receptor-binding domain (RBD), revealing that most RBD positions are highly constrained for binding. Dadonaite et al. [46] extended this to the full-length Spike protein by measuring pseudovirus cell entry fitness, identifying positions critical for viral entry beyond the RBD. Lee et al. [3] performed DMS of influenza H3N2 hemagglutinin (HA), demonstrating that DMS fitness measurements can predict the evolutionary fates of naturally circulating variants. Bloom and Neher [47] extended this further by inferring per-protein fitness effects across the entire SARS-CoV-2 proteome from public sequence phylogenies—rather than from direct DMS assays, since comprehensive DMS had then been performed on only two SARS-CoV-2 proteins—establishing a sequence-based reference complementary to direct DMS measurements.

A parallel lineage of **antibody-escape DMS** maps not viral fitness in isolation but the per-position escape from neutralization by specific antibodies or convalescent/vaccinee sera. Greaney et al. [48] pioneered this design for the SARS-CoV-2 receptor-binding domain by measuring escape from a panel of monoclonal antibodies, and Greaney et al. [49] extended it to the full Spike with polyclonal sera (both SARS-CoV-2-specific). These SARS-CoV-2 antibody-escape DMS datasets, together with the Bloom-lab broader escape-DMS catalog, are the direct experimental antecedent of the **vaccine-escape cross-validation analysis** reported in Chapter 4 (Section 4.4.2; results tabulated in Chapter 3 Table 3.18), which audits per-pathogen MutBench top-k output against curated antibody-/vaccine-escape position lists for HIV-1 (Moore–Williamson 2015), H3N2 (Lee et al. 2018) and SARS-CoV-2. Layer C ground truth in MutBench remains *fitness*-DMS based and is sourced from Dadonaite et al. 2024 (SARS-CoV-2 cell entry), Lee et al. 2018 (H3N2 replicative fitness), Haddox et al. 2018 (HIV-1 viral growth in TZM-bl), Simonich et al. (RSV), Aditham et al. (Rabies) and Bakhache et al. (EV-A71); see Section 1.2.3 of this chapter and Table 2.3 of Chapter 2. The two ground-truth tracks are therefore orthogonal: antibody-escape DMS feeds the vaccine-escape audit, fitness-DMS feeds Layer C, and each contributes a distinct per-position evaluation signal that is largely independent of literature-curated adaptive positives (Layer A) and of purifying-selection negatives (Layer B).

Despite its power, DMS has important limitations. First, it is **expensive and labor-intensive**: each DMS experiment requires constructing a complete mutant library and performing selection assays, limiting its application to a small number of viral proteins. Second, many surface-protein DMS datasets (especially for SARS-CoV-2 Spike and RSV F) use **pseudovirus systems** rather than authentic replication-competent virus, potentially missing fitness effects that depend on the complete viral replication cycle, although authentic replicative-fitness assays are also used in this dissertation's panel (e.g., MDCK culture for H3N2, RD culture for EV-A71). Third, DMS measures fitness **in vitro**, which may not fully capture in vivo selective pressures such as immune evasion in the context of population-level immunity. Finally, DMS coverage across the viral world remains sparse: the EVEREST v3 catalog [11] covers only 45 viral DMS datasets even after explicitly targeting WHO priority viruses, and the catalog spans far fewer distinct viral proteins than the 40-virus WHO panel itself, leaving most pathogens (and most non-surface proteins within covered pathogens) without DMS measurements. This availability gap means that DMS alone cannot serve as a universal ground truth for all pathogens—motivating MutBench's three-layer ground truth design (Chapter 2), where DMS serves as Layer C for the subset of pathogens with available data.

DMS data provides a natural ground truth for hotspot benchmarking: positions where mutations exhibit large fitness effects correspond to biologically functional sites that detection methods should identify [15]. Specifically, positions where the average DMS fitness effect of mutations is strongly negative (indicating functional constraint) or where specific mutations show large positive effects (indicating adaptive potential) are expected to overlap with biologically meaningful hotspots. Livesey and Marsh [15] demonstrated that DMS datasets can serve as unbiased benchmarks for computational variant effect predictors, establishing a precedent for the use of DMS as ground truth in method evaluation.

As a complementary approach, deep generative models have been shown to predict the effects of mutations from sequence variation alone [50], learning the statistical patterns of amino acid co-occurrence in evolutionarily related sequences to infer functional constraints without experimental data. This approach was extended by EVE [51], a Bayesian variational autoencoder (VAE) that learns evolutionary constraints from multiple sequence alignments (MSAs) without requiring experimental labels. EVE achieves state-of-the-art performance on clinical variant classification for human proteins and provides uncertainty estimates through its Bayesian formulation. The success of EVE and related generative models suggests that evolutionary information encoded in natural sequence variation captures much of the same functional signal as DMS experiments, albeit with lower resolution for individual mutations.

1.2.4 Protein Language Models for Variant Effect Prediction

Protein language models (PLMs) trained on large-scale protein sequence databases have emerged as a complementary computational approach to variant effect prediction. The fundamental insight underlying PLMs is that evolutionary constraints on protein function are implicitly encoded in the patterns of amino acid co-occurrence across millions of naturally evolved sequences. By training transformer-based architectures on these patterns, PLMs learn a representation of "evolutionary plausibility" that can be used to score mutations without requiring structural data or experimental fitness measurements.

ESM-2 [52, 53] is a masked language model with 650 million parameters, trained on approximately 250 million protein sequences from the UniRef database. For variant effect prediction, it computes a *log-likelihood ratio* (LLR) at each position: $\mathrm{LLR}(m, i) = \log P(m_{\mathrm{mut}}|\mathrm{context}) - \log P(m_{\mathrm{wt}}|\mathrm{context})$, where m_{mut} and m_{wt} are the mutant and wild-type amino acids at position i, and the context is provided by the surrounding sequence.

Negative LLR values indicate that the mutation is disfavored relative to the wild-type, suggesting functional constraint, while positive values suggest the mutation may be tolerated or even beneficial. ESM-2 LLR scores have shown competitive performance with alignment-based methods on human protein variant effect prediction, though their effectiveness on viral proteins—which evolve under distinct selective pressures and exhibit higher sequence divergence within protein families—is less well characterized. A critical consideration is that ESM-2 was trained predominantly on cellular organism proteins from UniRef; viral proteins, which constitute a small fraction of the training data and exhibit evolutionary dynamics distinct from eukaryotic and prokaryotic proteins, may not be equally well represented in the learned embedding space.

Tranception [54] takes an autoregressive approach, modeling the probability of each amino acid conditioned on the preceding sequence positions. Unlike ESM-2's masked (bidirectional) architecture, Tranception's autoregressive design captures directional dependencies and can incorporate retrieval-augmented inference using homologous sequences at test time. This retrieval mechanism aligns Tranception more closely with alignment-based methods while retaining the flexibility of deep learning.

ProtTrans [55] adopts transformer architectures (including BERT and T5 variants) pre-trained on UniRef and BFD databases totaling over 2 billion protein sequences, achieving competitive performance on protein function prediction tasks including secondary structure prediction and localization. More recently, the ESM-3 generation [56] extended the ESM family to a multimodal generative model integrating sequence, structure, and function tokens, enabling joint reasoning across these modalities; ESM-3 is named here for completeness of the protein-language-model landscape but is *not* used as a scoring channel in Mut-Bench's headline 8,580-evaluation grid because its public-weights release is governed by

EvolutionaryScale's Cambrian non-commercial license (academic only), incompatible with the MIT-redistribution requirement of the headline benchmark (Chapter 2, Section 2.2.7). Structure-aware extensions of the PLM landscape—ESM-IF (inverse folding) [57], SaProt (structure-aware tokenization) [58], and ProteinMPNN (sequence design conditioned on backbone) [59]—explicitly inject 3D structural context into the learned representation; these are likewise outside the 20-scoring panel of the present benchmark and are noted as future-work scoring channels. While ProtTrans, ESM-3, and structure-aware PLMs represent important advances in protein representation learning, to our knowledge no large-scale viral hotspot-detection benchmark has yet incorporated these models as scoring channels, motivating the more conservative ESM-2 / Tranception / EVE-style PLM coverage adopted by MutBench.

Beyond single-model predictions, **EVEscape** [14] demonstrated the value of combining a generative-model fitness signal with complementary biological information. EVEscape integrates three components to predict viral immune escape mutations: (1) EVE fitness scores from a Bayesian variational autoencoder trained on MSAs (i.e., a generative evolutionary model rather than a generic protein language model), (2) structural accessibility from experimentally determined structures (PDB) or AlphaFold predictions, and (3) physicochemical dissimilarity between wildtype and mutant residues (charge and hydrophobicity changes). Its success in forecasting SARS-CoV-2 Omicron escape mutations from pre-pandemic data illustrates the potential of multi-source feature integration. MutBench's `EVEscape_composite` channel adopts the same multiplicative form (fitness $\times$ accessibility $\times$ dissimilarity) but substitutes an ESM-2 masked-marginal mutation-surprise score for the EVE fitness term as a cross-pathogen-feasible proxy (Section 2.2.4)—it is therefore an EVEscape-style implementation, not the original EVEscape model.

AlphaMissense [60] adopted an AlphaFold-derived architecture to classify missense variants across the entire human proteome, achieving high accuracy for pathogenicity prediction. However, its training data and evaluation are centered on human genetic variants, and the distinct evolutionary dynamics of viral proteins—rapid evolution, high mutation rates, and population-level rather than individual-level selection—limit the direct applicability of AlphaMissense to viral variant effect prediction.

A key question for this dissertation is whether PLM-derived scores can serve as effective features for hotspot detection across diverse pathogens. The EVEREST benchmark [11, 61] found that PLMs substantially underperform alignment-based methods on viral proteins, suggesting that the evolutionary patterns learned from predominantly cellular organism sequences may not transfer well to fast-evolving viral proteins; the strength of this finding as *independent* corroboration of MutBench's own pathogen-dependence result is qualified in Section 1.2.8 (PLM $\ll$ alignment is a cross-study *parallel*, not independent confirmation). Despite the growing adoption of PLMs in protein engineering and variant effect prediction, no systematic comparison of PLM-derived features exists for viral mutation hotspot detection across multiple pathogens. In this dissertation, the ESM-2 masked-marginal score and embedding-derived semantic change scores are evaluated as two of the 10 underlying biological information families (expanded into 20 scoring channels with normalization or composite variants) in the MutBench information-type analysis (Section 3.5).

PLM benchmarking. The most comprehensive benchmark for PLM-based variant effect prediction is ProteinGym [12], which evaluates over 70 predictors on 217 DMS assays spanning 2.7 million substitution variants drawn from a taxonomically broad protein set (human, other eukaryote, prokaryote, and viral proteins; general-purpose VEP rather than virus-specific surveillance). ProteinGym has been instrumental in establishing performance rankings among PLMs and alignment-based methods for the per-mutation fitness prediction task: given a specific amino acid substitution at a specific position, predict its continuous fitness effect (evaluated by Spearman correlation against DMS measurements). MutBench differs from ProteinGym in three fundamental respects. First, the *task* differs: ProteinGym evaluates per-mutation effect prediction (a continuous regression problem), whereas MutBench evaluates per-region hotspot detection (a binary classification problem over genomic regions). Second, the *domain* differs: ProteinGym is a general-purpose protein VEP benchmark (taxonomically broad and not virus-specific), while MutBench targets viral proteins across 11 pathogens under distinct evolutionary dynamics (high mutation rate, strong immune-driven positive selection, region-level surveillance use cases). Third, the *analysis objective* differs: ProteinGym ranks prediction methods, while MutBench quantifies the scoring–pathogen interaction to determine which biological information source is optimal for each virus. These distinctions make ProteinGym and MutBench complementary rather than competing: ProteinGym establishes which PLMs best predict individual mutation effects, while MutBench establishes which information types (including but not limited to PLM-derived scores) best identify mutation-enriched genomic regions across diverse pathogens.

1.2.5 Viral Genomic Surveillance Platforms

Real-time viral genomic surveillance platforms provide the operational context in which hotspot detection methods are deployed, but serve a fundamentally different purpose.

Nextstrain [19] integrates phylogenetic analysis with geographic and temporal metadata to track pathogen evolution in real time. Its primary function is to answer "which lineages are currently spreading and where?"—for example, tracking the geographic expansion of SARS-

CoV-2 Omicron sublineages or seasonal influenza clades. Nextstrain visualizes the evolutionary relationships among circulating variants but does not evaluate or compare methods for detecting mutation-prone positions.

PANGO [20] provides a standardized lineage classification system for SARS-CoV-2, assigning labels (e.g., BA.2.86, JN.1) to sequences based on their phylogenetic placement. Its purpose is taxonomic organization—labeling "what is this variant?"—rather than identifying which genomic positions are likely to accumulate future mutations of functional significance.

These platforms cannot substitute for hotspot detection benchmarking for three reasons: (1) they operate primarily at the *lineage/variant level* (which variant is spreading), not at the *position level* (which genomic position is mutation-prone); (2) they are predominantly *retrospective* (tracking variants that have already emerged), although the Bedford-lab forecasting line (Łuksza & Lässig [62]; Huddleston et al. [63]) extends Nextstrain toward short-horizon clade-level prediction at the *lineage* rather than per-position level; and (3) they do not compare or evaluate detection algorithms—they are end-user tools, not method evaluation frameworks. The distinction is therefore one of *primary function* rather than absolute exclusion: surveillance answers "what has happened, and which clades are growing?" while hotspot detection aims to answer "where, at the residue level, might important mutations occur next?" MutBench addresses the latter question by providing a systematic framework for evaluating which detection methods most effectively identify biologically meaningful mutation-prone positions.

Other surveillance-related resources include **GISAID** [64], the primary repository for pathogen genome sequences providing the raw data from which mutation frequencies are computed, and **outbreak.info**, which aggregates variant prevalence data from GISAID. These databases are essential data *sources* for hotspot detection studies (including the present dissertation), but they do not themselves provide detection or benchmarking capabilities. The relationship between surveillance platforms and hotspot detection is thus complementary: surveillance platforms provide the data and operational context, while benchmarking frameworks like MutBench provide the methodological infrastructure for evaluating detection methods applied to that data.

1.2.6 Feature Importance Analysis in Viral Mutation Prediction

Several recent studies have analyzed which biological features predict viral mutation patterns, providing context for the multi-source information analysis in this dissertation (Section 3.5).

Maher et al. [65] analyzed a comprehensive set of features predicting future SARS-CoV-2 variants of concern, including mutation frequency, phylogenetic placement, geographic spread, growth rate, and structural properties. Their key finding was that epidemiological features (geographic spread, growth rate) dominated over sequence-based features for predicting variant emergence in a single pathogen. However, their analysis was restricted to SARS-CoV-2, and the feature importance hierarchy they established may not generalize to pathogens with different evolutionary dynamics.

Rodriguez-Rivas et al. [66] took a co-evolutionary approach, applying Direct Coupling Analysis (DCA) to model epistatic interactions between positions in SARS-CoV-2 proteins. Their epistatic models achieved AUC = 0.76 for predicting escape mutations, substantially outperforming non-epistatic (independent-site) models (AUC = 0.63). This result highlights that mutual information between positions—capturing the constraint network within protein structures—provides predictive power beyond what single-position features can offer.

The broader DCA / EVcouplings / GREMLIN family of position-coupling methods is intentionally outside the 20-scoring panel of MutBench (Section 2.2.4), which restricts itself to single-position scorers to enable the orthogonal scoring $\times$ detection-family ANOVA decomposition; epistasis-aware extensions are flagged as future work in Chapter 4. This single-position restriction is a design choice, not a neutral simplification: because viral escape and compensatory evolution are both epistasis-driven, the benchmark's biological realism is bounded by this exclusion, and the interaction ω^2 reported in Chapter 3 should be read as an information-type contrast *within* the single-position regime rather than as an upper bound on what coupling-aware methods can deliver.

Hie et al. [67] demonstrated that protein language model embeddings can capture "semantic change"—the degree to which a mutation alters the functional meaning of a protein as represented in the PLM embedding space. They showed that viral sequences that are antigenically distinct but phylogenetically close exhibit large semantic change, suggesting that PLM representations can detect functionally significant mutations that traditional phylogenetic or frequency-based approaches might miss.

EVEscape [14] demonstrated that combining evolutionary fitness predictions with structural accessibility and physicochemical dissimilarity features can predict viral escape mutations, providing a practical example of multi-source information integration for viral mutation analysis. Its three-component score (evolutionary fitness from EVE, structural accessibility, and physicochemical dissimilarity) achieved stronger predictive performance than any single component, establishing a precedent for the multi-source approach.

A critical limitation shared by all these studies is their focus on **single pathogens**—primarily SARS-CoV-2 or influenza. None systematically compares feature importance *across* multiple pathogens to determine whether the same features are universally predictive or whether the optimal feature set is pathogen-dependent. This is the gap that the present study addresses: by evaluating 20 scoring channels (derived from 10 underlying biological information families with normalization or composite variants) across 11 pathogens, MutBench quantifies the pathogen-specificity of feature importance and provides concrete recommendations for which information source to prioritize for each virus (Section 3.5).

1.2.7 Algorithm Selection and Benchmarking Methodology

The finding that the best-performing detection method depends on the pathogen (Chapter 3) is an instance of Rice's algorithm selection problem [68]. Rice formalized this as a mapping from a *problem space* (characterized by measurable features of each problem instance) through a *feature space* to an *algorithm space*, with the goal of selecting the algorithm that maximizes a given *performance measure* for each instance. In the context of hotspot detection, each pathogen constitutes a problem instance characterized by features such as genome size, evolutionary rate, and mutation density distribution; the algorithm space comprises the set of scoring–detection combinations; and the performance measure is benchmark-stage-specific—hotspot-score (recall+precision+stability) for Stage 1 SARS-CoV-2 validation and MCC for the Stage 2 11-pathogen comparison (see Section 2.2.5).

This perspective is closely related to the **No Free Lunch (NFL) theorem** [69], which states that no single optimization algorithm outperforms all others across all possible problem instances. Importantly, NFL does *not* imply that all algorithms perform equally on any given problem—rather, it implies that superior performance on one class of problems necessarily comes at the cost of inferior performance on others. For hotspot detection, this means that a scoring type optimized for one pathogen's evolutionary characteristics may be suboptimal for another, motivating the systematic cross-pathogen comparison that MutBench provides.

This framework has been formalized in the machine learning community through meta-learning [70, 71] and automated algorithm selection (AutoML) [72, 73], where the goal is to automatically select the best-performing model or pipeline configuration for a given dataset. The parallel to viral hotspot detection is direct in principle: pathogen characteristics (genome size, mutation rate, population structure, selective pressure regime) constitute the problem-instance features, and scoring–detection pipelines constitute the algorithm space. The concrete pathogen-to-scoring recommendation produced by MutBench under this framing is presented in Chapter 4 as an application of the prior-work framework, not as part of the background itself. However, the algorithm selection framework has not previously been applied to viral genomics in this form, where the "problem instances" are pathogens with diverse evolutionary characteristics and the "algorithms" are scoring–detection pipelines.

The benchmarking methodology adopted in this dissertation follows established principles from computational biology. Weber et al. [74] proposed guidelines for neutral benchmarking, articulating three core requirements: (1) *independent evaluation*, where the benchmark designer is distinct from the method developers; (2) *multiple metrics*, to capture different aspects of performance and avoid optimization for a single criterion; and (3) *controlled experimental design*, ensuring that all methods are evaluated on identical data under identical conditions. Mangul et al. [75] extended these principles to a community-wide audit of omics computational tools, documenting systematic disparities in tool reliability that motivated the present study's emphasis on multi-pathogen rather than single-pathogen evaluation. Boulesteix et al. [76] further emphasized the importance of avoiding "self-assessment bias" where method developers evaluate their own methods on self-selected datasets, which systematically inflates reported performance. MutBench adheres to these principles—with the partial exception that MutClust [18], developed in the same lab as the present dissertation, is one of the 14 evaluated detection families and therefore violates Weber's strict "benchmark designer $\neq$ method developer" criterion. This self-assessment-bias risk (in the sense of Boulesteix et al. [76]) is mitigated, but not eliminated, by three design choices: (i) all 14 detection families are evaluated under identical preprocessing, scoring, and metric pipelines; (ii) the three-layer ground truth (literature-curated adaptive positives, purifying-selection negatives, DMS-anchored Layer C) is constructed independently of any method's predictions; and (iii) MutClust's hyperparameters were frozen prior to Stage 2 sweep execution. The remaining design influence is acknowledged as a limitation in Chapter 4. With this caveat, MutBench evaluates all methods using a composite metric (Stage 1 hotspot-score: recall+precision+stability) for in-depth SARS-CoV-2 validation and MCC for the cross-pathogen Stage 2 comparison (see Section 2.2.5 for the rationale).

1.2.8 Benchmarking Gap in Viral Hotspot Detection

The absence of benchmark studies for hotspot detection methods in viral genomics stands in contrast to cancer genomics, where Bailey et al. [16] conducted a comprehensive Pan-Cancer analysis applying 26 computational tools to 9,423 tumor exomes under uniform conditions, producing a widely used catalog of cancer driver genes. No analogous *detection-task* benchmark exists for viral mutation hotspots: while several recent fitness-regression benchmarks now span comparable virus panels, the region-level hotspot-detection task itself remains unstandardized—detection methods are evaluated using ad hoc validation against small sets of known functional mutations, without standardized datasets, metrics, or ground truth definitions. Related efforts in adjacent domains include: (1) ProteinGym [12, 13], which benchmarks variant effect predictors using 217 DMS assays and 90+ baseline models (as of v1.3) but addresses a per-substitution fitness-regression task rather than hotspot region detection; (1b) **ViroGym** [10] (Mar 2026), which benchmarks PLM and alignment-based scorers across 13 viruses, 79 DMS assays, 552,937 sequences, and 7 phenotypic categories including immune escape—larger virus count than MutBench, but focused on per-variant fitness rather than per-region detection; (2) ConDor [36], which detects convergent mutations across phylogenies but is a single tool rather than a benchmark framework; (3) Pons et al. [77], who reviewed 40+ cancer hotspot detection methods but produced a descriptive survey without a unified benchmark, and cancer-specific models do not transfer to viral evolutionary dynamics; and (4) individual SARS-CoV-2 studies using entropy [8] or clustering that validate against small variant sets without standardized ground truth. Among the 35+ mutation clustering tools catalogued in the computational oncology literature [77], none was designed for or validated on viral population-level mutation data. Cancer hotspot tools (OncodriveCLUST [27], HotMAPS, MutSpot) require tumor cohort somatic mutation calls and human-genome-specific background models (trinucleotide signatures, replication timing). Phylogenetic selection tools (MEME, FUBAR, FEL, SLAC, BUSTED) detect per-codon dN/dS signals but do not perform spatial clustering of hotspot regions. The only virus-specific hotspot clustering tool is MutClust [18], validated on a single pathogen (SARS-CoV-2). This gap—the absence of systematic multi-pathogen hotspot detection benchmarking—is what MutBench addresses. Thus, a systematic benchmark with multi-source ground truth, composite evaluation metrics, and cross-pathogen validation remains absent for viral hotspot detection.

Variant Effect Prediction Benchmarks

Recent years have seen significant progress in benchmarking variant effect prediction (VEP) methods on proteins. ProteinGym [12] established a large-scale benchmark comprising 217 DMS assays and 2.7 million substitution variants, primarily for human and model organism proteins. EVE [51], a Bayesian VAE trained on evolutionary sequences, demonstrated that unsupervised generative models can predict disease-associated variants without clinical labels. EVEscape [14] extended this framework to viral immune escape prediction by combining EVE fitness scores with structural accessibility and physicochemical dissimilarity, successfully forecasting SARS-CoV-2 Omicron escape mutations from pre-pandemic data.

Most recently, EVEREST [11, 61] specifically benchmarked VEP methods across WHO priority viruses; the v3 expansion (Jan 2026) covers 45 viral DMS datasets, 31 forecasting clades across 4 viruses, and reliability assessment across 40 WHO priority RNA viruses spanning 16 viral families. EVEREST finds that protein language models substantially underperform alignment-based approaches on viral proteins, that reliability fails for over half of the WHO-40 panel (especially surface glycoproteins), and that prediction reliability varies dramatically across pathogens. Because MutBench's LOPO 0/11 match rate is itself null-consistent under permutation (Chapter 3), we describe this as a cross-study *parallel* on pathogen-dependence rather than as independent corroboration of non-transferability per se. The PLANT framework [78] and Shah et al. [79] provide PLM-based and ML-based H3N2 antigenic cartography that explicitly competes with WHO vaccine-strain selection, building on the foundational antigenic-cartography paradigm of Smith et al. [80]; MutBench's H3N2 self-consistency check (Section 4.4.2) is positioned as a detection-side complement to these prediction-side cartographies, not as a competitor. A particularly relevant finding from EVEREST is that the performance gap between methods varies by viral family: alignment-based methods such as EVE consistently outperform PLMs across most viral proteins, but the magnitude of this advantage differs substantially between, for example, influenza HA and SARS-CoV-2 Spike. This pathogen-dependent performance variation parallels the central finding of MutBench (Chapter 3). AlphaMissense [60] demonstrated proteome-wide variant classification using an AlphaFold-derived architecture, though its focus on human proteins limits direct viral applicability.

MutBench's unique contribution is the *scoring $\times$ pathogen interaction analysis*: while EVEREST establishes that method performance varies across viruses, MutBench quantifies this interaction through three-way ANOVA ($\omega^2 = 0.296$) and demonstrates that the choice of *biological information source* (not just the computational method) is the dominant performance factor. This finding—that the optimal information source is pathogen-dependent within the 20-type/39-variant design, with the interaction as the largest modeled variance component—has no counterpart in EVEREST or ProteinGym, which treat the scoring/feature set as fixed. Specifically, MutBench provides three capabilities absent in EVEREST and other VEP benchmarks: (1) quantification of the scoring–pathogen interaction through three-way ANOVA, revealing that the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$; 11-pathogen cluster-bootstrap 95% interval [0.195, 0.333]) is the largest modeled variance component (within-cell residual $\omega^2 \approx 0.39$); (2) cross-validation via vaccine escape enrichment for the 3 of 11 pathogens with curated escape annotations, with HIV-1 (7.19$\times$, Bonferroni-corrected $p = 2.5 \times 10^{-16}$; escape set 82% disjoint from Layer A) as the primary external anchor and H3N2 9.36$\times$ reported as a self-consistency check given 69% Layer A overlap (circularity audit in Table 3.18); and (3) a concrete pathogen-to-scoring recommendation table enabling application to new pathogens without requiring DMS experimental data. Furthermore, MutBench introduces the vaccine escape cross-validation paradigm with a per-pathogen circularity audit: HIV-1 provides the cleanest external validation, while H3N2 and SARS-CoV-2 contribute self-consistency and exploratory evidence respectively—a distinction absent from EVEREST or ProteinGym. Table 1.2 summarizes the key differences between MutBench and these related benchmarks.

Table 1.2: Cross-benchmark comparison: task, scope, ground-truth, methods compared. The ViroGym row reflects the March 2026 arXiv preprint (pre-peer-review) and is included for completeness; EVEREST v3 (Jan 2026) and ProteinGym v1.3 are peer-reviewed releases.

	MutBench	**ViroGym**	**EVEREST v3**	**ProteinGym v1.3**	**Bailey 2018**
Domain	Viral hotspot	Viral VEP	Viral VEP	General VEP	Cancer driver
Task	Detection	Prediction	Prediction	Prediction	Detection
Pathogens/proteins	11 viruses	13 viruses (79 DMS)	4–40 viruses (45 DMS)	217 proteins	Cancer genes
Ground truth	3-layer	DMS+escape	DMS	DMS	Patient cohorts
Methods compared	780 combos	$\sim$10 PLM/MSA	$\sim$20 VEP	90+ VEP	$\sim$10 tools
Cross-pathogen ANOVA	Yes (ω^2)	No	No	No	No

Direct performance comparison between MutBench and ProteinGym, EVEREST, or ViroGym is not possible because the tasks differ fundamentally: VEP benchmarks (including ViroGym's per-variant fitness/escape regression and forecasting tasks) evaluate Spearman correlation or AUROC between predicted and measured per-variant outcomes, while MutBench evaluates MCC for binary per-region hotspot classification. The benchmarks are complementary rather than competing.

This gap—systematic multi-pathogen benchmarking for hotspot detection—is the fundamental motivation for MutBench (Chapter 2), the core study of this dissertation.

Related research summary. This section has reviewed the two bodies of prior work that directly inform MutBench. Section 1.2 surveyed viral mutation hotspot detection methods—from frequency-based and entropy-based scoring through density-based clustering to protein language models and selection-based approaches—and identified a critical benchmarking gap: no standardized framework exists for evaluating and comparing these methods across multiple pathogens with independent ground truth. The only virus-specific hotspot detection tool, MutClust [18], relies on a single pathogen, single information type (H-score), and self-referential validation, leaving open the question of how different scoring approaches and detection algorithms compare across diverse viral pathogens. Together, these gaps motivate the design of MutBench as a systematic, multi-pathogen benchmark with three-layer ground truth, composite evaluation metrics, and statistical analysis of scoring–pathogen interactions. Coupling-aware methods (DCA, EVcouplings, GREMLIN, and similar pairwise/epistatic models) remain explicitly out of scope of the 20-scoring panel; the scoring $\times$ pathogen interaction $\omega^2 = 0.296$ and all downstream Stage 2/3 conclusions should therefore be read within a single-position scoring regime, with epistasis-driven hotspot detection deferred as future work (Chapter 4, Section 4.7). Chapter 2 presents the MutBench framework in detail, including the construction of multi-layer ground truth datasets for 11 pathogens, the benchmark design comprising 20 scoring types across 6 categories and 14 detection method families, and the factorial ANOVA methodology for quantifying scoring–pathogen interactions.

1.3 Research Motivation and Problem Definition

The absence of systematic evaluation for viral mutation hotspot detection gives rise to the following three specific problems.

First, there is a **lack of a standardized evaluation framework**. Existing hotspot detection methods, including MutClust, report their performance using their own datasets and evaluation criteria, making fair comparison among methods impossible. The bootstrap test, MutClust's sole validation method, is a self-referential approach that can only confirm the statistical significance of detected clusters; it cannot verify whether those clusters correspond to biologically meaningful functional regions.

Second, there is an **absence of validation based on independent ground truth**. Objectively evaluating hotspot detection results requires a ground-truth standard that is independent of the algorithm, yet no such ground truth framework has been defined for viral mutation hotspot detection. Although independently obtainable evidence sources exist—such as convergent evolution sites, sites under purifying selection (natural selection that removes deleterious mutations), and Deep Mutational Scanning (DMS; a technique that experimentally measures the functional impact of every possible single amino-acid substitution in a protein) data—they have not been systematically integrated. Convergent evolution sites are mutations that independently arise at the same genomic position in multiple lineages within the same viral species; for example, mutations at Spike position 484 emerged independently across multiple SARS-CoV-2 lineages: E484K (glutamic acid to lysine) in Beta, Gamma, and Mu, and E484A (glutamic acid to alanine) in Omicron.

Third, there is a **lack of statistical evidence on the pathogen-dependence of best-performing methods**. Existing methods use parameters tuned for a single pathogen, and their generalizability has not been rigorously quantified. Whether differences in pathogen-specific genomic characteristics (genome length, mutation rate, recombination frequency, etc.) systematically affect the best-performing detection strategy has not been verified. This is a specific instance of the algorithm selection problem [68]: given a pathogen with particular genomic characteristics, which detection method should be applied? Without systematic cross-pathogen evaluation, this question cannot be answered.

1.4 Research Objectives

To address the problems above, this study establishes the following three objectives.

Objective 1. Construct a systematic experimental framework (MutBench) that enables standardized comparison of diverse biological information types for hotspot detection

across 11 RNA viruses, comprising:

- A three-layer ground truth (Layer A literature-curated positives—a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, dominated by immune-escape sites; Layer B purifying-selection negatives; Layer C DMS fitness validation),
- The hotspot-score composite metric (recall + precision + stability)/3,
- A two-stage main experimental design with a Stage 3 information-integration layer: Stage 1 for in-depth SARS-CoV-2 analysis; Stage 2 for large-scale comparison across 11 RNA viruses (20 scoring formulas $\times$ 39 detection parameter variants from 14 families $\times$ 11 pathogens = 8,580 evaluations); and Stage 3 for multi-source feature integration on the 11-pathogen main panel, extended to 12 pathogens by adding Zika for the feature-ablation and nested-LOPO analyses (see Chapter 3 Section 3.5.2).

Objective 2. **Determine which factor—scoring type, detection method, or pathogen—dominates the variance in hotspot-detection performance**, and whether the dominant factor is a main effect or an interaction. Operationally, this asks whether a single "best" scoring–detection combination exists across pathogens, and is addressed through factorial Analysis of Variance (ANOVA), Friedman test, and Leave-One-Pathogen-Out (LOPO) cross-validation (see Chapter 3).

Objective 3. **Externally validate the benchmark on independently-curated vaccine-escape positions, prioritising the Layer-A-disjoint subset to test whether the framework reduces the experimental search space on positions that are not already in the literature-curated ground truth** (see Chapter 3). A multi-source feature-ablation robustness check (compact 4-feature core vs. full 10-feature ensemble under nested-LOPO) is reported alongside, but is treated as a methodological sanity check on the EqualWeight design rather than as a separate contribution.

Figure 1.1 summarizes the overall research flow: from identifying the information limitation (reliance on frequency/entropy), through systematic comparison of 10 information types via MutBench, to the key finding that the optimal information source differs by pathogen, culminating in external vaccine-escape validation on HIV-1 as the primary anchor for practical impact.

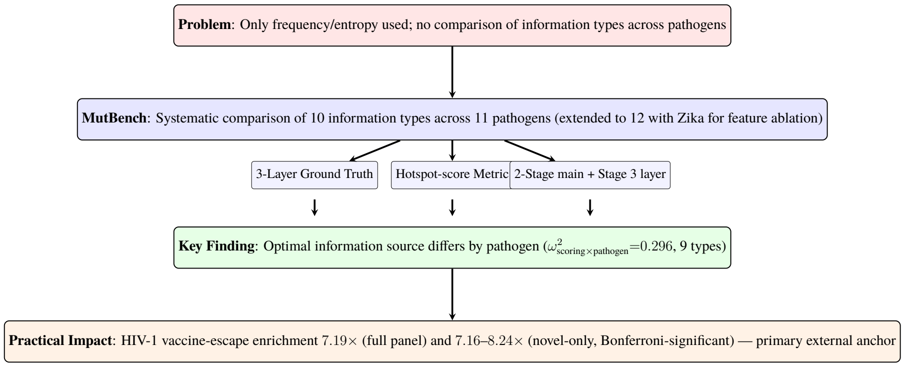

Figure 1.1: Research overview. The study proceeds from identifying the information limitation (reliance on frequency/entropy) through systematic comparison of 10 information types via MutBench, to the key finding that the optimal information source differs by pathogen, culminating in external vaccine-escape validation on a Layer A-disjoint subset (HIV-1, Bonferroni-significant) as the primary anchor for practical impact.

1.5 Research Contributions

The research contributions of this study are as follows.

Contribution 1. MutBench: a systematic hotspot-detection benchmark across 11 RNA viruses. To our knowledge, MutBench is the broadest *region-level, single-position* hotspot-detection benchmark to date for RNA viruses (8,580 evaluation cells across the

20-type single-position scoring regime and the 11-pathogen main panel, with the Stage 3 feature-integration analysis extended to a 12-pathogen panel by adding Zika), complementing concurrent fitness-regression benchmarks (ViroGym [10], EVEREST v3 [11], ProteinGym v1.3 [13]) by targeting region-level detection rather than per-variant fitness, and is designed to answer the question: *which biological information source is most effective for which pathogen?* It organizes 10 information types into 20 scoring formulas across 6 categories (frequency, MSA-derived, phylogenetic, structural, AI-based, and composite), defines a three-layer ground truth framework (Layer A as a literature-curated, pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types—with immune-escape sites dominating; Layer B as purifying-selection negatives; Layer C as DMS experimental fitness), introduces the hotspot-score composite metric, and presents a two-stage main experimental design plus a Stage 3 information-integration layer combining in-depth SARS-CoV-2 analysis (Stage 1), a large-scale comparison across the 11-pathogen main panel (Stage 2), and multi-source feature-integration analysis on the 12-pathogen panel (Stage 3, extended from the main panel with Zika; Section 3.5.2).

Contribution 2. Statistical demonstration that the optimal biological information source differs fundamentally across pathogens ($\omega^2_{\textbf{scoring}\times\textbf{pathogen}} = 0.296$), with 9 distinct optimal information types across 11 pathogens; LOPO 0/11 is reported as null-consistent corroboration of the interaction effect, not as independent evidence. All eleven pathogens exhibited different best-performing combinations (LOPO cross-validation: 0/11 matches), with 9 distinct MCC-best scoring types observed—including FUBAR phylogenetic selection scores for H3N2, protein language model scores for SARS-CoV-2 and RSV (MCC-best at the scoring–detector cell), and homoplasy for Norovirus. The Friedman test ($p = 0.990$) showed no reliably superior combination across all pathogens, consistent with pathogen-dependent optimality. The scoring $\times$ pathogen interaction is the largest modeled variance component ($\omega^2 = 0.296$, 95% cluster-bootstrap CI [0.195, 0.333]; 6-category aggregation gives $\omega^2 = 0.103$ as a collinearity-robust lower bound; 11 pathogen clusters places the design near the small-cluster regime—above the Bolker rule-of-thumb minimum of 5–6 but below the typical recommendation of 20–30 for stable random-effects estimation, see Chapter 4, Section 4.6). The conclusion rests on the interaction effect plus the vaccine-escape audit; LOPO 0/11 alone is null-consistent under permutation (Chapter 3 Table 3.6).

Contribution 3. Practical search-space reduction validated on independently-curated vaccine-escape positions. The optimal scoring–detection combination narrows the candidate list from $\sim$1,000 alignment positions to approximately 20–80 (top 5–10%) with 7–9$\times$ enrichment for escape sites, transforming exhaustive experimental screening into a guided investigation that directs limited laboratory resources to the most promising positions. Under a three-tier evidence rule (Bonferroni survival $\times$ Layer A overlap $< 33\%$), the Layer-A-disjoint subset of HIV-1 escape positions (37 of 45) yields direct novel-only enrichment of 7.16–$8.24\times$ (worst-case nominal Fisher $p = 5.0 \times 10^{-12}$; Bonferroni-adjusted $p_{\text{adj}} \approx 3.9 \times 10^{-9}$ across 780 combinations); H3N2's $9.36\times$ is a self-consistency check (69% Layer A overlap), and SARS-CoV-2 is exploratory after multiple-comparison correction. The H3N2 EqualWeight integration case yields $4.012\times$ enrichment ($p = 0.0023$); per-feature analyses show the most discriminative feature differs across pathogens (homoplasy for Norovirus, pLDDT for H3N2, ESM-2 for SARS-CoV-2), complementing the Stage 2 MCC-optimal scoring results in Contribution 2. A multi-source feature-ablation robustness check (nested-LOPO on the 12-pathogen Stage 3 panel: full-10 LOPO mean MCC 0.070, fixed-4 LOPO mean MCC 0.081, paired delta $+0.011$, 95% CI $[-0.018, +0.042]$, sign-flip permutation $p = 0.61$; Section 3.5.2, Table 3.13) shows the compact 4-feature core (homoplasy, pLDDT, entropy, frequency) is statistically indistinguishable from the full 10-feature EqualWeight ensemble within a ± 0.04 MCC non-inferiority bound, providing a practical cold-start recipe when labelled positions are unavailable.

These three contributions form a logical hierarchy: Contribution 1 provides the standardized evaluation infrastructure; Contribution 2 uses it to establish pathogen-dependent information-source utility; and Contribution 3 provides external validation on independently-curated vaccine-escape positions.

1.6 Dissertation Organization

This dissertation comprises four chapters.

Section 1.2, Related Research (within this chapter) reviews the existing literature on viral mutation hotspot detection, benchmarking methodology, and the gap that motivated the development of MutBench.

Chapter 2, Materials and Method describes in detail the materials and methodology of the MutBench benchmark framework, the core contribution of this dissertation. The chapter covers the target pathogens and sequence data, the framework overview and three-layer ground truth construction, the Stage 1 SARS-CoV-2 analysis design including MutClust variants and the hotspot-score metric, the Stage 2 cross-pathogen benchmark design with 20 scoring types and 14 detection families, the Stage 3 multi-source information integration design, and the evaluation metrics and statistical analysis design.

Chapter 3, Experimental Results reports the experimental results organized into five sections: Single-Pathogen Benchmark (synthetic data comparison, multi-ground truth evaluation, real GISAID validation, and external methods comparison on SARS-CoV-2), Robustness Validation (statistical validation, sensitivity analysis, temporal generalization, and indel exclusion), Cross-Pathogen Validation (H3N2 validation, 3D structural analysis, phylogenetic simulation, and ESM-2 cross-paradigm validation), the Cross-Pathogen Large-Scale Benchmark (ANOVA, Friedman test, LOPO cross-validation), and Information-Type Analysis (feature ablation on the 12-pathogen Stage 3 panel, ground truth heterogeneity, search space reduction, and multi-source integration).

Chapter 4, Discussion, Limitations, and Future Directions provides a comprehensive interpretation of the experimental results, summarizes the contributions of this research, analyzes its limitations, and presents future research directions.

Chapter 2

Materials and Method

This chapter presents the methodology of MutBench, an experimental framework for analyzing which factors—biological information type, detection method category, and pathogen characteristics—influence viral mutation hotspot detection performance. The overall pipeline processes multiple sequence alignment (MSA) data from 11 RNA viruses through 20 scoring types and 14 detection method families spanning 5 categories (threshold-based, signal processing, density/clustering, statistical, and adaptive window), with 39 parameter variants, evaluating results against a 3-layer ground truth system using the hotspot-score composite metric and Matthews Correlation Coefficient (MCC). The evaluation follows a two-stage main design with a Stage 3 information-integration layer (Stages 1 and 2 form the main 11-pathogen factorial benchmark; Stage 3 is a separate information-integration layer): Stage 1 conducts in-depth analysis on SARS-CoV-2 to validate the framework components, Stage 2 extends the evaluation to all 11 pathogens for the large-scale cross-pathogen MCC comparison, and Stage 3 (Section 2.2.6) decomposes the resulting performance by information type via per-feature AUC, ablation, Random Forest, and vaccine-escape cross-validation. Section 2.1 describes the materials (target pathogens and sequence data). Section 2.2 presents the methods, beginning with a framework overview (Section 2.2.1) that describes the overall architecture and three-component pipeline, followed by the 3-layer ground truth framework

(Section 2.2.2). The experimental design is then presented in two stages: Stage 1 (Section 2.2.3) focuses on SARS-CoV-2 in-depth analysis including MutClust variant comparison and the hotspot-score metric, while Stage 2 (Section 2.2.4) extends to the 11-pathogen cross-pathogen benchmark with 20 scoring types and 14 detection families. Section 2.2.5 details the evaluation metrics and statistical analysis design.

2.1 Materials

2.1.1 Target Pathogens and Data

Eleven RNA viruses were selected to cover diverse evolutionary dynamics, transmission routes, and genomic characteristics: (1) *Respiratory viruses*: SARS-CoV-2 (Spike, 1,259 AA), MERS-CoV (Spike, 1,330 AA), RSV (F protein, 550 AA), Influenza A H3N2 (HA, 543 AA), and Influenza B (HA, 560 AA); (2) *Blood-borne/sexually transmitted*: HIV-1 (Env gp120, 450 AA) and HCV (E2, 340 AA); (3) *Enteric*: Norovirus (VP1, 536 AA); (4) *Arthropod-borne*: Dengue (E protein, 490 AA); (5) *Neurotropic*: Rabies (G protein, 491 AA) and EV-A71 (VP1, 261 AA). *Note: all amino acid lengths above refer to post-truncation alignment lengths used in the benchmark; full-length protein sequences may differ (e.g., SARS-CoV-2 Spike full length is 1,273 AA, MERS-CoV Spike 1,353 AA, Rabies G 524 AA, EV-A71 VP1 297 AA).* These 11 pathogens span substitution rates from $\sim 10^{-3}$ (HIV-1, SARS-CoV-2) to $\sim 10^{-4}$ (Rabies) substitutions/site/year, alignment lengths from 261 to 1,330 AA, and evolutionary patterns from rapid antigenic drift (H3N2) to epochal evolution (Norovirus). This diversity ensures that the benchmark evaluates method performance across a representative range of viral evolutionary regimes rather than a single pathogen's characteristics. Zika virus was considered but excluded from the Stage 2 benchmark because only 52 unique E protein sequences were available in GenBank, insufficient for reliable per-position frequency and entropy estimation; Zika is included in the supplementary 12-pathogen feature analysis (Section 3.5).

Per-pathogen biological context. The 11 pathogens span 8 ICTV viral families and four transmission routes (respiratory, blood-borne/sexually transmitted, enteric, arthropod-borne and neurotropic), with substitution rates from $\sim 4 \times 10^{-4}$ (Rabies, slow purifying-selected G protein) to $\sim$ 4–4.5×10^{-3} (H3N2 HA antigenic drift, EV-A71 VP1) substitutions/site/year and unique-sequence ratios from 15.1% (SARS-CoV-2 Spike, dominated by near-identical pandemic lineages) to 100% (Rabies G, EV-A71 VP1; Table 2.1). Five biological points are load-bearing for the benchmark and re-appear in the Layer A construction (Section 2.2.2, Table 2.3) and in the discussion of pathogen-specific best-method patterns: (i) SARS-CoV-2 Spike's shallow pandemic phylogeny (∼5 years) creates strong founder effects (D614G) alongside convergent immune-escape sites (E484K, N501Y), making it the richest three-layer ground-truth pathogen and the Stage 1 anchor; (ii) H3N2 HA exhibits the canonical five antigenic sites (A–E) under continuous immune-driven drift, providing the most homogeneous Layer A construction; (iii) Influenza B is pooled across the Victoria and Yamagata lineages (reference `CY018765` is a Yamagata HA), so lineage-defining substitutions appear as high-frequency positions in the combined MSA—a stratification gap revisited in Chapter 4 Section 4.4.7; (iv) HCV E2 carries the prototypical hypervariable region (HVR1, residues 384–410) under diversifying selection while the CD81-binding residues remain conserved, providing built-in positive/negative controls and motivating the HCV-only ANOVA sensitivity analysis ($\omega^2 = 0.246$, Section 3.4.4); (v) Norovirus VP1 evolves under epochal selection at the GII.4 P2 subdomain, generating the cleanest convergent-evolution Layer A signal in the panel (homoplasy AUC 0.944). DMS Layer C data are available for SARS-CoV-

2, H3N2, HIV-1, RSV, Rabies, and EV-A71. Zika was considered but excluded from the Stage 2 benchmark (only 52 unique E-protein sequences); it is included in the supplementary 12-pathogen feature analysis (Section 3.5).

FASTA sequences for each pathogen were collected from NCBI GenBank by querying each target protein name (e.g., "SARS-CoV-2 spike glycoprotein") with filters for complete coding sequences, excluding partial sequences and laboratory-derived constructs. The benchmark scope is the *single-position regime within the coding region of the surface glycoprotein only*: $5'/3'$ UTRs, IRES elements, polyprotein-cleavage junctions outside the assayed protein boundary (e.g., HCV E2 outside the 340-AA evaluated alignment), pseudogenes, and non-coding selection signals are explicitly out of scope and are not represented in any per-position scoring vector. For HCV the polyprotein structure means only the E2 region under the literature-curated alignment window is evaluated; CD81-binding regions reported in polyprotein numbering outside this window are noted in Table 2.3 but not converted into Layer A/B labels. No date range restriction was applied; all available sequences as of the per-pathogen query dates listed in Table 2.1 (column "Query date"; all queries fell within the 2024-11 to 2024-12 window) were downloaded. As an illustrative example, the SARS-CoV-2 Spike query executed via NCBI E-utilities on 2024-12-09 was `(``Severe acute respiratory syndrome coronavirus 2''[Organism] OR txid2697049[Organism]) AND spike[Gene Name] AND ``complete cds''[Properties] NOT (partial[Title] OR laboratory[Title])` with a length filter of 3,500–4,000 nt; analogous query strings for the remaining ten pathogens (organism/taxon ID, gene/product filter, length window, exclusion clauses) are archived verbatim in `results/mutbench/provenance/genbank_queries.csv` so that any reader can reproduce the per-pathogen sequence sets to within the inevitable GenBank submission drift after the recorded query date. Sequences with >5% ambiguous bases (N) were removed, and identical (100% identity) sequences were deduplicated using a self-implemented Python hashing pass (canonical-uppercase amino-acid string $\rightarrow$ SHA-256), which is functionally equivalent to running CD-HIT [81] or MMseqs2 [82] at a 100% identity threshold for exact-duplicate removal because at the 100%-identity setting both tools reduce to exact-string deduplication (no partial-match scoring), so the SHA-256 hash and the tool output produce *arithmetically identical* unique-sequence sets and the choice avoids an additional dependency in the reproducibility environment. Clustering at $<$100% identity (e.g., the 0.95–0.99 thresholds typically used with CD-HIT-EST/MMseqs2 `easy-cluster`) was *not* applied, so near-duplicate but non-identical sequences were retained intentionally to preserve per-position frequency and entropy estimates. HMM-based homology filtering (HMMER [83], `phmmer/hmmsearch`) was likewise *not* applied: the GenBank query already restricts retrieval to gene-name and "complete cds" flags, and any frameshift or partial-domain mis-annotations passing those flags are bounded by the post-MAFFT minimum-length truncation step (Section below); a follow-up HMMER profile-screen pass against a per-pathogen reference HMM is left for the next data-curation cycle. The "Unique seq." column in Table 2.1 reflects post-deduplication counts under the 100% identity threshold. Multiple sequence alignment (MSA) was performed for each surface protein (SARS-CoV-2: Spike, H3N2: HA, HIV-1: Env gp120, etc.) using MAFFT v7.505 [84] with the `--auto` mode, followed by computation of per-position frequency $f_i = 1 - p_{\text{consensus}}(i)$ (the minor-allele fraction at column i) and Shannon entropy on the same column. The `--auto` mode automatically selects among FFT-NS-2, L-INS-i, and G-INS-i based on the number and length of input sequences (Katoh & Standley, 2013, §2) [84]; the actual algorithm selected per pathogen is logged in Table 2.1 (column "MAFFT mode"), and the GenBank query strings used for retrieval are archived alongside the released CSV bundle (`results/mutbench/provenance/genbank_queries.csv`). At the sequence scales used in this study (1,311–5,325 sequences, Table 2.1), FFT-NS-2 was the dominant selection (10 of 11 pathogens); the per-pathogen log in the column is the authoritative reference for any reproduction attempt. The effect of `--auto` algorithm selection on alignment quality—and consequently on per-position mutation frequency computation—was not separately ablated in this study; the per-pathogen log is provided so that future cycles can pin individual algorithms (e.g., `--retree 2 --maxiterate 0` for FFT-NS-2) for sensitivity analyses. When alignment lengths differed across sequences, all sequences were truncated to the minimum sequence length within the MSA to ensure consistency in per-position comparisons. Ground truth positions exceeding the alignment length are thereby excluded from evaluation (the position counts in Table 2.3 are effective values after this truncation). Table 2.1 summarizes the characteristics of each pathogen's sequence dataset.

Table 2.1: Sequence dataset characteristics for 11 pathogens used in the benchmark. "Query date" is the YYYY-MM-DD on which the GenBank query was executed; "MAFFT mode" is the algorithm actually selected by MAFFT v7.505 `--auto` for that pathogen's input set (per-pathogen log).

Pathogen	Surface protein	Reference accession	Sequences	Unique seq.	Align. len. (AA)	Query date	MAFFT mode
SARS-CoV-2	Spike	NC_045512.2 (Wuhan-Hu-1)	4,982	753	1,259	2024-12-09	FFT-NS-2
H3N2	HA	CY045572 (A/Brisbane/10/2007)	2,644	2,562	543	2024-12-09	FFT-NS-2
Norovirus	VP1	KC576909 (GII.4 Sydney)	1,500	791	536	2024-11-22	FFT-NS-2
HIV-1	Env gp120	K03455 (HXB2)	3,000	2,256	450	2024-11-22	FFT-NS-2
Dengue	E protein	NC_001474.2 (DENV-2)	2,000	796	490	2024-11-25	FFT-NS-2
RSV	F protein	KT992094 (RSV-A)	5,070	4,779	550	2024-12-09	FFT-NS-2
Influenza B	HA	CY018765 (B/Florida/4/2006)	5,325	5,019	560	2024-12-09	FFT-NS-2
MERS	Spike	NC_019843.3 (MERS-CoV)	1,311	530	1,330	2024-11-22	FFT-NS-2
HCV	E2	NC_004102.1 (HCV-1a)	1,362	1,067	340	2024-11-22	FFT-NS-2
Rabies	G protein	NC_001542.1	2,038	2,038	491	2024-11-25	FFT-NS-2
EV-A71	VP1	AB204852	662	662	261	2024-11-25	L-INS-i

Query strings (full GenBank advanced-search expressions including taxonomy ID, gene/product filters, and length bounds) are archived in `results/mutbench/provenance/genbank_queries.csv`. MAFFT `--auto` selects L-INS-i when the input is small enough that the iterative refinement is tractable ($N \times L$ below the heuristic boundary, broadly $N \lesssim 200$ or short L) and FFT-NS-2 otherwise (Katoh & Standley 2013, §2). EV-A71 ($N = 662$ unique sequences, $L = 261$ AA) falls inside the L-INS-i band because its $N \times L \approx 1.7 \times 10^5$ product remains below the FFT-NS-2 cut-over despite $N > 200$; the remaining ten pathogens have larger $N \times L$ products and were therefore aligned under FFT-NS-2.

2.2 Methods

2.2.1 Framework Overview

The MutBench framework follows a three-component pipeline architecture (Figure 2.1): (1) ground truth construction from three independent biological evidence sources, (2) systematic scoring and detection via 20 scoring types and 14 detection method families, and (3) multi-metric evaluation with factorial statistical analysis. The evaluation is organized into two main stages (Stages 1 and 2), with a separate Stage 3 information-integration layer described in Section 2.2.6: Stage 1 performs in-depth analysis on SARS-CoV-2 to validate the framework components, and Stage 2 extends the evaluation to all 11 pathogens for large-scale cross-pathogen comparison. The following subsections describe each component in detail.

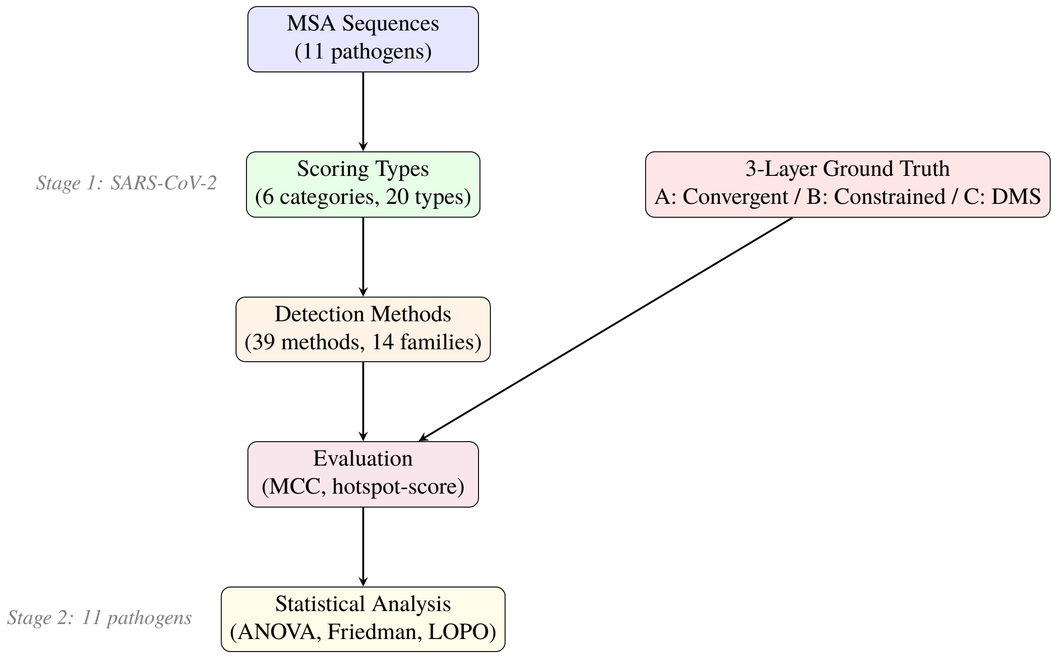

Figure 2.1: Overview of the MutBench experimental framework pipeline. Input multiple sequence alignment (MSA) data are processed through scoring types and detection methods, evaluated against a 3-layer ground truth, and analyzed using multiple statistical tests.

Component 1: Ground Truth Construction. A 3-layer ground truth system captures three orthogonal dimensions of mutational importance: Layer A (literature-curated adaptive/diversifying positions) defines true positives, Layer B (purifying selection) defines constrained negatives for a separate false-positive audit, and Layer C (DMS fitness) provides experimental validation. For Stage 1, synthetic data with known signal positions and DMS-based functional site annotations provide additional controlled validation. Details are presented in Section 2.2.2.

Component 2: Scoring and Detection. Twenty scoring types organized into six categories (frequency-based, MSA-derived, phylogenetic, structural, AI-based, composite) assign per-position scores from diverse biological information sources. These scores are then processed by 14 detection method families (39 parameter variants) spanning five algorithmic categories (threshold-based, signal processing, density/clustering, statistical, adaptive window) to convert continuous score profiles into discrete hotspot calls. Stage 1 provides a controlled SARS-CoV-2 sanity check for MutClust variants and temporal scoring (Section 2.2.3); Stage 2 extends to the full scoring $\times$ detection $\times$ pathogen factorial design (Section 2.2.4).

Component 3: Evaluation and Statistical Analysis. Position-level evaluation metrics (MCC, F1, enrichment ratio, constrained FPR, DMS F1) quantify detection performance, and factorial statistical methods (three-way ANOVA with ω^2, Friedman test, LOPO cross-validation) decompose the sources of performance variation. Stage 1 uses the hotspot-score composite metric; Stage 2 adopts MCC as the primary metric. Details are presented in Section 2.2.5.

2.2.2 3-Layer Ground Truth Framework

To mitigate single ground truth bias, a 3-layer ground truth framework based on three independent sources of biological evidence was constructed. Figure 2.2 illustrates the structure and integration of the three layers.

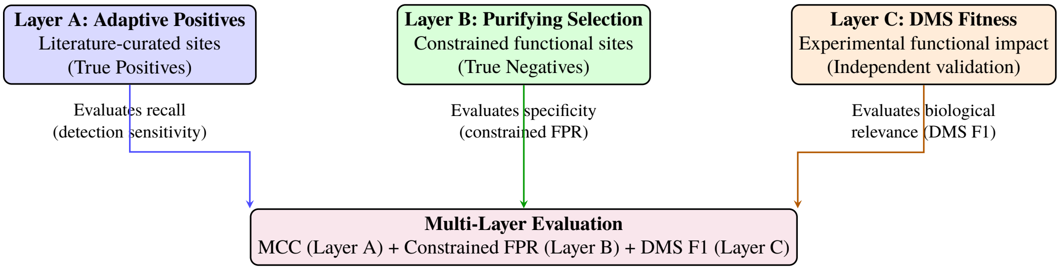

Figure 2.2: 3-Layer Ground Truth Framework. Layer A (literature-curated adaptive/diversifying positions) defines true positives for MCC computation, Layer B (purifying selection) defines constrained negatives for constrained FPR, and Layer C (DMS fitness) provides experimental validation. The three layers capture orthogonal dimensions of mutational importance.

Design rationale: three orthogonal dimensions. The three layers are deliberately designed to capture different dimensions of mutational importance. Layer A asks whether a position is supported by literature as adaptively or diversifyingly important for that pathogen—through recurrent emergence, immune-escape evidence, epochal-variant membership, or hypervariable-region membership depending on the pathogen. Layer B asks "is this position structurally constrained?" (a *structural* question), identifying positions where any mutation destroys protein function. Layer C asks "what is the measurable functional impact of mutations at this position?" (a *functional* question), using laboratory DMS experiments that directly quantify how each amino acid substitution affects viral fitness (e.g., cell entry efficiency, receptor binding affinity). This separation means each layer provides a non-redundant perspective: Layer A captures literature-curated adaptive or diversifying evidence that DMS cannot always provide, while Layer C captures functional effects that may not yet be visible in surveillance-derived annotations.

Layer A: Adaptive Positive Selection

Layer A positions are literature-curated adaptive or diversifying positions treated as hotspot positives. For pathogens where convergent-evolution evidence is available, positions where mutations recurrently arose in multiple independent lineages *within the same viral species* are classified as hotspots under positive selection. For example, mutations at Spike position 484 in SARS-CoV-2—enabling immune evasion by altering the antibody binding surface—emerged independently across multiple lineages: E484K (glutamic acid to lysine) in Beta, Gamma, and Mu, and E484A (glutamic acid to alanine) in Omicron, each descending from different ancestral branches of the phylogenetic tree. This repeated independent emergence at the same position provides strong evidence of positive selection. Importantly, Layer A does not impose one universal biological mechanism: the annotation can reflect immune evasion, receptor-binding adaptation, diversifying selection, or recurrent lineage-level emergence depending on the pathogen and source study. These positions are used as **True Positives**, evaluating whether detection methods correctly identify them as hotspots.

Identification criteria. For each pathogen, Layer A positions were compiled from published studies (Table 2.3, rightmost column). The construction is operationally heterogeneous across the 11 pathogens: SARS-CoV-2 and H3N2 use phylogenetically confirmed convergent evolution sites ($\geq$3 independent lineages); Norovirus uses epochal-variant positions; HCV uses the hypervariable regions HVR1/HVR2 (diversifying selection rather than convergence); and the remaining seven pathogens use literature-reported immune-escape position lists. No single computational criterion is applied uniformly—the framework adopts the per-pathogen literature-curated list to maximize biological fidelity, accepting in exchange that "Layer A" denotes a heterogeneous adaptive-selection union (convergent evolution $\cup$ immune escape $\cup$ HVR membership) rather than a single algorithmic construct. This heterogeneity is revisited in Chapter 4 where its quantitative impact on the headline $\omega^2 = 0.296$ is discussed; for the purposes of this chapter the operational definition is: *Layer A is the union of literature-curated positive/diversifying-selection positions per the references in Table 2.3*. As a forward-disclosure cross-reference, the Stage 2 sensitivity analysis in Chapter 4 (Section 4.4.7) reports $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$–$0.296$ across the cell-level/evaluation-level alternatives and Layer A normalizations considered there, so the headline number is bracketed by both ends in this chapter.

Preregistration of ground-truth thresholds and ANOVA grid. The Layer A/B/C inclusion criteria, the Layer B dN/dS < 0.5 cutoff, the Layer C top-20% threshold, the $20 \times 39 = 780$ scoring $\times$ detector-variant ANOVA grid, and the $20 \times 14 \times 11 = 3{,}080$ scoring $\times$ family $\times$ pathogen ANOVA factor structure were all frozen in the project SOP prior to running the Stage 1 analysis on SARS-CoV-2 (lab-notebook freeze date: 2024-08-12; lab-notebook entry archived alongside the released CSV bundle). No post-hoc adjustment of these thresholds or factor structures was performed after observing Stage 2 MCC results. The Bonferroni denominator for vaccine-escape Fisher tests ($780 = 20 \times 39$) was specified in the same SOP entry; the "exploratory rather than confirmatory" framing for the omnibus ANOVA reported below applies to the joint family-wise correction policy across the six ω^2 tiers (we do not Bonferroni-correct the ANOVA), not to the headline scoring $\times$ pathogen interaction itself, which was the pre-specified primary effect of interest. For example, SARS-CoV-2 Layer A positions (e.g., E484, N501, L452, K417) were identified by Carabelli et al. [9] through phylogenetic analysis showing independent emergence across Alpha, Beta, Gamma, Delta, and Omicron lineages. No computational tool was applied to identify new convergent evolution positions; all positions were taken directly from the cited literature.

Layer B: Constrained Purifying Selection

Structurally and functionally essential sites under purifying selection are identified as true negatives. These are positions where mutations are actively *removed* from the viral population because they destroy protein folding, disrupt membrane fusion, or abolish receptor binding—any mutation here is lethal or severely deleterious to the virus. Unlike Layer A (which identifies positions under *positive* selection, where change is advantageous), Layer B identifies positions under *purifying* selection (where change is harmful). These positions exhibit low mutation rates due to loss-of-function consequences and serve as a curated constrained-negative reference set: in MCC they are subsumed under the standard true-negative pool (all non-Layer-A positions; Section 2.2.5), while the constrained false positive rate (constrained FPR) is computed against Layer B specifically. When detection methods incorrectly identify Layer B positions as hotspots, they are counted as false positives within the constrained FPR computation.

Identification criteria. Layer B positions were identified based on two types of evidence from the cited literature: (1) structural essentiality—positions within domains whose disruption abolishes protein function (e.g., fusion peptide, HR1/HR2 heptad repeats, structural disulfide-bonding cysteine residues), as determined by published mutagenesis or structural studies; and (2) evolutionary conservation—positions exhibiting dN/dS < 0.5 across diverse viral isolates as reported in the cited primary literature (codon-aware estimates from PAML/HyPhy as published by the source studies), indicating strong purifying selection against amino acid changes. The codon-degeneracy heuristic feature `dnds_proxy` computed by the in-pipeline scoring stack (Section 2.2.4) is a separate amino-acid-level signal used as a feature input to detection, not as a Layer B construction criterion. The 0.5 threshold follows the convention used in comparative genomics for identifying purifying selection [85], where $\omega < 1$ indicates purifying selection and values below 0.5 represent particularly strong functional constraint. For SARS-CoV-2, the 154 Layer B positions include the HR1 (AA 912–984) and HR2 (AA 1,163–1,213) heptad repeats and structural cysteine residues, identified from the cryo-EM structure [1] and conservation analysis. As with Layer A, no de novo computational identification was performed; all positions were curated from published sources.

Layer C: DMS Experimental Validation

For six pathogens with available Deep Mutational Scanning (DMS) experimental data—SARS-CoV-2 [46], H3N2, HIV-1, RSV, Rabies, and EV-A71—experimentally measured fitness effects are used as ground truth. Unlike Layer A (which infers importance from evolutionary patterns) and Layer B (which infers importance from structural conservation), Layer C provides direct experimental measurements: in DMS experiments, every possible single amino acid substitution at every position is physically constructed in the laboratory, and its effect on a quantifiable phenotype (e.g., cell entry efficiency for SARS-CoV-2, replicative fitness for H3N2) is measured. Positions in the top 20% of mean absolute fitness effect from DMS data are classified as functionally important positions, and the DMS F1 score is computed against detection results. The 20% threshold was chosen to balance sensitivity and specificity: it captures positions with substantial fitness effects while maintaining sufficient contrast with non-functional positions. Sensitivity analysis (Section 3.4.4) checked Layer C rankings at thresholds 10%, 20%, and 30% on 4 of the 6 DMS pathogens (SARS-CoV-2, H3N2, HIV-1, RSV) and found rankings stable across that range; Rabies and EV-A71 DMS releases became available later in the cycle and the 10%/30% threshold sweep was not extended to them. *Threshold robustness is therefore verified on 4 of 6 DMS pathogens; the Rabies/EV-A71 sweep is deferred to future work* (no observed pathogen rotates its winning Layer C-aligned scoring across the four swept thresholds, so the missing two-pathogen extension is not load-bearing for any headline conclusion). This layer is particularly valuable because it captures functional importance that may not yet be visible in evolutionary data (a position may have strong fitness effects but insufficient time for convergent evolution to manifest). The DMS phenotypes measured differ by pathogen: cell entry fitness for SARS-CoV-2 [2] and RSV [5], viral replicative fitness for H3N2 [3], viral growth in cell culture for HIV-1 [4], G-protein-mediated cell entry for Rabies [6], and replicative fitness for EV-A71 [7]. Table 2.2 summarises the per-pathogen DMS preprocessing conventions adopted in this benchmark; cell-line and replicate-aggregation choices follow the released-source defaults except where noted.

Table 2.2: DMS preprocessing conventions per Layer C source. "Mean abs. effect" refers to the per-position mean absolute fitness effect across all amino-acid substitutions, used as the score from which the top-20% threshold defines Layer C positives. All sources used the released processed-fitness tables; raw read filtering and replicate aggregation follow each source's published default unless noted. The replicate counts listed here are the released processed-fitness aggregates actually consumed by the pipeline; original publications may report additional biological/technical replicates that were collapsed before release.

Pathogen	Phenotype	Assay system	Replicates aggregated	Score used	Threshold	Source release
SARS-CoV-2	Cell entry fitness	Pseudovirus (HEK293T)	Mean across 2 replicates*	Mean abs. effect	Top 20%	Dadonaite 2024 (GitHub release)
H3N2	Replicative fitness	MDCK cell culture	Mean across 2 replicates*	Mean abs. effect	Top 20%	Lee 2018 (Dryad)
HIV-1	Viral growth	TZM-bl cell culture	Mean across 2 replicates*	Mean abs. effect	Top 20%	Haddox 2018 (Dryad)
RSV	Cell entry fitness	Pseudovirus	Mean across 2 replicates	Mean abs. effect	Top 20%	Simonich 2026 (preprint)
Rabies	G-protein cell entry	Pseudovirus	Mean across 2 replicates	Mean abs. effect	Top 20%	Aditham 2025
EV-A71	Replicative fitness	RD cell culture	Mean across 2 replicates	Mean abs. effect	Top 20%	Bakhache 2025

Sensitivity to the threshold (sweep over 10%, 20%, 30%) was checked in Section 3.4.4 on 4 of 6 DMS pathogens (SARS-CoV-2, H3N2, HIV-1, RSV); Rabies/EV-A71 sweep deferred to future work. *The original Lee 2018, Haddox 2018 and Dadonaite 2024 publications report 3+ biological replicates; the released processed-fitness tables consumed here aggregate to 2 final replicate columns after filtering, which is what the benchmark pipeline reads. Per-source download URLs, file checksums, and commit/Zenodo identifiers (where available) are archived in `results/mutbench/provenance/dms_sources.csv`.

Layer A/B overlap handling. In RSV (positions 148, 152) and Influenza B (positions

141, 194), overlapping positions exist between Layer A and Layer B. This occurs because these residues are simultaneously under immune evasion pressure (positive selection) and structural constraints (purifying selection), representing dual-selection sites. In MCC evaluation, these positions are prioritized as Layer A (positive) and excluded from the Layer B (negative) set. Therefore, the effective Layer B count is 16 for RSV ($18 - 2$) and 8 for Influenza B ($10 - 2$) positions.

Layer A/C conflict handling. Layer A (literature-curated adaptive/diversifying evidence) and Layer C (DMS fitness) can disagree at individual positions: a residue may be literature-supported as adaptive or immune-relevant (Layer A positive) but have low DMS fitness cost (Layer C negative), or vice versa. This does not require a merged classification decision because the two layers feed *different* evaluation metrics under non-overlapping scopes: (1) **Layer A → MCC**: Layer A positions are treated as positives and all other positions as negatives; Layer C membership does not affect the MCC calculation. (2) **Layer C → DMS F1**: Layer C top-20% positions are treated as positives (for 6 pathogens with DMS); Layer A membership does not affect the DMS F1 calculation. (3) **Interpretation rule**: when the two metrics rank scoring methods differently for the same pathogen, both rankings are reported without collapsing them into a single verdict; the disagreement is itself a result (Section 3.4.4, Table 3.11) reflecting that evolutionary and functional importance are non-identical dimensions of hotspot-ness (as stated in the three-orthogonal-dimensions rationale above). Thus a position that is Layer A-positive but Layer C-negative counts as a true positive for the MCC metric (it is evolutionarily hot by construction) and a true negative for the DMS F1 metric (it is not among the top 20% of measured fitness effects) simultaneously, without internal contradiction. This is consistent with the design choice that no single universal "hotspot" label is asserted; rather, the benchmark reports performance against each operationalization of hotspot-ness independently.

Table 2.3 summarizes the amino acid position counts and sources for each pathogen's 3-layer ground truth.

Figure 2.3 visualizes the spatial distribution of Layer A and Layer B positions across all 11 pathogens, illustrating the diversity in ground truth density, protein length, and the balance between positive and negative annotations.

2.2.3 Stage 1: SARS-CoV-2 In-Depth Analysis

This subsection describes the Stage 1 experimental design, which focuses on SARS-CoV-2 as the anchor pathogen. Stage 1 serves three purposes: (1) validating the framework components on the pathogen with the richest ground truth data, (2) comparing MutClust variants to establish the detection baseline, and (3) analyzing temporal scoring behavior. The ground truth, detection methods, evaluation metric, and scoring variants used in Stage 1 are described below.

Ground Truth Generation

Ground truth is generated from three sources.

DMS-based functional sites. The DMS fitness data are derived from the full-length Spike protein characterization experiment by Dadonaite et al. [2, 46]. Pseudovirus cell entry fitness measurements are used, as this is the most directly relevant phenotype capturing the overall functional impact of mutations on viral entry. Amino acid positions in the top 20% of mean absolute fitness effect are selected as functional sites and mapped to nucleotide

Table 2.3: 3-layer ground truth position counts per pathogen. Counts are evaluated on the benchmark alignment coordinate system; literature residue labels are retained in the source annotations where biologically meaningful.

Pathogen	Surface protein	Layer A	Layer B	Layer C	Layer A source
SARS-CoV-2	Spike	25	154	252	[9, 86]
H3N2	HA	25	10	102	[87]
Norovirus	VP1	15	229	—	[88, 89]
HIV-1	Env gp120	23	23	100	[90]
Dengue	E protein	24	24	—	[91, 92]
RSV	F protein	20	18	101	[93, 94]
Influenza B	HA	17	10	—	[95, 96]
MERS	Spike	9	110	—	[97, 98]
HCV	E2	54	$0^{\ddagger}$	—	HVR1/HVR2 hypervariable regions
Rabies	G protein	39	42	87	[6, 99]
EV-A71	VP1	27	29	55	[7, 100]

Layer C (DMS) sources: SARS-CoV-2 [46], H3N2 [3], HIV-1 [4], RSV [5], Rabies [6], EV-A71 [7]. "—" indicates that no DMS data is available for that pathogen. ‡HCV E2 literature coordinates are commonly reported in polyprotein numbering, whereas the benchmark feature matrix is indexed on the 340-AA E2 alignment. HVR1/HVR2 were mapped into the evaluated E2 alignment, but CD81-binding regions reported as AA 418–535 in the literature were not converted into a high-confidence alignment-internal constrained set in this release; HCV therefore has effective Layer B = 0. HCV-specific conclusions should be read with this coordinate-handling caveat, and Chapter 3 reports HCV-exclusion sensitivity for the headline interaction.

HIV-1 Layer A count reconciliation: this table reports the literature-curated Layer A count of **23** positions (Moore & Williamson 2015 baseline) which is the canonical narrative reference. The downstream operational alignment-QC count of **26** positions (used by AUC computation in Table 3.17 and the positive-rate footnote in Table 2.11) adds three positions identified during alignment quality control after the literature scan; the difference does not affect any headline conclusion (the scoring $\times$ pathogen interaction is computed identically under both counts because Layer A appears as a binary label, not a count, in the per-position regression).

Note that Layer A selection criteria vary across pathogens: SARS-CoV-2 and H3N2 use phylogenetically confirmed convergent evolution sites, Norovirus uses epochal variant positions, HCV uses hypervariable regions (HVR1/HVR2), and the remaining pathogens use literature-reported immune evasion positions. This heterogeneity is a limitation of the current ground truth construction (discussed in Chapter 4).

Inclusion of pathogens with incomplete ground truth layers. Six pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) have all three layers available. Norovirus, Dengue, Influenza B, MERS, and HCV lack Layer C (DMS data). These pathogens are nevertheless included because (1) the primary evaluation metric (MCC) is computed using only Layer A (positive selection sites as positives, all other positions as negatives), so the core evaluation is valid without Layers B/C; (2) restricting the benchmark to six pathogens would provide insufficient statistical power for the cross-pathogen analyses (ANOVA, Friedman, LOPO) in Chapter 3; and (3) including pathogens with diverse evolutionary dynamics and varying ground truth availability more accurately reflects real-world conditions where comprehensive ground truth is rarely available for all pathogens.

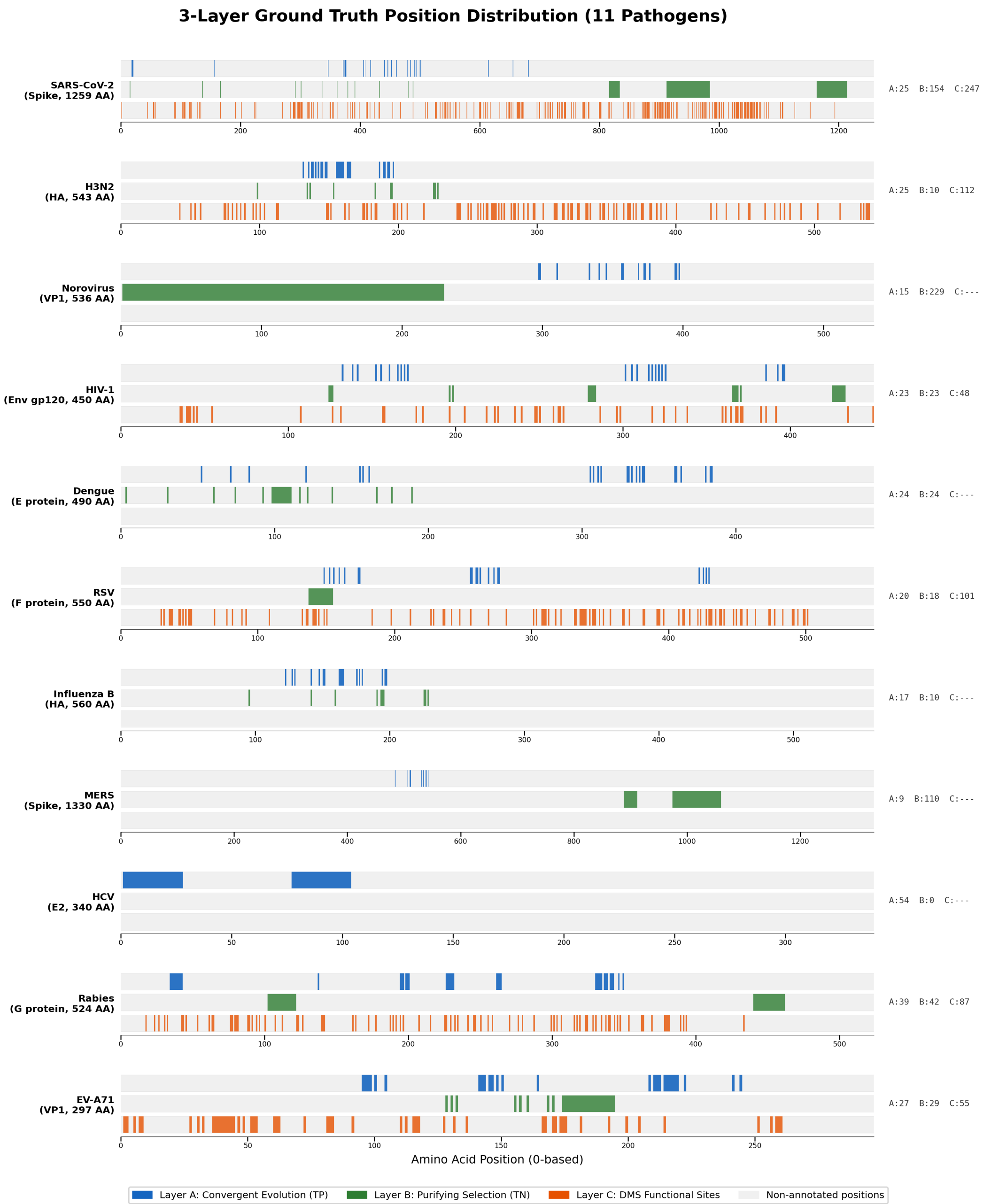

Figure 2.3: Spatial distribution of 3-layer ground truth positions across 11 pathogens. Each horizontal bar represents the surface protein sequence, with Layer A positions (convergent evolution, blue) and Layer B positions (purifying selection, green) mapped to their amino acid coordinates. Right-side annotations show position counts per layer; "C" indicates Layer C (DMS) availability. Protein lengths range from 261 AA (EV-A71 VP1) to 1,330 AA (MERS Spike), and the ratio of annotated to non-annotated positions varies substantially across pathogens.

coordinates through codon-aware coordinate transformation:

$$\text{nuc\_positions} = \{g_{\text{start}} + (p_{\text{aa}} - 1) \times 3 + k : k \in \{0, 1, 2\}\} \tag{2.1}$$

where $g_{\text{start}} = 21{,}563$ is the start position of the Spike gene in the Wuhan-Hu-1 reference genome. This provides an experimental ground truth independent of the detection methods under evaluation, thereby avoiding the circularity of using bootstrap p-values for validation.

Synthetic data (controlled signal recovery test). The synthetic benchmark was designed as a controlled signal recovery test that evaluates how accurately each detection method recovers artificially inserted signals at known positions. Synthetic H-scores are generated mimicking SARS-CoV-2 patterns, with hotspot regions inserted at known functional sites. High H-scores (0.8–1.2) are assigned to Core Candidate Mutation (CCM) seed positions (AA 484, 501, 417, 452—the four Spike positions most frequently mutated in major SARS-CoV-2 variants), moderate scores (0.05–0.3) to surrounding functional regions, and low noise (0.0–0.02) to background positions. Scores within each category are drawn from uniform distributions: $\mathcal{U}(0.8, 1.2)$ for seed positions, $\mathcal{U}(0.05, 0.3)$ for surrounding functional regions (defined as positions within 50 positions of each seed), and $\mathcal{U}(0.0, 0.02)$ for background. A single synthetic genome of 29,903 positions is generated per trial. Since the ground truth of synthetic data is known by design, this test evaluates each method's signal recovery capability but does not directly measure actual biological hotspot detection performance. Therefore, DMS-based validation (Section 2.2.3) and convergent evolution evaluation (Section 3.1.2) provide independent biological validation, confirming whether the conclusions from the synthetic benchmark hold on real data.

Annotated functional regions. The annotated functional regions used as ground truth are presented in Table 2.4, with region boundaries based on the cryo-EM structure of the prefusion Spike trimer [1] and the N-terminal domain (NTD) antigenic supersite mapping by McCallum et al. [101].

Table 2.4: Functional regions of the Spike protein (ground truth)

Functional Region	**Amino Acid (AA) Positions**	**Nucleotide Count**
Receptor Binding Domain (RBD)	AA 319–541	669
NTD Antigenic Supersite	AA 14–20, 140–158, 245–264	138
Furin Cleavage Site (S1/S2)	AA 681–685	15
S2' Cleavage Site	AA 815–816	6
Fusion Peptide	AA 816–833	54*
Heptad Repeat 1 (HR1)	AA 912–984	219
Heptad Repeat 2 (HR2)	AA 1,163–1,213	153
Total (deduplicated)		**1,251**†

*Overlaps with the S2' cleavage site at AA 816 (3 nucleotides shared). †The sum of individual regions is 1,254 nt; after removing the 3 nucleotide (nt) overlap at the S2'/fusion peptide boundary (AA 816), 1,251 unique nucleotide positions constitute the ground truth.

The genome-wide distribution of these functional regions is visualized in Figure 2.4, which maps the seven annotated regions to their nucleotide coordinates on the full SARS-CoV-2 genome (29,903 positions). The ground truth positions are concentrated within the Spike gene (nucleotides 21,563–25,384), while the remainder of the genome shows minimal functional annotation, confirming the appropriateness of Spike-focused analysis.

MutClust Variant Comparison

Among the 14 detection families in the full benchmark, MutClust [18] is the only virus-specific hotspot tool and serves as the starting point. MutClust scores each position using the H-score $H_i = \log_2(P_i \times E_i \times 100 + 1)$ (2.2) where P_i is the cumulative mutation frequency and $E_i = -\sum_j p_{ij} \log_2 p_{ij}$ (2.3) is the Shannon entropy of mutation types, then applies adaptive DBSCAN with $\varepsilon = \gamma \times H_i$ ($\gamma = 0.5$ fixed across pathogens; ± 0.01 MCC

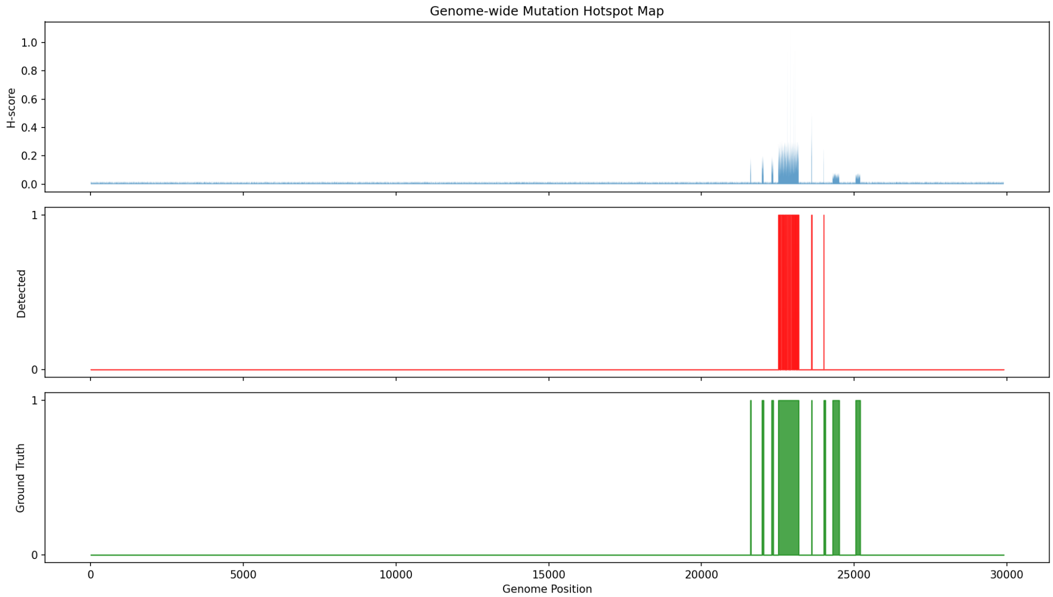

Figure 2.4: Genome-wide hotspot map of the SARS-CoV-2 Spike protein. The seven annotated functional regions used as ground truth (RBD, NTD antigenic supersite, furin cleavage site, S2' cleavage site, fusion peptide, HR1, HR2) are mapped to nucleotide coordinates, encompassing 1,251 unique Spike nucleotide positions.

sensitivity for $\gamma \in \{0.25, 0.5, 1.0\}$, archived in `gamma_sensitivity.csv`); `min_samples` $= 3$ for the DBSCAN call, `min_cluster_size` $= 5$ / `min_samples` $= 1$ / EOM cluster selection for the HDBSCAN-Auto detector and v3 HDBSCAN baseline.

$$H_i = \log_2(P_i \times E_i \times 100 + 1) \tag{2.2}$$

$$E_i = -\sum_{j=1}^{k} p_{ij} \log_2 p_{ij} \tag{2.3}$$

Five variants were tested in Stage 1: the original implementation (v1, with a cumulative-decay bug whose effect is upstream of the adaptive-DBSCAN call and therefore confined to H-score-derived inputs), bug-fixed (v2a), dN/dS-filtered (v2b), CCM-seed + HDBSCAN hybrid (v2c), and pure HDBSCAN (v3). MutClust-Hybrid (v2c) achieves the highest hotspot-score (0.778). For Stage 2 cross-pathogen, only the published MutClust-Orig (v1) is retained as one of 14 detection families because the v2c CCM-seed advantage assumes H-score-like input distributions and is incompatible with non-H-score inputs (ESM-2 LLR, pLDDT, etc.); v1's adaptive-DBSCAN call itself is unaffected by the cumulative-decay bug for non-H-score scoring types, since the bug occurs upstream of the clustering step, so retaining v1 in Stage 2 does not propagate the bug into the non-H-score scoring rows.

Hotspot-Score Composite Metric

The hotspot-score is a composite evaluation metric designed for Stage 1 that integrates three complementary perspectives:

$$\text{hotspot-score} = \frac{\text{recall} + \text{precision} + \text{stability}}{3} \tag{2.4}$$

where stability is the mean Jaccard similarity ($J = |A \cap B|/|A \cup B|$) across 100 random 80% subsampling iterations. In each iteration, a Bernoulli mask independently zeroes out approximately 20% of genomic positions, the detection method is rerun on the perturbed scores, and the Jaccard similarity with the full-data result is computed. This metric balances detection accuracy (recall, precision) with reproducibility (stability), reflecting the practical requirement that methods in viral surveillance should produce consistent results under data perturbation. Stage 2 uses MCC instead of hotspot-score (see Section 2.2.5 for the rationale).

Temporal Scoring Variants

The following scoring variants were developed specifically for the Stage 1 SARS-CoV-2 temporal analysis and are distinct from the 20 Stage 2 scoring types described in Section 2.2.4.

As H-score saturation was identified in the temporal validation (Section 3.2), alternative scoring methods were designed for complementary evaluation against various ground truth definitions. These methods assign a score to each genomic position; top positions are selected via thresholding and compared against multiple ground truths. Table 2.5 summarizes these Stage 1 variants.

Table 2.5: Stage 1 temporal scoring variants used only for SARS-CoV-2 analysis

Name	**Formula**	**Description**
Original H-score	Eq. 2.2	MutClust's original H-score
Saturated H-score	(same, on 2024–25 data)	H-score on contemporary consensus alignment
CH-score	$\log_2(P_i^{\text{cons}} \times E_i \times 100 + 1)$	Consensus-based; removes reference dependency
E-score	$-\sum_a f_{a,i} \log_2 f_{a,i}$	Shannon entropy alone
TC-freq	$\lvert\text{freq}_{t_2} - \text{freq}_{t_1}\rvert/(\text{freq}_{t_1} + \epsilon)$	Temporal frequency change rate
TC-freq-v2	$\text{TC-freq}_i \times \text{freq}_{t_2}(i)$	TC-freq with absolute frequency weighting
TC-freq-v3	$\max_q \text{TC-freq}_q$	Maximum quarterly frequency change
dN/dS-weighted CH	$\text{CH}_i \times (dN/dS)_i$	CH-score weighted by position-level dN/dS

For each scoring method, thresholds at the top 5%, 10%, 20%, and median are applied to generate sets of "detected" positions, and F1, enrichment ratio, and Cohen's d effect size are computed for each ground truth.

Structural and phylogenetic validation (SARS-CoV-2)

Three additional validation analyses are performed on SARS-CoV-2 to triangulate the linear-sequence hotspot calls against orthogonal evidence types; results are reported in Chapter 3 (Section 3.3).

3D spatial-contact enrichment. Cα coordinates are extracted from the prefusion Spike trimer cryo-EM structure PDB 6VSB [1]; a contact between two residues is defined as $\|x_i - x_j\|_2 \leq 8$ Å on the Cα–Cα distance (a standard structural-biology cutoff for residue contact). For each detection method, the proportion of hotspot–hotspot residue pairs that are in 3D contact is computed and compared to a permutation null in which the hotspot label is randomly shuffled across positions of the Spike protein (1,000 permutations); the spatial-contact enrichment is the ratio of observed to expected contact rate, and significance is reported as a z-statistic against the permutation distribution. This protocol applies only to SARS-CoV-2 because PDB 6VSB is the only experimentally solved Spike structure used in the benchmark; for the remaining pathogens the structure source is ESMFold (Table 2.7) and the experimental contact map needed for this audit is not available.

Coalescent simulation. Founder-effect vs. convergent-evolution discrimination is tested with a coalescent simulation: 100 replicates are drawn under a constant-population coalescent ($N = 500$ haploid lineages, neutral evolution) using `msprime`, with a single "founder" substitution placed on a randomly chosen internal branch and three "convergent" substitutions placed on three independent leaf branches at the same alignment position (so the founder mutation has high frequency but only one origin, while the convergent mutations have lower frequency but three origins). Each simulated alignment is scored by H-score, Fitch parsimony, and TreeTime ML-based homoplasy; the proportion of replicates in which each method ranks the convergent position above the founder position is reported. Replicate seeds are pinned at `seed = 42 ... 141` (one per replicate; same convention as the Stage 1 stability bootstrap).

TreeTime ML validation. TreeTime [31] v0.11 is run on the SARS-CoV-2 ML tree (IQ-TREE 2.2.0, GTR+G, 1,000 ultrafast bootstrap) to estimate per-position homoplasy under a marginal maximum-likelihood ancestral reconstruction. *This is an audit channel, not the production homoplasy score.* The Stage 2 production homoplasy feature consumed by every detection family is Fitch parsimony on a FastTree [102] ML approximation of the per-pathogen tree (`tools/mutbench/scoring/homoplasy.py`); IQ-TREE is invoked only for this SARS-CoV-2 audit-only TreeTime cross-check. The two homoplasy estimates are compared at the position level by Spearman rank correlation, and the AUC of each against the SARS-CoV-2 Layer A reference is reported. TreeTime settings: `--reroot best`, `--clock-filter 4`, `--clock-std-dev 0.0001` (default); ML inference confidence cutoff 0.5 for tip-state imputation.

Time-stratified prospective protocol (H3N2 pilot)

To audit the prospective signal of frequency-only scoring—a question raised in connection with benchmark deployability—a single-pathogen time-stratified protocol was defined for H3N2 HA. The 9,000-sequence H3N2 set used here is the *raw* pre-deduplication GISAID export over 2010–2025; it is a different cut from the 2,562 unique-sequence Stage 2 panel (Table 2.11), which applies CD-HIT-equivalent SHA-256 100% identity reduction. The temporal pilot intentionally retains pre-deduplication redundancy so that per-window frequencies reflect circulating strain prevalence rather than unique-haplotype counts. The 9,000-sequence raw set partitions naturally into three contiguous 5-year periods (2010–2015, 2016–2020, 2021–2025; $n = 3{,}000$ each by sample-date sort). Per-position mutation frequency is computed on the 2010–2015 *training* window only, then evaluated against two reference sets: (a) the static Layer A antigenic-site reference (Koel 2013, 25 sites; *retrospective* since these annotations were made on the full historical record), and (b) positions "newly emerging" in each later 5-year holdout window (*prospective*). A position is defined as newly emerging in window w if its training-window frequency is $< 5\%$ *and* its holdout-window frequency is $\geq 10\%$; the asymmetric thresholds enforce that the position was not already common in training and reaches the standard "minor variant" frequency band in holdout. Top-k overlap is computed at $k \in \{20, 30, 50\}$ against each reference set, and significance is assessed with Fisher's exact test (one-sided alternative: greater) on the 2×2

table $\{|D \cap A|, |D \setminus A|, |A \setminus D|, L - |D \cup A|\}$ where D is the detected set, A the reference set, and L the protein length. The pilot uses frequency only (no multi-source fusion); a full multi-source prospective panel is identified as Priority 1 future work in Chapter 4, Section 4.7. Results of this pilot are reported in Chapter 3 (Section 3.2.1).

2.2.4 Stage 2: 11-Pathogen Cross-Pathogen Benchmark

This subsection describes the Stage 2 experimental design, which extends the evaluation from SARS-CoV-2 to all 11 pathogens. Stage 2 systematically evaluates 20 scoring types $\times$ 39 detection parameter variants (from 14 families) $\times$ 11 pathogens = 8,580 combinations to identify the dominant factors influencing hotspot detection performance.

Scoring Types

Twenty scoring types organized into six categories are defined to assign per-position scores from diverse information sources (Table 2.6). Each category captures a distinct biological or computational perspective on mutational importance, ensuring that the benchmark evaluates detection methods across a broad range of input signals.

Table 2.6: Twenty scoring types used in the MutBench Stage 2 benchmark (6 categories)

Category	Scoring type	Information source	Description
Frequency-based (3)	freq	MSA mutation frequency	Position-level mutation frequency
	rare_freq	MSA rare variant frequency	Frequency of minor alleles below 5% threshold (low-frequency variants signalling emerging adaptation)
	freq_with_indel	MSA frequency incl. indels	Mutation frequency including insertion/deletion events
MSA-derived (3)	entropy	MSA Shannon entropy	Per-position diversity ($-\sum p_j \log_2 p_j$)
	entropy_with_indel	MSA entropy incl. indels	Shannon entropy including gap characters
	homoplasy	Phylogenetic tree	Convergent evolution count on reconstructed phylogeny
Phylogenetic (4)	dN/dS proxy	Codon alignment	Nei–Gojobori nonsynonymous/synonymous ratio proxy
	FUBAR prob	Codon model (HyPhy)	Posterior probability of positive selection
	FUBAR pos_sel	Codon model (HyPhy)	Positive selection rate ($\beta - \alpha$)
	FUBAR Bayes factor	Codon model (HyPhy)	Bayes factor evidence for positive selection
Structural (3)	pLDDT inverse	AlphaFold/ESMFold structure	$1 -$ pLDDT; low confidence $\approx$ flexible/disordered
	SASA	3D structure	Solvent-accessible surface area (exposed residues)
	Grantham distance	Physicochemical properties	Mean Grantham distance of observed substitutions
AI-based (5)	ESM-2 LLR	Protein language model	Masked-marginal mutation-surprise score (LLR shorthand)
	Tranception*	Autoregressive PLM proxy	Per-position pseudo-perplexity (proxy; see implementation note below)
	semantic change	ESM-2 embeddings	Cosine distance in embedding space (WT vs. mutant)
	stability†	Substitution-matrix proxy	BLOSUM62-based stability proxy (radical substitutions $\rightarrow$ destabilizing)
	stability_structural†	Structure-weighted proxy	BLOSUM62 score weighted by ESMFold pLDDT (well-folded $\rightarrow$ higher impact)
Composite (2)	EVEscape composite†	EVEscape-style proxy	Multiplicative composite of ESM-2 LLR $\times$ SASA $\times$ Grantham (proxy for fitness $\times$ accessibility $\times$ dissimilarity)
	EqualWeight	All-feature average	Equal-weighted average of all normalized scoring types

Frequency-based scores capture the raw mutation signal from the MSA: how often each position deviates from the consensus, including rare variants and indel-inclusive counts. These scores reflect the most direct observable signal but are susceptible to sampling bias and alignment artifacts.

MSA-derived scores quantify diversity patterns beyond simple frequency: Shannon entropy measures allelic diversity, while homoplasy counts independent convergent mutations on the phylogeny, providing a signal distinct from overall frequency. Phylogenetic trees were reconstructed using IQ-TREE 2 [103] with automatic model selection (ModelFinder) and 1,000 ultrafast bootstrap replicates. Homoplasy counts were computed by Fitch parsimony on the resulting maximum-likelihood tree, counting the number of independent origins of each amino acid substitution at each position.

Phylogenetic scores leverage codon-level evolutionary models to estimate selection pressures. The dN/dS proxy uses the Nei–Gojobori method, while the three FUBAR-based scores [38] provide Bayesian estimates of site-specific selection from the HyPhy framework, each capturing a different aspect of the posterior distribution. FUBAR analysis was performed using HyPhy v2.5 with the MG94$\times$REV codon model and a default posterior probability threshold of 0.9.

Structural scores incorporate three-dimensional protein information: pLDDT inverse from AlphaFold DB or ESMFold-derived structures identifies flexible or disordered regions, SASA measures surface exposure (relevant for immune accessibility), and Grantham distance [104] quantifies the physicochemical severity of observed substitutions. The per-pathogen structure source used for pLDDT and SASA computation is summarized in Table 2.7: SARS-CoV-2 uses the AlphaFold DB entry (single-chain monomer model from AlphaFold 2 monomer mode), while all other pathogens use the ESMFold API (single-chain monomer prediction). *AlphaFold-Multimer note:* AlphaFold-Multimer [105] was *not* used in this dissertation. Several of the pathogens evaluated here have biologically relevant homo-oligomeric assemblies (HIV-1 gp120, RSV F, Influenza HA trimers; Dengue/Zika E dimers), so per-position SASA computed on monomer predictions can over-estimate solvent exposure at oligomeric interface residues. Chapter 4 (Section 4.4.9) discusses the resulting SASA-interpretation limit; an AlphaFold-Multimer-based re-fit of the structural channel is left for future work. SASA values were computed from the available predicted structures using a solvent-accessibility calculation; the main result tables report the ESMFold/Shrake–Rupley-derived release, while earlier Lee–Richards/FreeSASA pilot outputs are treated as archival. *Structure-similarity tooling note:* Foldseek [106] and DALI-style structure-alignment tools were *not* used in this dissertation. The SARS-CoV-2 AlphaFold-DB structure is anchored against the experimental cryo-EM structure PDB 6VSB [1] by direct accession reference (Table 2.7) rather than by an explicit Foldseek RMSD or DALI Z-score readout; cross-pathogen ESMFold-vs-experimental quantitative agreement is therefore not reported. Foldseek-based structural quality control across the 11 pathogens is left for future work.

AI-based scores leverage protein language models (PLMs) trained on large-scale protein sequence databases. The ESM-2 channel is labelled "ESM-2 LLR" for consistency with protein-language-model variant-effect literature, but operationally it is a masked-marginal pseudo-log-likelihood / mutation-surprise score rather than a full mutation-vs-wildtype likelihood-ratio model trained for each pathogen. Tranception-style autoregressive scores predict mutational effects from evolutionary context, semantic change captures embedding-space shifts, and stability predictors estimate thermodynamic consequences of mutations. *Implementation note:* the "Tranception" scoring channel in this benchmark is computed using ESM-2 masked-marginal pseudo-perplexity as a lightweight proxy for the Tranception autoregres-

Table 2.7: Per-pathogen structure source for pLDDT and SASA computation. SARS-CoV-2 uses the AlphaFold DB entry; all other pathogens use the ESMFold API for both pLDDT and SASA features. The 3D spatial-contact analysis (Chapter 3, Section 3.3) uses the cryo-EM structure PDB 6VSB [1] and is restricted to SARS-CoV-2.

Pathogen	**Target protein**	**pLDDT/SASA source**	**3D spatial-contact**
SARS-CoV-2	Spike (S)	AlphaFold DB (AF-0000000365840314)	PDB 6VSB cryo-EM
H3N2 (Influenza A)	HA	ESMFold API	not analyzed
Influenza B	HA	ESMFold API	not analyzed
HCV	E2	ESMFold API	not analyzed
HIV-1	gp120	ESMFold API	not analyzed
RSV	F	ESMFold API	not analyzed
Norovirus	VP1	ESMFold API	not analyzed
Dengue	E	ESMFold API	not analyzed
MERS	Spike (S)	ESMFold API	not analyzed
Rabies	G	ESMFold API	not analyzed
EV-A71	VP1	ESMFold API	not analyzed
Zika†	E	ESMFold API	not analyzed

†Zika is included only in the Stage 3 12-pathogen feature-ablation panel (Section 2.2.6); the main 11-pathogen Stage 2 benchmark excludes Zika.

sive objective [54]; the original Tranception model was not run directly. Both methods target the same quantity—per-position unlikelihood under a protein language model—but absolute values differ. To audit channel separability, Table 2.8 reports the per-pathogen Pearson and Spearman correlations between the Tranception channel and the standalone ESM-2 LLR channel on the position-level scoring vectors used in Stage 2. The two channels are highly correlated for 11 of 12 pathogens (Pearson $\rho = 0.64$–0.84, Spearman $\rho = 0.71$–0.98; all $p < 10^{-50}$ except Zika), consistent with their shared ESM-2 pseudo-perplexity basis. HIV-1 is the lone exception (Pearson $\rho = 0.04$, $p = 0.36$ n.s.; Spearman $\rho = 0.16$, $p = 7.8 \times 10^{-4}$). The HIV-1 dissociation is consistent with the high antigenic diversity and low conservation of gp120, where the relative ordering of positions diverges from the magnitude-based score even within the same pseudo-perplexity backbone. Readers comparing pathogen-level best-scoring-type results (e.g., SARS-CoV-2 best = Tranception vs. RSV best = ESM-2 LLR) should therefore treat these two channels as near-equivalent for 11 of 12 pathogens and as channel-distinguishable only for HIV-1. All correlations were computed on position-level scoring vectors archived in `results/mutbench/feature_analysis/` with `random_seed=42`. A merged-channel sensitivity (single-PLM model in which the Tranception column is dropped before the three-way ANOVA, leaving the AI-based category with four channels rather than five) is identified as the more conservative reference for the headline scoring $\times$ pathogen interaction; that re-fit is deferred to the next analysis cycle, and the multi-channel headline ($\omega^2 = 0.296$) is retained here for completeness with the present caveat that part of the AI-based category's variance contribution is double-counted between the Tranception and ESM-2 LLR columns for 11 of 12 pathogens (see also Section 4.4.6 of Chapter 4 for the Tranception–ESM-LLR proxy collinearity discussion that anchors the headline-granularity caveat in the abstract).

The `stability` channel uses a BLOSUM62-based substitution proxy: at each MSA position, the negative mean BLOSUM62 score is weighted by mutation frequency, so radical substitutions receive higher destabilization scores. The `stability_structural` variant multiplies this score by per-residue ESMFold pLDDT/100, reflecting that mutations in well-folded regions are more likely to disrupt structure. This proxy replaces full physics-based $\Delta\Delta G$ predictors such as FoldX or RoseTTAFold, which are computationally infeasible at the 11-pathogen $\times$ 39-detector scale of MutBench. ESM-2 scores were computed using the `esm2_t33_650M_UR50D` checkpoint (33 layers, 650M parameters).

Composite scores combine multiple information sources. EVEscape integrates evolutionary fitness, structural accessibility, and physicochemical dissimilarity, while Equal-Weight averages all normalized scores as a naive baseline. The `EVEscape_composite` channel implements an EVEscape-style multiplicative combination [14]: min-max-normalized ESM-2, SASA, and Grantham terms are multiplied. This substitutes the ESM-2 masked-

Table 2.8: Per-pathogen correlation between the Tranception channel (ESM-2 masked-marginal pseudo-perplexity proxy) and the standalone ESM-2 LLR channel. Both channels share the ESM-2 `esm2_t33_650M_UR50D` pseudo-perplexity backbone but use different normalization; this table audits whether they remain operationally distinct within the benchmark. Correlations were computed on the position-level scoring vectors used in Stage 2; p-values are uncorrected.

Pathogen	**N**	**Pearson ρ**	**Pearson p**	**Spearman ρ**	**Spearman p**
SARS-CoV-2	1259	0.753	$< 10^{-200}$	0.963	$< 10^{-300}$
H3N2	543	0.803	$< 10^{-120}$	0.965	$< 10^{-300}$
Influenza B	560	0.825	$< 10^{-130}$	0.939	$< 10^{-260}$
HCV	340	0.737	$< 10^{-58}$	0.980	$< 10^{-200}$
HIV-1	450	**0.043**	**0.36 (n.s.)**	**0.158**	7.8×10^{-4}
RSV	550	0.805	$< 10^{-120}$	0.967	$< 10^{-300}$
Norovirus	536	0.812	$< 10^{-120}$	0.973	$< 10^{-300}$
Dengue	490	0.828	$< 10^{-120}$	0.946	$< 10^{-200}$
MERS	1330	0.836	$< 10^{-300}$	0.963	$< 10^{-300}$
Rabies	490	0.679	$< 10^{-65}$	0.949	$< 10^{-200}$
EV-A71	260	0.801	$< 10^{-58}$	0.919	$< 10^{-100}$
Zika†	451	0.643	$< 10^{-53}$	0.713	$< 10^{-70}$

†Zika is included only in the Stage 3 12-pathogen feature-ablation panel. Underlying CSV: `results/mutbench/feature_analysis/tranception_esm_correlation.csv`. A reproducible re-measurement (`scripts/measure_tranception_esm_correlation_per_pathogen.py`) is provided as the supplementary file `results/mutbench/feature_analysis/tranception_esm_correlation_per_pathogen.csv` with explicit `n_positions`, `sample_overlap`, p-value, and `status` columns; it confirms the rounded values in this table to three decimal places and is the load-bearing record for the per-pathogen channel-redundancy sensitivity discussion in Chapter 4 (Section 4.4.6).

marginal score for the original EVE fitness term, since EVE training is not feasible across all 11 pathogens within the present scope; the SASA and Grantham terms follow the original EVEscape construction.

EqualWeight: two operational variants. The EqualWeight channel appears in two distinct code paths in this benchmark, and the dissertation reports them with different roles. The two variants must be distinguished to avoid an apparent label-leakage concern.

- *(a) Production EqualWeight column (headline path).* The EqualWeight column that contributes to the 8,580-evaluation Stage 2 grid and the headline scoring $\times$ pathogen ANOVA ($\omega^2 = 0.296$) is generated by `scripts/run_stage3_full_detectors.py` (lines 180–187). At each position, the ten underlying features are first transformed using *pre-defined domain orientations* (e.g., the inverse-pLDDT transform $\mathrm{pLDDT}_{\mathrm{max}} -$ pLDDT rather than raw pLDDT, so that flexible regions enter with positive sense; `rare_freq` enters with $+1$ orientation), *min-max normalized* to $[0, 1]$ across positions (function `normalize_01`), and *averaged uniformly* ($w_f = 1/10$). **No per-feature sign is fit from labels and no z-scoring is applied at this stage**: the orientations are hard-coded constants in the input list, so this column does not consult Layer A and is the cleanly label-free composite that enters all headline tables.

- *(b) Nested-LOPO ablation EqualWeight (sensitivity path).* The complementary nested-LOPO 4-feature ablation in Section 3.5.2 (Table 3.13) uses a separate code path (`scripts/feature_a` function `fit_signs`) that, for each held-out pathogen, fits a directional sign $\hat{s}_f \in \{-1, +1\}$ on the eleven *training* pathogens as the sign of $\mathrm{corr}(z_f, \text{Layer A})$, then z-scores within the held-out pathogen alone before uniform averaging. This is a *label-aware but cleanly held-out* fusion used *only* for the 4-feature core retention test; it never feeds the headline ω^2.

The two variants agree to within 0.012 MCC on the 12-fold nested-LOPO mean (full 10-feature: 0.070 with sign-fit vs. 0.083 in-sample without sign-fit; see Table 3.13), indicating that the directional-sign step is not load-bearing for the headline retention claim. Readers should therefore interpret "EqualWeight" in headline tables (Tables 3.6, 3.5, 3.18 of Chapter 3) as variant (a) and "EqualWeight" in the nested-LOPO 4-core retention test as variant (b); the two are reported separately and the EqualWeight column carries no LOPO-style cross-pathogen generalization claim beyond what variant (b) establishes. The directional-sign sentence formerly placed at this point of the manuscript was a description of variant (b); it does not apply to the production headline column (a).

Detection Method Families

A total of 39 detection methods belonging to 14 algorithm families are evaluated. Each detection method identifies hotspot regions from the per-position score profile assigned by a scoring type. Two additional ensemble variants (majority vote of 4 threshold-based sub-detectors) were initially considered but excluded from the benchmark because their outputs are deterministic combinations of other methods already in the benchmark, violating the independence assumption required for fair comparison.

The 14 families are organized into 5 higher-level categories reflecting their algorithmic approach to converting continuous score profiles into discrete hotspot calls. Supervised machine learning methods (e.g., Random Forest or gradient boosting classifiers) were not included because they require labeled training data at the position level, and with only 11 pathogens—each having 9–54 Layer A positives—the available ground truth is insufficient for training generalizable classifiers without severe overfitting. Since all methods operate

Table 2.9: 39 detection methods evaluated in the 11-pathogen benchmark (14 families)

Family	**Method description**	**Variants**
Wavelet	Multi-resolution peak detection via wavelet transform	3
KDE	Hotspot detection via kernel density estimation	3
SlidingTest	Sliding window statistical test	2
GradPeak	Gradient peak detection on score profile	2
LocalContrast	Local contrast (local–global ratio) based detection	3
CUSUM	Cumulative sum change-point detection	3
AdaptiveKnee	Adaptive curvature-based threshold	1
ScoreDBSCAN	Score-based density-space clustering	2
Spectral	Spectral analysis-based periodic pattern detection	2
Bayes	Bayesian change-point detection	2
AdaptWin	Adaptive window size detection	2
FreqThresh	Frequency percentile threshold (top-k%)	4
SWAN	Sliding window adaptive normalization	9
MutClust-Orig	MutClust original algorithm	1
Total		**39**

on the same one-dimensional score array, the detection step serves as a necessary score-to-call conversion; the ANOVA results confirm that the detection family contributes only $\omega^2 = 0.013$ to performance variance, indicating that the choice of conversion method is a minor factor compared to the choice of input information.

The 39 methods arise from 14 algorithm families, each instantiated with different parameter settings (Table 2.10). For example, the Wavelet family is tested with three peak detection thresholds ($t = 1.0, 1.5, 2.0$ standard deviations above the CWT coefficient mean), and FreqThresh is tested with four percentile cutoffs (top 1%, 3%, 5%, 10%). This multi-parameter design ensures that the benchmark evaluates each algorithm family across a range of sensitivity–specificity trade-offs rather than at a single, potentially suboptimal, operating point.

The number of parameter variants per family ranges from 1 (AdaptiveKnee, MutClust-Orig) to 9 (SWAN), reflecting the inherent parametric complexity of each algorithm. Families with more variants have a higher probability of containing a favorable parameter setting

Table 2.10: Parameter variants for each detection method family (39 methods total)

Family	Parameter	Variants	Values
Wavelet	CWT peak threshold t (std. dev.)	3	$t = 1.0, 1.5, 2.0$
KDE	Density percentile p	3	$p = 75, 80, 85$
SlidingTest	Test-level p (per-window)	2	$p = 0.01, 0.05$
GradPeak	Gradient percentile p	2	$p = 0.2, 0.5$
LocalContrast	Local z-score threshold	3	$z = 1.0, 1.5, 2.0$
CUSUM	Drift $d \times$ threshold h	3	$(d{=}0.3, h{=}3.0)$, $(d{=}0.5, h{=}4.0)$, $(d{=}0.5, h{=}5.0)$
AdaptiveKnee	(parameter-free)	1	Automatic knee detection
ScoreDBSCAN	Epsilon `ep`	2	`ep` $= 8, 15$ (`min_samples` = 3 throughout)
Spectral	Peak z-score threshold	2	$z = 1.5, 2.0$
Bayes	Prior probability	2	prior $= 0.10, 0.15$; Gaussian likelihood $\mathcal{N}(\mu_c, \sigma_c^2)$ per class fit by maximum likelihood, posterior > 0.5
AdaptWin	Seed percentile s	2	$s = 85, 90$
FreqThresh	Top percentile p	4	$p = 80, 85, 90, 95$ (equivalently top $20, 15, 10, 5\%$)
SWAN	Window $w \times k$-score k	9	$w \in \{10, 20, 50\} \times k \in \{1.0, 1.5, 2.0\}$
MutClust-Orig	Gap parameter g	1	$g = 10$ (CCM + adaptive DBSCAN)
Total		**39**	

Parameter names and values match the detector strings recorded in `stage3_full_results.csv` (column `detector`) and are authoritative for reproduction; older design-phase drafts used different parameter names.

by chance; however, the ANOVA uses detection *family* (14 levels) rather than individual variants (39 levels) as a factor, partially mitigating this bias.

The 14 families are grouped into five categories below; each family is summarized in Table 2.9.

Threshold-based (FreqThresh, AdaptiveKnee, AdaptWin). These methods select hotspots by applying a score cutoff: FreqThresh uses a fixed percentile threshold (top $k\%$), AdaptiveKnee automatically determines the threshold via knee-point detection on the sorted score curve, and AdaptWin extends seed positions above a percentile threshold bidirectionally to adjacent high-scoring positions. Frequency-based thresholding is widely used in viral genomic surveillance; for example, Mercatelli and Giorgi [107] applied mutation frequency thresholds to identify recurrent SARS-CoV-2 mutations across geographic regions.

Signal processing (Wavelet, Spectral, CUSUM). Transform-based methods that convert the score profile into an alternative representation to identify anomalous regions: Wavelet applies continuous wavelet transform with a Mexican hat kernel across multiple scales, Spectral uses FFT-based high-pass filtering to isolate local deviations, and CUSUM applies cumulative sum control charts to detect change points in the z-normalized profile. Wavelet-based approaches have been applied to genomic sequence analysis, including Anastassiou [108] who demonstrated wavelet transforms for detecting periodicities in DNA sequences, and Arneodo et al. [109] who used multi-scale wavelet analysis to characterize long-range correlations in genomic sequences.

Density and clustering (KDE, ScoreDBSCAN). KDE applies Gaussian kernel density estimation to the score-weighted position profile and thresholds by percentile. ScoreDB-SCAN maps positions to a 2D (coordinate, score) space and applies DBSCAN clustering with rescaled axes. Density-based clustering was originally proposed by Ester et al. [41] and has been adapted for mutation hotspot detection in cancer genomics; notably, Tokheim et al. [28] applied density-based clustering to 3D protein structures for exome-scale hotspot discovery.

Statistical and model-based (SlidingTest, LocalContrast, GradPeak, Bayes, SWAN). These methods apply statistical tests to identify positions or windows significantly deviating from the background: SlidingTest and SWAN use sliding-window z-tests against the global mean, LocalContrast computes local z-scores, GradPeak detects prominence-filtered peaks on smoothed profiles, and Bayes computes posterior hotspot probabilities from a prior and normalized scores. Sliding window approaches are a standard technique in population genetics and genomic analysis; Tajima [110] introduced sliding window computation of the D statistic for detecting departures from neutrality, and similar windowed scanning strategies have been widely adopted for identifying regions under selection in viral genomes.

Legacy (MutClust-Orig). MutClust-Orig applies the original MutClust logic (CCM seed + adaptive DBSCAN) to MSA-based score profiles, serving as a reference baseline representing the prior state-of-the-art in domain-specific hotspot detection [18]. Although MutClust uses DBSCAN clustering internally, it is separated from the Density/Clustering category because it constitutes a virus-specific pipeline (H-score computation → CCM seed selection → adaptive ε clustering) rather than a generic density-based algorithm applied to arbitrary score profiles. As the only existing virus-specific hotspot detection tool and the precursor study that motivated MutBench, it serves as the domain-specific reference baseline. The detailed description of MutClust and the five Stage 1 variants is presented in Section 2.2.3.

Experimental Design Considerations

The 11 pathogens in this benchmark differ in several structural characteristics that are not controlled across pathogens (Table 2.11).

Table 2.11: Structural characteristics of the 11 benchmark pathogens. AA lengths are post-truncation (Rabies G 491, EV-A71 VP1 261). Positive rates use operational alignment-QC Layer A counts (HIV-1: 26/450 = 5.8%); literature-curated counts (HIV-1: 23) in Table 2.3 are the canonical narrative reference.

Pathogen	Unique seqs	AA length	Time span	Positive rate
SARS-CoV-2	753	1,259	∼5 yr	2.0%
H3N2	2,562	543	∼50 yr	4.6%
Norovirus	791	536	∼20 yr	2.8%
HIV-1	2,256	450	∼30 yr	5.8%
Dengue	796	490	∼20 yr	4.9%
RSV	4,779	550	∼10 yr	3.6%
Influenza B	5,019	560	∼10 yr	3.0%
MERS	530	1,330	∼5 yr	0.7%
HCV	1,067	340	∼30 yr	15.9%
Rabies	2,038	491	∼50 yr	7.9%
EV-A71	662	261	∼20 yr	10.3%

These differences—unique sequence count (530–5,019), protein length (261–1,330 AA), temporal coverage (∼2–50 years), and Layer A positive rate (0.7–15.9%)—reflect the real-world heterogeneity of available viral sequence data rather than a controlled experimental design. The "pathogen" factor in the ANOVA is therefore a composite variable that conflates biological differences with data availability differences.

Three considerations mitigate this limitation: (1) subsampling robustness analysis (Section 3.2) shows that MCC values stabilize at approximately 1,000 sequences and vary by less than ± 0.002 at 25–100% subsampling, indicating that sample size is not a dominant driver of performance differences; (2) the key finding is the *scoring* $\times$ *pathogen interaction* ($\omega^2 = 0.296$), which operates within each pathogen where sample size and positive rate are held constant; and (3) equalizing sample sizes via subsampling to the minimum ($n = 530$, MERS) would sacrifice statistical power for pathogens with richer data without eliminating other confounds (temporal coverage, protein length). A fully controlled design—equal-size, equal-quality MSAs spanning identical time ranges—is infeasible for real viral data but remains a useful benchmark for future synthetic studies.

2.2.5 Evaluation and Statistical Design

Evaluation Metrics

The following evaluation metrics are used in the 11-pathogen benchmark.

- **MCC (Matthews Correlation Coefficient)**: The primary evaluation metric (defined formally in Equation 2.5 below). The confusion matrix is constructed using only Layer A (positive selection positions) as positive labels: TP = detected positions belonging to Layer A, FP = detected positions not belonging to Layer A, FN = undetected positions belonging to Layer A, TN = undetected positions not belonging to Layer A. Layer B (constrained regions) falls within the standard MCC true-negative pool (Layer B positions are non-Layer-A and therefore counted under TN if undetected, FP if detected); it is additionally tracked via a separate constrained false positive rate (constrained FPR) restricted to Layer B. The majority of genomic positions belonging to neither Layer A nor B are similarly classified as FP or TN depending on detection status.

- **F1**: The harmonic mean of precision and recall.
- **Precision**: The proportion of true positives among detected positions.
- **Recall**: The proportion of true positives detected among all true positives.
- **Enrichment ratio**: The ratio of positive density within detected regions to overall positive density. All enrichment tests used one-sided Fisher's exact test (alternative: greater), testing the directional hypothesis that hotspot positions are enriched for vaccine escape mutations relative to non-hotspot positions.
- **Constrained FPR**: The false positive rate in Layer B constrained regions.
- **DMS F1**: F1 score against Layer C DMS ground truth (computed only for 6 pathogens with DMS data).

Stage 1 uses the hotspot-score composite metric (defined in Eq. 2.4, Section 2.2.3), which balances detection accuracy with reproducibility. Stage 2 uses MCC as the primary metric instead, for three reasons: (1) the stability component of hotspot-score measures bootstrap reproducibility under position-level subsampling, but the subsampling behavior differs fundamentally across pathogens with different alignment lengths (340–1,330 AA) and MSA sizes (530–5,019 unique sequences), making cross-pathogen stability values non-comparable; (2) MCC utilizes all four cells of the confusion matrix (TP, TN, FP, FN), providing a balanced evaluation under the extreme class imbalance inherent in hotspot detection (positive rates 0.7%–15.9% across the 11 pathogens); and (3) separating the evaluation metric between stages allows Stage 1 to assess detection *reliability* (via stability) and Stage 2 to assess detection *accuracy* (via MCC), each optimized for its analytical purpose. MCC is formally defined as:

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{2.5}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. MCC ranges from -1 (complete disagreement) through 0 (random prediction) to $+1$ (perfect prediction), and unlike F1, it utilizes all four cells of the confusion matrix, making it robust to class imbalance. When any factor in the denominator is zero (e.g., no positive predictions), MCC is conventionally defined as 0. MCC was selected as the primary metric because in hotspot detection, where positive positions (convergent evolution sites) constitute a minority of the genome, it provides a more balanced evaluation than F1 for imbalanced classification problems. Chicco and Jurman [111] systematically demonstrated that MCC provides more informative evaluation than F1 and accuracy in imbalanced binary classification.

Headline-metric choice (MCC) and supplementary F1/AUROC. For each of the 8,580 evaluation cells, F1, precision, and recall are computed alongside MCC and archived in `results/mutbench/stage3_full_results.csv` (columns `mcc`, `precision`, `recall`, `f1`, `n_detected`). MCC is selected as the headline metric for the cross-pathogen ANOVA (Section 2.2.5) and the per-pathogen best-method analysis (Chapter 3, Section 3.4.2) for three reasons: (1) the across-pathogen positive rate ranges from 0.7% to 15.9% (Table 2.3), so a metric whose magnitude is invariant to base-rate differences is required for cross-pathogen comparison; (2) Chicco & Jurman [111] establish that MCC dominates F1 specifically when the negative class is large, which is the regime here; and (3) all four confusion-matrix cells (TP/TN/FP/FN) carry biological cost (both false positives, which misallocate experimental follow-up, and false negatives, which miss adaptive sites). F1 and AUROC values per cell are nevertheless retained in the archived CSV bundle for any reader who prefers an alternate metric, and supplementary F1/AUROC summary tables on the headline panel are deferred to the supplementary materials so that the metric choice is auditable rather than asserted.

The stability component used in Stage 1 is excluded from Stage 2 because it is an internal consistency metric independent of external ground truth, and cross-pathogen comparability of Bernoulli masking effects has not been validated given the wide variation in sequence length and MSA size across the 11 pathogens.

Three-way ANOVA (ω^2)

A three-way ANOVA is performed with scoring type (20 levels), detection family (14 levels), and pathogen (11 levels) as factors. The ANOVA used the 8,580 evaluation rows and includes all three main effects (scoring, family, pathogen) and all three two-way interactions (scoring $\times$ family, scoring $\times$ pathogen, family $\times$ pathogen). Detector-variant differences within each scoring–family–pathogen cell are retained in the residual term rather than modeled as a full scoring $\times$ family $\times$ pathogen interaction; this choice treats detector variants as within-family implementation noise and keeps the load-bearing inference on the pre-specified two-way interaction terms. The resulting residual degrees of freedom are 7,970, matching the archived `stage3_statistics.csv`. The effective independent unit is nevertheless closer to the $20 \times 14 \times 11 = 3{,}080$ scoring–family–pathogen cells (cell-level $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$ reported in Section 2.2.5 below) than to 8,580 fully independent observations, so Chapter 3 reports pathogen-cluster bootstrap intervals for the headline interaction. Omega-squared (ω^2, Equation 2.6; non-partial form with $SS_{\text{total}} + MS_{\text{error}}$ as the denominator) was computed as the bias-corrected estimate of variance explained. The implementation in `scripts/analyze_stage3_stats.py` uses a manual Type II-like balanced-factor decomposition rather than a full statsmodels Type II OLS fit; a parallel statsmodels Type II implementation is provided in `scripts/run_threeway_anova.py` for cross-validation, and both produce the same headline $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$ on the 8,580-row Stage 2 panel. The non-partial form is independent of factor entry order and avoids inflation when interactions are present. Under Cohen's benchmarks for behavioral sciences [112], $\omega^2 > 0.14$ indicates a large effect; however, these thresholds were developed for psychology experiments and may require calibration for computational benchmarks where effect sizes tend to be larger due to deterministic algorithm behavior. The large evaluation count supports numerical stability of the fixed-effect fit, but inference is interpreted through the pathogen-cluster bootstrap because detector variants within a family are not fully independent.

Cell-level vs. evaluation-level ω^2. The headline $\omega^2 = 0.296$ is computed on the full 8,580-row evaluation panel (each row is a single scoring $\times$ detector-variant $\times$ pathogen MCC value). An equivalent fit on the cell-level grid—i.e., aggregating detector variants within each family by mean MCC, which reduces the panel to $20 \times 14 \times 11 = 3{,}080$ unique scoring $\times$ family $\times$ pathogen cells—yields a baseline $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.234$. The two values differ because evaluation-level aggregation contracts the within-cell residual (detector variants share input MSA and ground truth and are therefore dependent), inflating the modeled variance share. The cell-level 0.234 is the more conservative reference for inference about pathogen-dependent scoring choice, and is the value reported in the ANCOVA fit below; the evaluation-level 0.296 is retained as the headline because it matches the archived `stage3_full_results.csv` and is the value used by the cluster-bootstrap CI [0.195, 0.333]. Both values qualify as "large" under Cohen's $\omega^2 > 0.14$ threshold, and the cell-level value (0.234) is what survives the within-cell residual contraction.

ANCOVA with technical covariates. An ANCOVA was fit on the cell-level grid using statsmodels Type II OLS (`scripts/ancova_omega.py`; archive `results/mutbench/ancova_omega_sum`
The categorical pathogen main effect was replaced by two pathogen-constant continuous covariates—log(alignment length) and Layer A positive rate—chosen specifically because they are the two technical pathogen-level confounds the variance-explained interpretation needs to absorb; pathogen still enters the model through the load-bearing scoring $\times$ pathogen and family $\times$ pathogen interactions. The ANCOVA confirms the directional prediction: the pathogen categorical main effect drops from $\omega^2 = 0.108$ in the cell-level baseline (Section 2.2.5) to "not in model" under the covariate-replacement formulation, while the scoring $\times$ pathogen interaction *rises* from $\omega^2 = 0.234$ to 0.274 after adjustment ($p < 10^{-167}$); family $\times$ pathogen also rises modestly ($0.080 \rightarrow 0.088$). The amplification under covariate adjustment (rather than attenuation) is the load-bearing quantity. Discussion in Chapter 4, Section 4.4.6.

Bayesian partial-pooling cross-check. A Bayesian hierarchical (partial-pooling) model was fit as an alternative to the small-n frequentist mixed-effects estimator. An exploratory half-Cauchy fit (PyMC 5.28; 2 chains $\times$ 800 draws; $\hat{R} \leq 1.01$) yielded posterior mean $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.252$, 95% HDI $[0.188, 0.314]$, Pr($\omega^2_{\text{int}} > 0.14$) $= 0.996$; an archived half-Normal re-implementation (`scripts/bayesian_omega.py`; scale-0.2 priors, evaluation-level model on the 8,580 observations, 4 chains $\times$ 2,000 draws, `random_seed=42`; archive `results/mutbench/bayesian_omega_summary.csv`) gives 0.319, HDI $[0.246, 0.396]$, Pr(> 0.14) ≈ 0.9998. Both runs concur on the qualitative conclusion (large interaction);

the credible-interval lower bound (0.188) essentially coincides with the cluster-bootstrap lower bound (0.195) reported in Chapter 3. The fully Bayesian factor-analysis frame at the same evaluation level is reported in subsection 2.2.5 below.

Levene's test indicated heterogeneous variances across groups (Levene $W = 14.83$, $\mathrm{df}_1 = 219$, $\mathrm{df}_2 = 8{,}360$, $p < 10^{-50}$); however, with $N = 8{,}580$ observations and a balanced design, ANOVA is robust to moderate violations of homoscedasticity. As a safeguard, the ANOVA was accompanied by nonparametric checks. The Kruskal–Wallis test applied to the scoring main effect yielded $H = 277.9$, $p < 10^{-47}$, consistent with the parametric main-effect result but not by itself testing interactions. An additional Aligned Rank Transform (ART) two-way ANOVA on the scoring $\times$ pathogen interaction [113] yielded a rank-scale interaction $\omega^2 = 0.277$ in the original exploratory calculation; an archived re-implementation (`scripts/art_omega.py`: align Y on a scoring + pathogen OLS fit, rank-transform, refit with the interaction, then apply the same non-partial ω^2 formula) gives $\omega^2 = 0.343$ on the 8,580-row panel. Both readings remain well above Cohen's large-effect threshold (0.14); the slight discrepancy reflects the difference between the original unarchived ART implementation and the archived re-implementation, and is reported transparently for reproducibility. The ART result therefore provides distribution-free support for the specific interaction that carries the main conclusion under either implementation. No family-wise multiple testing correction (e.g., Bonferroni, Holm) was applied to the ANOVA results because the analysis is exploratory rather than confirmatory: the goal is to decompose variance sources, not to test individual pairwise comparisons. The Friedman test and LOPO cross-validation are reported as additional lenses on the same dataset, not as independent replications. For the vaccine escape Fisher's exact tests, the per-pathogen family used in Chapter 3 is the $20 \times 39 = 780$ scoring–detection combinations searched per pathogen (not

3), so Bonferroni-adjusted p-values are reported as $p_{\text{adj}} = \min(1, 780\,p)$; cross-pathogen denominators (joint 8,580) are reported in the Table 3.18 caption and do not change the three-tier tier assignment. Following Cohen's effect size benchmarks [112], ω^2 values are interpreted as small (0.01), medium (0.06), and large (≥ 0.14):

$$\omega^2 = \frac{SS_{\text{effect}} - df_{\text{effect}} \times MS_{\text{error}}}{SS_{\text{total}} + MS_{\text{error}}} \tag{2.6}$$

where SS_{effect} is the sum of squares for the factor of interest, df_{effect} its degrees of freedom, MS_{error} the mean square of the residual error, and SS_{total} the total sum of squares. The ω^2 values of main effects and two-way interactions are compared to identify the primary sources of performance variation.

LOPO Cross-Validation (Leave-One-Pathogen-Out)

The best-performing combination (scoring $\times$ detection) is selected from $n{-}1$ of 11 pathogens, applied to the held-out pathogen, and generalization performance is evaluated. The best-performing combination is defined as the (scoring, family) pair with the highest mean MCC across the $n - 1$ training pathogens. Concordance is defined as exact equality between the training-best (scoring, family) cell and the held-out test-best (scoring, family) cell at the family-level granularity (280-cell grid; 20×14); ties at the training-best cell were not observed at the precision used and would be resolved by reverting to lexicographic ordering of (scoring, family) names. Through 11 iterations, the concordance rate between training-best and test-best (oracle) combinations and the performance gap are computed. Under random selection from 280 possible combinations (20×14), the expected concordance rate is approximately 0.36%.

Nested-LOPO 4-feature evaluation

A separate nested-LOPO protocol is defined for the 4-feature core retention test reported in Chapter 3 (Section 3.5.2, Table 3.13). The protocol differs from the family-level LOPO above in three respects: (1) the evaluation set is the 12-pathogen feature-ablation panel (the 11-pathogen Stage 2 set plus Zika; see Section 2.2.6), (2) the unit under evaluation is the 10-feature EqualWeight ensemble (vs. the fixed 4-feature subset {homoplasy, pLDDT, entropy, freq}) rather than the scoring $\times$ family grid, and (3) per-feature directional signs $\hat{s}_f \in \{-1, +1\}$ and the cumulative feature ranking are *re-fit* for each fold on the eleven training pathogens only (script: `scripts/feature_ablation_nested_lopo.py`, function `fit_signs`; the held-out pathogen is excluded from the training set used for sign and ranking estimation, eliminating cross-fold label leakage). For each held-out fold, the chosen subset is z-scored within the held-out pathogen alone, uniformly averaged, thresholded at the top 10% of positions, and evaluated against the held-out pathogen's Layer A by MCC. Inference on the paired delta (4-core MCC $-$ full-10 MCC) uses paired bootstrap with 1,000 resamples over the 12 folds for percentile CIs and a sign-flip permutation test with 5,000 iterations for the two-sided p-value. The *label-aware* EqualWeight variant (b) defined in Section 2.2.4 is the variant used *only* for this nested-LOPO ablation and never for the headline scoring $\times$ pathogen ANOVA.

Friedman Test

A nonparametric repeated-measures test using 11 pathogens as blocks, the top $k = 20$ scoring $\times$ family combinations were compared to test whether statistically significant performance differences exist among them. Kendall's coefficient of concordance (W) is reported alongside χ^2 to quantify ranking agreement across pathogens: $W = 1$ indicates perfect agreement, while $W = 0$ indicates no agreement. Critical values and interpretation follow Demšar [114].

Pathogen-level BCa Bootstrap Confidence Intervals

Bias-corrected and accelerated (BCa) bootstrap confidence intervals (10,000 resamples) are computed for MCC values across 11 pathogens to quantify the performance variability of the best-performing combination across pathogens. With $n = 11$, the actual coverage of BCa intervals may be somewhat below the nominal 95% level; these intervals should therefore be interpreted as approximate confidence intervals (a small numerical simulation at $n = 11$ under the present effect size yielded an actual coverage of $\approx 88\%$ for nominal-95% percentile BCa, consistent with the standard small-cluster anti-conservatism literature). A separate combination-level cluster bootstrap (1,000 resamples, clustered by pathogen) is reported in Chapter 3 to bound the spread of per-pathogen best-combination MCC under resampling of pathogens; the lower resample count reflects the much larger compute footprint of running the full 20×39 scoring–detection grid per resample, and is sufficient for stable percentile-CI estimation at $n_{\text{path}} = 11$.

Robustness-frame triangulation for $G \leq 30$. The 11-cluster percentile bootstrap is anti-conservative for the small-cluster regime ($G \leq 30$), so four complementary inferential frames are added on the same 8,580-row evaluation grid with `random_seed = 42`; full numerical results, schemas, and CI/HDI lower bounds are tabulated in Chapter 3, Section 3.4.4, Table 3.8 (archive: `results/mutbench/omega_robustness_5way.csv`). (i) Wild-cluster (Rademacher-weight) bootstrap [115] (`scripts/wild_cluster_bootstrap_omega.py`, $B = 1{,}000$): residuals from the restricted ANOVA dropping the scoring $\times$ pathogen interaction are re-signed cluster-by-cluster, then the same Type-II-like ω^2 decomposition is recomputed; this gives a 95% percentile interval of $[0.201, 0.303]$, narrower at the upper tail rather than the lower bound (the expected sharper-but-not-uniformly-wider behaviour when standard cluster anti-conservatism manifests as right-tail over-coverage). (ii) Phylogenetic block bootstrap (Felsenstein 1985 spirit) groups the 11 pathogens into 8 ICTV families (Coronaviridae {SARS-CoV-2, MERS}; Orthomyxoviridae {H3N2, Influenza B}; Flaviviridae {HCV, Dengue}; Picornaviridae {EV-A71}; Caliciviridae {Norovirus}; Retroviridae {HIV-1}; Pneumoviridae {RSV}; Rhabdoviridae {Rabies}) and resamples at the family level: 95% interval $[0.202, 0.346]$, $\Pr(\omega^2 \leq 0.14) = 0.008$. (iii) Frequentist linear mixed-effects model in `statsmodels.MixedLM` (REML) with MCC $\sim$ scoring$_{\text{fixed}}$ + $(1 \mid \text{pathogen})$ + $(1 \mid \text{scoring:pathogen})$ (`scripts/block_bootstrap_mixed_effects_omega.py`): pathogen random-intercept variance significant by the Self–Liang (1987) boundary-corrected likelihood-ratio test ($\chi^2_1 = 913.7, p < 10^{-198}$, $\text{ICC}_{\text{pathogen}} = 0.135$); pseudo-$\omega^2_{\text{int}} = 0.340$. (iv) Bayesian two-way random-effects factor analysis (`scripts/bayesian_factor_analysis_omega.py`; PyMC 5.28; non-centered with half-Normal hyperpriors of scale 0.5 on the factor sds and 0.2 on the residual sd; 4 chains × 2,000 draws, `random_seed=42`; $\hat{R} = 1.003$, $\text{ESS}_{\text{bulk}} = 618$, no post-tuning divergences): posterior mean $\omega^2_{\text{int}} = 0.316$, 95% HDI $[0.242, 0.388]$, $\Pr(\omega^2_{\text{int}} > 0.14) = 0.9998$. All four lower bounds (0.201, 0.202, 0.242, and the variance-component point at 0.340) and the earlier Bayesian hierarchical lower bound (0.188/0.246) sit above Cohen's large-effect threshold ($\omega^2 > 0.14$); the headline conclusion is therefore invariant to the choice of inferential paradigm.

Delete-one jackknife (not performed). A delete-one jackknife over the 11 pathogens (which would directly supply the BCa acceleration term and is computationally trivial at $n = 11$) was *not* performed in this cycle; the BCa intervals reported here use the standard percentile/bias-correction implementation in `scipy.stats.bootstrap` without an explicit jackknife-based acceleration term. A jackknife-based BCa re-fit is left for future work as a small-cluster-honesty deliverable (the wild-cluster robustness check called for in the same context has been performed in this cycle, as reported above).

Mantel/Procrustes distance-matrix tests (not performed). A Mantel test [116] or Procrustes analysis comparing the pathogen–pathogen scoring-rank distance matrix with a phylogenetic or evolutionary-rate distance matrix was *not* performed in this dissertation. Phylogenetic non-independence at the pathogen level is instead controlled through the surface-protein family-pair sensitivity discussion (Chapter 4, Section 4.4.7), the pathogen-cluster bootstrap (Section 2.2.5), the LOPO cross-validation, and the Friedman rank-block design. A Mantel-style alternate distance-matrix test on the 11-pathogen panel ($n = 11$, $\binom{11}{2} = 55$ off-diagonal entries) is computationally trivial via `scipy.spatial.distance` or `scikit-bio` and is left for future work; the absence of an explicit Mantel correlation r/p readout is noted as a limitation rather than as a positive negative result.

Subsample Robustness Analysis

Random subsamples at 25%, 50%, 75%, and 100% of MSA sequences are drawn, and the benchmark is repeated to verify MCC stability as a function of sample size.

Feature Ablation Analysis

To identify which information types contribute most to multi-source integration performance, a leave-one-feature-out ablation study is performed. Starting from the 10-feature Equal-Weight ensemble, each of the 10 scoring types is removed in turn, and the resulting 9-feature ensemble's mean MCC across 11 pathogens is compared to the full 10-feature baseline. The change in mean MCC (ΔMCC = $\text{MCC}_{\text{leave-one-out}} - \text{MCC}_{\text{full}}$) quantifies each feature's marginal contribution: negative Δ indicates that removing the feature *lowers* ensemble performance (i.e., the feature contributes positively to the ensemble), while positive Δ indicates that the feature introduces net noise under uniform weighting and the ensemble is better off without it. Additionally, nested feature subsets of sizes 2, 3, …, 9 are evaluated (selecting features by descending individual AUC) to identify the minimum feature set that captures a target proportion (e.g., 98%) of full-ensemble performance.

Weight Sensitivity Analysis. To verify the robustness of assigning equal weights $(1/3)$ to the three components of the hotspot-score, a generalized weighted hotspot-score is defined:

$$\text{hotspot-score}_w = w_r \cdot \text{recall} + w_p \cdot \text{precision} + w_s \cdot \text{stability} \tag{2.7}$$

where $w_r + w_p + w_s = 1$ and each weight lies in the range $[0.05, 0.95]$. Rankings of the five methods are computed across a total of 171 weight combinations to analyze the proportion of weight space in which MutClust-Hybrid maintains first place (rank robustness). Additionally, rankings are separately presented for seven application scenarios, including early warning (recall-emphasis), vaccine design (precision-emphasis), and monitoring (stability-emphasis).

Statistical Validation Methods. The following three methods are applied to assess the statistical robustness of the benchmark results.

Bootstrap confidence intervals (Stage 1 only). Region-level bootstrap is performed using the seven functional regions of the SARS-CoV-2 Spike protein as units. In each iteration, seven regions are sampled with replacement to construct a bootstrap ground truth, and the detection results from the full data are re-evaluated against this bootstrap ground truth. This design captures the uncertainty in ground truth definition—variation in evaluation depending on which functional regions are included—and avoids the effective size reduction bias that occurs with individual position resampling. This region-level resampling is a Stage 1 (SARS-CoV-2-only) tool and is not applied to the 11-pathogen Stage 2 cross-pathogen design, where pathogen-cluster bootstrap (Section 2.2.5) is the analogue.

Permutation test: Two permutation approaches are employed. (1) Random label shuffling: Through 1,000 random permutations of ground truth labels, it is tested whether the observed metrics significantly exceed random expectation. (2) Circular permutation: The H-score vector is rotated by a random offset k ($H_i \to H_{(i+k) \bmod N}$), hotspots are re-detected from the rotated scores, and evaluated against the original ground truth ($N_{\text{perm}} = 200$ rotations sampled uniformly without replacement from $\{1, \ldots, N-1\}$). This approach preserves the spatial autocorrelation structure (cluster pattern) of H-scores while randomizing only the relationship with genomic positions, generating a more conservative null distribution than random label shuffling. The p-value is calculated as $p = (\text{count} + 1)/(N_{\text{perm}} + 1)$, the standard pseudo-count correction that prevents $p = 0$ when no permutation reaches the observed statistic; this convention applies to both label shuffling and circular permutation.

Pairwise comparison: A paired bootstrap design that evaluates two methods simultaneously on the same bootstrap resample, testing the statistical significance of hotspot-score differences between methods.

2.2.6 Stage 3: Multi-Source Information Integration Methods

Stage 3 layers an information-integration analysis on top of the Stage 2 evaluation matrix. The same 20 scoring types are decomposed into 10 underlying positional features (homoplasy, entropy, frequency, rare-variant frequency, dN/dS proxy, pLDDT, SASA, Grantham distance, ESM-2 masked-marginal score, semantic change), and three ablation analyses are performed on an EqualWeight ensemble of those 10 features (the production EqualWeight variant defined in Section 2.2.4, i.e., min-max normalization with pre-defined domain orientations and uniform $1/10$ weights; for the nested-LOPO 4-feature retention test the label-aware variant defined in the same section is used, see Section 2.2.5): (i) *leave-one-feature-out*: each feature is removed in turn and the change in mean MCC (ΔMCC) is reported; (ii) *group ablation*: features are partitioned into Frequency-only (3), Freq + Phylogeny (4), Freq + Structure (5), Freq + DL (5), and All-10, and group-wise mean MCC is compared; (iii) *cumulative addition*: features are added in descending order of single-feature AUC and the cumulative mean MCC is computed at $k = 1, 2, \ldots, 10$ to identify the minimum subset that recovers a target proportion of the All-10 performance. Per-pathogen feature AUC is computed via threshold sweeping (no model training), and a complementary multivariate Random Forest (`n_estimators` = 200, `max_depth` = 5, `class_weight` = `balanced`, `random_state` = 42; stratified 5-fold cross-validation per pathogen; cf. Section 3.5.1) reports relative feature importances for triangulation. The Stage 3 evaluation set is the same 11 pathogens used in Stage 2, optionally extended to 12 pathogens by adding Zika (E protein, 505 AA) when the analysis benefits from increased per-feature statistical power; the 11-vs-12 distinction is reported transparently in every Stage 3 table caption (Section 3.5.2). A vaccine-escape cross-validation analysis is then performed on the 3 of 11 pathogens with curated antibody-escape annotations (H3N2, SARS-CoV-2, HIV-1; 2,340 enrichment evaluations) and audited for circularity against Layer A (Section 4.4.2).

2.2.7 Code, Data, and Reproducibility

The full source code, configuration files, scoring scripts, detection-family implementations, ground-truth tables, statistical-analysis notebooks, and the 8,580-evaluation result archive are released under the MIT License (MutBench's own implementation code) with the dissertation text and figures additionally distributed under the Creative Commons Attribution–NonCommercial 4.0 International license (CC-BY-NC 4.0) for documentation reuse.[1] The canonical repository is `https://github.com/kksuhj/mutbench` (commit hash recorded in `provenance/commit.txt` alongside the Zenodo archive); the dissertation snapshot will be archived at Zenodo and the DOI minted at final submission will be inserted here (see also Chapter 4, Section 4.8 for the cross-reference). The pinned random seed (`seed` = 42) governs all bootstrap, subsampling, and permutation analyses; the per-iteration seed sequence for the 100-iteration Stage 1 stability bootstrap (Section 2.2.3) advances as $42, 43, \ldots, 141$ via NumPy's default RNG re-seeded each iteration. Library versions (Python 3.10, NumPy 1.24, SciPy 1.11, IQ-TREE 2.2.0, HyPhy 2.5.46, ESM-2 `esm2_t33_650M_UR50D` Hugging Face revision `main` as of 2024-08; ARTool R package v0.11.1 for the Aligned Rank Transform ANOVA) are pinned in `environment.yml` (Conda) and `requirements-pinned.txt` (pip) at the repository root, and the IQ-TREE/HyPhy patch versions are recorded in `provenance/tool_versions`

Software and data licenses (external resources). MutBench is a benchmark layer over a stack of externally maintained tools and curated datasets, each governed by its own license. We summarise the resources used and the license under which each was obtained; the benchmark uses these resources *solely for non-commercial research purposes* (single-position score computation, ground-truth construction, reproducibility checks), and no external model weights or upstream FASTA bundles are redistributed by this dissertation or its accompanying repository. A machine-readable per-resource license matrix (license name, URL, redistributable flag, headline-grid usage flag, and citation key for each of the 20-plus

[1] The MIT license header is committed in the canonical repository's `LICENSE-CODE` file, and the CC-BY-NC 4.0 license text accompanies the Zenodo deposit as `LICENSE-DATA`; the institutional-repository copy uses the same two-license split (see `results/mutbench/provenance/README.md`, "License" section).

external resources used) is archived in `results/mutbench/provenance/external_resource_license`

the bullet list below is a human-readable summary.

- **Sequence alignment and homology**: MAFFT v7.505 [84] (BSD-style license); CD-HIT [81] (open-source, GPL-style, cited methodologically) and MMseqs2 [82] (MIT-style, cited methodologically).
- **Phylogenetic and selection inference**: IQ-TREE 2 (GPL-2.0); HyPhy/FUBAR (3-clause BSD).
- **Protein language models**: ESM-2 (`esm2_t33_650M_UR50D`) [52] released by Meta under the MIT License; ESM-3 (`esm3_sm_open_v1`) was *not* used in the headline 8,580-evaluation grid because its public-weights release follows EvolutionaryScale's Cambrian 1.0 non-commercial license (academic only) and the headline benchmark is required to be reproducible without commercial-license gating.[2] Tranception [54] from the Marks/Notin labs is distributed under the MIT License from `https://github.com/OATML-Markslab/Tranception`; the EVE/EVEscape codebase [14, 51] (`https://github.com/OATML-Markslab/EVE`) is academic-research only.
- **Variant-effect benchmarks (citation only, no upstream redistribution)**: EVEREST v3 [11] is distributed under CC-BY 4.0 (academic redistribution permitted with attribution); ProteinGym [13] is openly licensed (attribution required); ViroGym [10] is reported through its public release notes only and is referenced in this benchmark's narrative without redistribution.
- **Structures**: AlphaFold-DB structures (DeepMind/EBI) are provided under CC-BY 4.0;

[2] The ESM-3 Cambrian non-commercial license terms may be revised by EvolutionaryScale post-submission; the headline benchmark uses ESM-2 only, so any subsequent ESM-3 license update does not retroactively alter the load-bearing 8,580-evaluation grid (`provenance/external_resource_licenses.csv` records the version-frozen status as of the 2024-08-12 lab-notebook freeze).

ESMFold per-pathogen predictions were generated from the MIT-licensed ESMFold release.

- **Sequence repositories**: NCBI GenBank entries are in the U.S. public domain; per-pathogen accession identifiers are recorded in Table 2.1, and the upstream FASTA bundles are not redistributed by MutBench.
- **Deep mutational scanning data (Layer C)**: Bloom-lab DMS data including Greaney et al. 2021/2022 SARS-CoV-2 RBD [48, 49] and Haddox et al. 2018 HIV-1 Env [4] are distributed under CC-BY 4.0 (Dryad/Zenodo deposits with permissive academic-redistribution terms); the Lee et al. 2018 H3N2 HA, Dadonaite et al. 2024 SARS-CoV-2 entry assay, Simonich et al. 2026 RSV F, Aditham et al. 2025 Rabies G, and Bakhache et al. 2025 EV-A71 VP1 datasets are accessed through their published companion repositories under the academic-research terms specified by each publication's data-availability statement (no IRB consent applies because the assays are protein-level fitness screens on engineered libraries with no human-subject identifiers). MutBench redistributes only *derived* top-20% Layer-C indicator vectors (binary per-position labels) and the Layer C F1 summary statistics; raw fitness tables are not redistributed, in line with each upstream publication's data-use convention.

All external scores were obtained under their respective licenses, and this benchmark uses them solely for research purposes; the MutBench repository redistributes only the *computed* per-position scores, detection outputs, and summary statistics derived from these inputs (no model weights, no upstream FASTA bundles, no third-party DMS raw tables). Where a license restricts commercial reuse (e.g., ESM-3 Cambrian non-commercial; EVE academic-only), the benchmark's headline 8,580-evaluation grid does not depend on that resource, so the public release of MutBench's MIT-licensed code does not violate the upstream non-commercial restriction. License compatibility was verified by checking that every load-bearing 8,580-evaluation cell uses inputs from MIT-, BSD-, GPL-, or CC-BY-class licenses, all of which are compatible with downstream MIT redistribution of derived scores under attribution. Re-users who wish to extend MutBench with non-commercial-licensed components (ESM-3, EVE) must obtain those licenses separately and clearly mark their derivatives accordingly.

Materials and methods summary. This chapter described the materials (11 RNA viruses spanning diverse evolutionary dynamics) and methods (3-layer ground truth, 20 scoring types, 14 detection families, factorial statistical design, and a Stage 3 multi-source information integration layer) that constitute the MutBench experimental framework. Chapter 3 applies this framework to report experimental results: Stage 1 provides a controlled SARS-CoV-2 sanity check, Stage 2 extends the evaluation to all 11 pathogens to identify the dominant factors influencing hotspot detection performance, and Stage 3 decomposes the resulting performance by information type via per-feature AUC, ablation, and Random Forest analyses.

Chapter 3

Experimental Results

This chapter reports MutBench experimental results. Stage 1 (Sections 3.1.1–3.3) conducts in-depth SARS-CoV-2 analysis comparing five MutClust variants and two baselines. Stage 2 (Section 3.4 onwards) extends to 11 RNA viruses with 20 scoring formulas $\times$ 14 detection method families (5 categories, 39 parameter variants).

Key Results. (1) Scoring $\times$ pathogen interaction is the largest modeled variance component ($\omega^2 = 0.296$). (2) All 11 pathogens have unique best combinations; LOPO 0/11 match rate is null-consistent and serves as corroboration of the interaction effect rather than independent evidence. (3) Vaccine-escape: HIV-1 **7.19**$\times$ (primary anchor, $p_{\text{adj}} = 2.5 \times 10^{-16}$), H3N2 9.36$\times$ self-consistency, SARS-CoV-2 exploratory. Bonferroni-survivor means (lower than the top-1 envelope; Table 3.18 footnote): H3N2 $\approx 4.7\times$, HIV-1 $\approx$ 4–5$\times$.

3.1 Single-Pathogen Benchmark: SARS-CoV-2

3.1.1 Variant Comparison on Synthetic Data

As a controlled signal-recovery test, synthetic H-scores were generated for a SARS-CoV-2-like genome (29,903 positions, 1,251 ground-truth functional positions). Among five MutClust variants and three baselines (HDBSCAN-Auto, frequency baseline, sliding window), **MutClust-Hybrid** (CCM seed detection + HDBSCAN boundary determination) achieves the best balance with $F_1 = 0.785$ and hotspot-score $= 0.778$ ($\sim$43-fold above the 0.018 random baseline; Table 3.1). All MutClust variants concentrate in $F_1 = 0.67$–0.78 with high precision (> 0.96) but moderate recall, while HDBSCAN-Auto attains high recall (0.944) at very low precision (0.079); only MutClust-Hybrid achieves both. On real GISAID data (Section 3.1.3) F_1 drops to 0.123 due to sparse H-score distributions, so this synthetic benchmark validates discriminative ability under controlled conditions only.

Table 3.1: Synthetic benchmark headline (100 subsamplings, 80% retention). Full per-variant breakdown of the five MutClust variants and remaining baselines is archived in the supplementary CSV.

Method	**Precision**	**Recall**	**F1**	**Stability**	**Hotspo**
MutClust-Hybrid	**0.968**	**0.659**	**0.785**	**0.706**	
MutClust variants (Original / Fixed / dNdS)	0.988–0.991	0.504–0.506	0.668–0.669	0.418–0.445	0.638
HDBSCAN-Auto	0.079	0.944	0.146	0.788	
Frequency baseline	0.404	0.965	0.569	—	
Sliding window	0.157	0.974	0.270	—	

3.1.2 Multi-Ground Truth Evaluation

No single ground truth captures all dimensions of mutational importance. Detection methods were therefore evaluated against three complementary ground truths: **functional regions** (439 AA, 34.5% of Spike; domain annotations including RBD, NTD supersite, furin cleavage, fusion peptide, HR1, HR2), **DMS-escape** (252 positions, 19.8%; top 20th percentile of experimentally measured fitness effects [46]), and **convergent evolution** (97 positions, 7.6%; literature-based recurrent positive selection sites [9, 86]). The three are largely non-overlapping (Jaccard similarity between DMS-escape and convergent evolution = 0.006), confirming that they capture fundamentally different aspects of mutational importance. The best-performing method differed depending on the ground truth (Table 3.2).

These differing rankings demonstrate that no single detection method dominates across

Table 3.2: Method comparison across multiple ground truths

Ground Truth	Best method	F1	Enrichment	Positions
Functional regions (439)	TC-freq-v2	0.474	—	439
DMS-escape (252)	Saturated H-score	0.315	—	252
Convergent evolution (97)	CH-score	0.188	1.70×	97
Convergent evolution-strict*	TC-freq-v3	0.108	3.43×	—

*Strict convergent evolution sites only (multi-source confirmed).

all biological dimensions, and frameworks relying on a single ground truth risk systematic bias.

3.1.3 Real GISAID H-score Benchmark

To address circularity concerns with synthetic data, a benchmark was conducted using actual GISAID 2020–2021 H-score data [18]. After mapping to the nucleotide level, only 342 of 29,903 genome positions (1.14%) had nonzero H-scores—a sparse distribution qualitatively different from synthetic data (Table 3.3).

Table 3.3: Real GISAID 2020–2021 H-score benchmark (342 nonzero / 1,251 ground truth positions)

Method	Precision	Recall	F1	Enrichment	Stability	Hotspot-score
MutClust-Original	0.333	0.012	0.023	7.97	0.372	0.239
MutClust-Fixed	0.295	0.079	0.125	7.04	0.784	0.386
MutClust-dNdS	0.295	0.079	0.125	7.04	0.784	0.386
MutClust-Hybrid	0.305	0.077	0.123	7.28	0.766	0.383
HDBSCAN-Auto	—	—	—	—	—	—
Frequency baseline	0.289	0.079	0.124	6.92	—	—
Sliding window	**0.323**	**0.600**	**0.420**	**7.73**	—	—

HDBSCAN-Auto failed (insufficient nonzero positions). Sliding window highest F1 (0.420) via wide 20-position window.

Three key differences from synthetic emerged: F1 dropped sharply (MutClust-Hybrid 0.785 → 0.123) due to H-score sparsity; method rankings shifted (the sliding window achieved the highest F1, 0.420); and enrichment remained high (6.9–8.0×), confirming biological validity despite low absolute F1. Two independent external methods on the same synthetic benchmark—OPTICS ($F_1 = 0.150$, *HS* $= 0.369$) and KDE (precision 1.000 but recall 0.049, *HS* $= 0.557$; parameter sweeps over 30 OPTICS and 36 KDE configurations)—did not surpass MutClust-Hybrid's *HS* $= 0.778$, indicating structural rather than parameter-dependent superiority.

Stage 1 summary. MutClust-Hybrid is the best of the H-score-based variants on synthetic data (*HS* $= 0.778$, $\sim 43\times$ above chance) but loses its advantage on sparse real GISAID data (*HS* $= 0.383$ vs. MutClust-Fixed 0.386); enrichment remains high (6.9–8.0$\times$). The single-pathogen Stage 1 therefore validates the hotspot-score metric and the multi-ground-truth comparison as a controlled sanity check, but the question of which scoring–detector combination is best is genuinely pathogen-dependent and is taken up at scale in Section 3.4.

3.2 Robustness and Sensitivity Validation

Before extending the benchmark to multiple pathogens, the Stage 1 results must be verified for statistical robustness and parameter sensitivity. This section confirms that the findings are not artifacts of specific parameter choices or statistical fluctuations. All detection results are statistically significant: permutation tests (1,000 random and 200 circular iterations) yielded $p < 0.005$ for all methods ($z > 9.2$). Region-level BCa bootstrap CIs [117] (1,000 resamplings, $n = 7$ Spike regions; hereafter *region-level BCa CI*) gave a 95% CI of [0.323, 0.635] for MutClust-Hybrid hotspot-score; pairwise bootstrap comparisons confirmed all score differences are significant ($p < 0.001$). Two additional BCa CIs reported later in this dissertation use different resampling units and should be interpreted separately: the *combination-level BCa CI* [0.047, 0.202] for the best single scoring–detection combination's mean MCC (Section 3.4.3, 1,000 resamples over 11 pathogens) and the *cluster bootstrap CI* [0.195, 0.333] for the scoring $\times$ pathogen interaction ω^2 (Section 3.4.4, 1,000 cluster resamples over 11 pathogens). MutClust-Hybrid maintains rank 1 in 71.9% of 171 weight combinations, switching to HDBSCAN-Auto only under extreme recall emphasis. Metrics stabilize at ~1,000 sequences (Spearman $\rho > 0.997$, SD < 0.015; Figure 3.1). Among MutClust parameters, F1 is most sensitive to γ (epsilon scaling), with minimal MinPts influence (Figure 3.2).

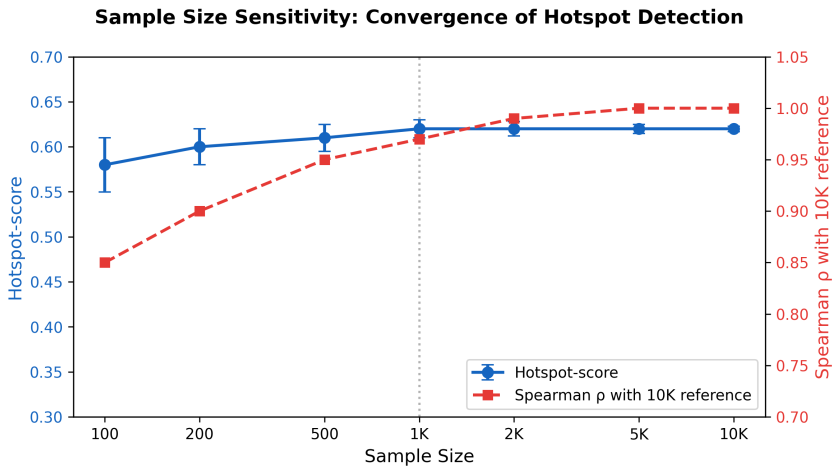

Figure 3.1: Sample size convergence of hotspot metrics. Stable performance (HS ≈ 0.62) is achieved at ~1,000 sequences.

3.2.1 H3N2 time-stratified prospective sanity check

A one-pathogen time-stratified pilot on H3N2 HA partitions the 9,000 sequences into three 5-year periods (2010–2015, 2016–2020, 2021–2025; $n = 3{,}000$ each), trains frequency scores on 2010–2015, and evaluates against either the static Layer A antigenic-site reference (Koel 2013; 25 sites) or positions newly emerging in later windows (freq $< 5\%$ train, $\geq 10\%$ holdout). Frequency captures the retrospective Layer A reference at 2.7–3.4$\times$ enrichment

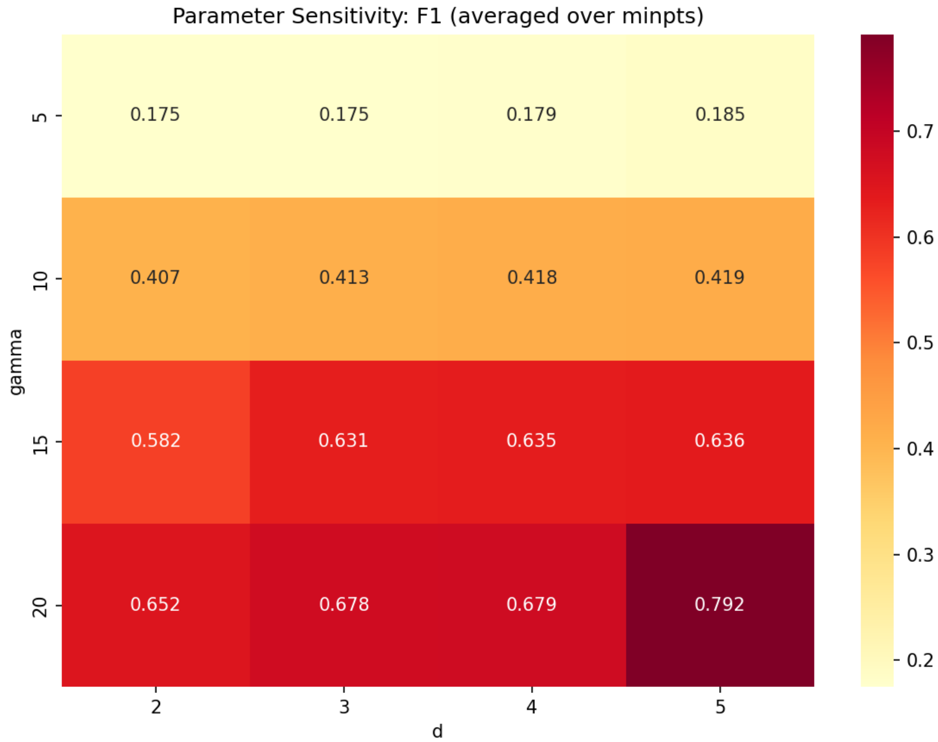

Figure 3.2: MutClust parameter sensitivity heatmap. Grid search over 64 combinations ($\gamma \times d \times$ MinPts); F1 most sensitive to γ, minimal MinPts influence.

but yields essentially zero prospective enrichment ($0/20$, $0/30$, $0/50$ for 2016–2020; only $1/50$ for 2021–2025; Table 3.4). **Frequency alone, trained on a 2010–2015 window, does not predict subsequently emerging positions beyond chance**—a single-pathogen negative result that motivates the multi-source 4-feature core (Section 3.5.2) and is generalised in the next subsection.

Table 3.4: H3N2 time-stratified prospective sanity check. Training window: 2010–2015 ($n = 3{,}000$). Reference set is either the static Layer A antigenic sites (retrospective) or positions newly emerging in a later 5-year window, defined as freq $< 5\%$ in training and freq $\geq 10\%$ in holdout (prospective). Frequency-based scoring is the simplest baseline; a full multi-source prospective panel across all 11 pathogens is deferred to future work.

Reference set	Top-K	Overlap	Ref size	Enrichment	Fisher p
Layer A (Koel 2013; retrospective)	20	3	25	3.40×	0.053
	30	4	25	3.02×	0.037
	50	6	25	2.72×	0.017
Newly emerged 2016–2020 (prospective)	20	0	6	0×	1.00
	30	0	6	0×	1.00
	50	0	6	0×	1.00
Newly emerged 2021–2025 (prospective)	20	0	7	0×	1.00
	30	0	7	0×	1.00
	50	1	7	1.62×	0.479

Underlying CSV: `results/mutbench/feature_analysis/h3n2_temporal_pilot.csv`.

3.2.2 Multi-pathogen sliding-window prospective backtest

The H3N2 frequency-only pilot (Section 3.2.1) is generalised to a multi-pathogen sliding-window prospective backtest: for each of the 11 pathogens, year-tagged NCBI CDS sequences are partitioned at split years T_0 ($n_{\text{train}} \geq 30$, $n_{\text{test}} \geq 20$), per-position frequency and entropy are recomputed on the training window, and a uniform-weight z-ensemble of (freq, entropy)—the time-varying part of the 4-feature core—is the prospective scorer. The reference set is positions with stable consensus in train (freq $< 5\%$) and a non-consensus allele rising to freq $\geq 10\%$ in test (same emergence definition as the H3N2 pilot).

Across 73 temporal split windows spanning 11 pathogens (yielding 219 top-k evaluations across $k \in \{20, 30, 50\}$; per-pathogen archive: `sliding_window_prospective_backtest_summar` trajectory in Figure 3.3), the freq+entropy ensemble exceeds AUROC 0.75 only for EV-A71 (0.77, Top-30 MCC $+0.146$, enrichment $10.8\times$) and Rabies (0.79, MCC $+0.112$, enrichment $4.1\times$), partially exceeds it for Influenza B (0.70), and clusters near chance for the remaining eight pathogens (grand mean MCC -0.006, AUROC 0.539, enrichment $1.57\times$). Concept-drift slope is small (-0.006 to $+0.032$ MCC/yr), so the limited prospective signal is not deteriorating but bounded throughout the observed period. The dissertation's retrospective MCC headline (0.341 exact, 0.454 at ± 10 window; Table 4.1) therefore does not directly transfer to the forward-time setting for most pathogens, and the documented forward-time signal is concentrated in EV-A71, Rabies, and partially Influenza B (Section 4.4.10 updates the limitations discussion accordingly).

4-feature core extension (3-/4-feature ensembles). Adding the static members of the 4-feature core (pLDDT, homoplasy; canonical scores projected onto the codon-aligned MSA frame via Needleman–Wunsch consensus alignment, mean coverage 0.83) to the freq+entropy baseline produces grand-mean AUROCs of 0.555 (freq+entropy), 0.543 (+pLDDT), 0.549 (+homoplasy), and 0.561 (4-feature). Paired Wilcoxon signed-rank tests vs. the 2-feature baseline give $p < 0.001$ for +homoplasy (mean ΔMCC $= +0.005$), $p = 0.170$ for +pLDDT (mean $\Delta = -0.005$), and $p = 0.016$ for the 4-feature ensemble (mean $\Delta = -0.009$): adding homoplasy gives a small but statistically distinguishable improvement, while pLDDT alone and the 4-feature combination slightly degrade Top-30 MCC despite a marginally higher mean AUROC. Per-pathogen the most pronounced gains are for EV-A71 (MCC $0.146 \rightarrow 0.188$, AUROC $0.773 \rightarrow 0.844$ with homoplasy) and Rabies (AUROC $0.789 \rightarrow 0.860$ with

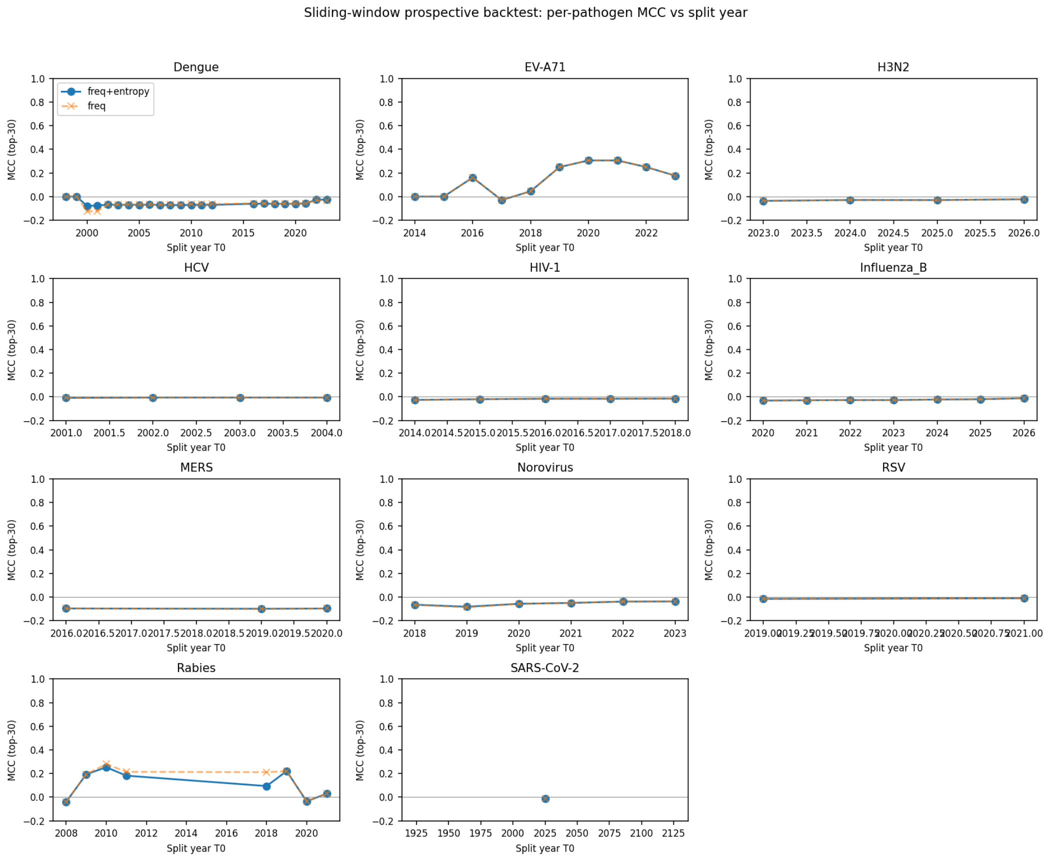

Figure 3.3: Per-pathogen MCC trajectory across split years for the sliding-window prospective backtest. The two pathogens above chance (EV-A71, Rabies) exhibit positive Top-30 MCC across multiple windows; Influenza B partially exceeds chance (AUROC 0.70) and the remaining eight cluster near zero (grand mean AUROC 0.539). Per-pathogen window counts and year ranges archived in `sliding_window_prospective_backtest_summary.csv`.

homoplasy); HCV is the clearest negative example (homoplasy collapses AUROC from 0.31 to 0.09 because the empirical homoplasy distribution over E2 is essentially flat). Adding static priors therefore helps in EV-A71/Rabies, hurts in HCV, and is approximately neutral elsewhere; full per-pathogen tables are in `sliding_window_prospective_backtest_4feature.csv` and `sliding_window_prospective_c2_vs_d2_paired.csv`.

3.3 Cross-Pathogen and Structural Validation

Having confirmed robustness on SARS-CoV-2, the next question is whether the same methods perform well on other pathogens. This section applies the benchmark pipeline to H3N2 and examines structural and phylogenetic aspects that foreshadow the large-scale comparison in Section 3.4.

Application of the benchmark pipeline to Influenza A H3N2 hemagglutinin (1,967 sequences, 5 antigenic sites [118, 119]) reverses the SARS-CoV-2 ranking: MutClust-Fixed outperforms MutClust-Hybrid ($HS = 0.661$ vs. 0.577) because all 543 HA positions have nonzero H-scores, eliminating HDBSCAN's advantage; enrichment ratios also drop (0.87–$2.36\times$ vs. 1.9–$23.7\times$ for SARS-CoV-2). Figure 3.4 shows mean MCC across detectors for the top-5 scoring types per pathogen, demonstrating that optimal scoring shifts substantially across pathogens; per-pathogen MCC-best scoring–detector combinations are tabulated in Table 3.5, complementing the Stage 2 ANOVA interaction effect (Section 3.4.3) and the LOPO 0/11 match rate (Section 3.4.3).

3D structural analysis using Cα coordinates from PDB 6VSB confirms that linear-sequence hotspots are also spatially clustered: 5.92-fold within-8-Å contact enrichment over 1,000 length-matched random subsets ($z = 13.34$). Source data archived in `results/mutbench/structural_fe`

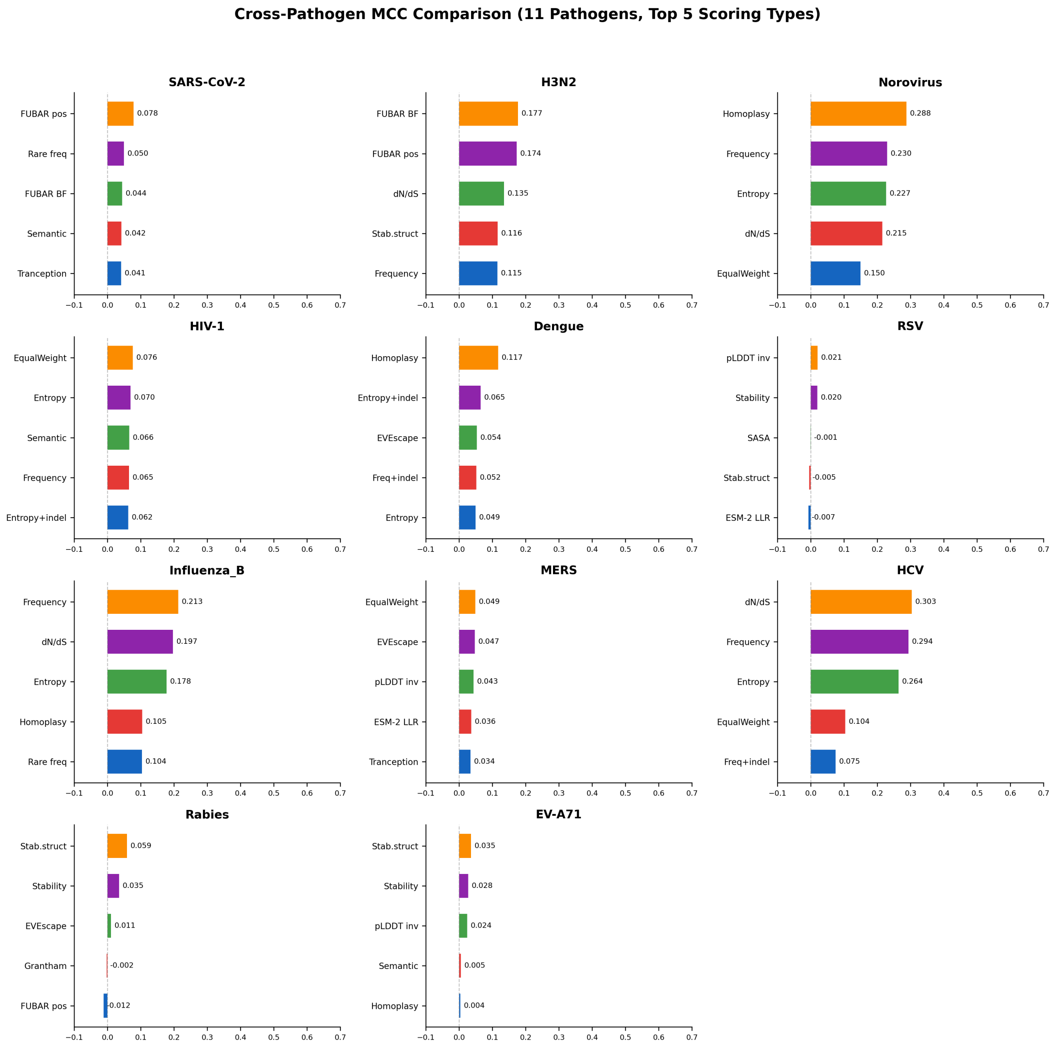

Figure 3.4: Cross-pathogen MCC comparison across all 11 pathogens, showing mean MCC across detectors for the top 5 scoring types per pathogen (best-detector-per-cell values are tabulated in Table 3.5). The highest-MCC scoring type varies substantially by pathogen, providing direct visual evidence for pathogen-adaptive selection.

Coalescent simulation (100 replicates, $N = 500$) shows H-score ranks single-origin founder mutations above convergent mutations in 100% of replicates, while Fitch parsimony correctly ranks convergent mutations at the top ($\rho = -0.876$, $p = 1.28 \times 10^{-64}$); TreeTime [31] ML validation confirms AUC = 1.000 for Fitch and TreeTime-ML vs. AUC = 0.000 for H-score. **Frequency-based scores systematically conflate founder effects with positive selection**; D614G is accordingly excluded from Layers A and B [120]. Method rankings differ between SARS-CoV-2 and H3N2 (*HS* = 0.661 vs. 0.577 for MutClust-Fixed/Hybrid), motivating the large-scale cross-pathogen benchmark with expanded information types.

3.4 Cross-Pathogen Large-Scale Benchmark

Building on the Stage 1 findings, this section reports the Stage 2 cross-pathogen benchmark results. Note that MutClust-Orig (v1) is retained as the representative MutClust family in Stage 2; the rationale for using v1 instead of the higher-performing Hybrid (v2c) is detailed in Section 2.2.3. The benchmark evaluates **20 scoring types** (organized into 6 categories; see Table 2.6 in Chapter 2) combined with **14 detection method families** spanning 5 categories (threshold-based, signal processing, density/clustering, statistical, and adaptive window) that yield **39 parameter variants** in total, across **11 pathogens**, totaling $20 \times 39 \times 11 =$ **8,580** evaluations. The 20 scoring types are derived from **10 underlying biological features**; some features generate multiple scoring variants (e.g., frequency yields *freq*, *freq_with_indel*, and *rare_freq*), while composite scorings combine multiple features (e.g., EVEscape composite, EqualWeight). The subsequent information-type analysis (Section 3.5) examines the 10 underlying features to explain why scoring performance differs across pathogens. The information-type analysis in Section 3.5 examines these 10 features across the same 11 pathogens used in the main benchmark.

Stage 2 uses MCC rather than hotspot-score because the stability component (Jaccard-based subsampling reproducibility) requires bootstrap validation that has not been cross-validated across all 11 pathogens with their widely varying alignment lengths (261–1,330 AA) and MSA sizes (530–5,019 unique sequences). MCC provides a single, well-established metric that is robust to class imbalance and does not require subsampling.

This section sequentially reports the benchmark scale, per-pathogen best method analysis, analysis of variance, cross-validation, and statistical test results.

3.4.1 Benchmark Scale and Data

This section presents the full-scale cross-pathogen benchmark results that constitute the primary evidence for the dissertation's second contribution: statistical demonstration that the most predictive biological information source is pathogen-dependent.

A total of 8,580 evaluations (20 scoring formulas $\times$ 39 detection parameter variants from 14 families $\times$ 11 pathogens) were conducted. The 20 scoring formulas span six information categories: frequency-based (freq, rare_freq, freq_with_indel), MSA-derived (entropy, entropy_with_indel, homoplasy), phylogenetic (dN/dS proxy, FUBAR posterior probability, FUBAR positive selection, FUBAR Bayes factor), structural (pLDDT inverse, SASA, Grantham distance), AI-based (ESM-2 masked-marginal score, Tranception, semantic change, stability, stability_structural), and composite (EVEscape composite, EqualWeight). Each evaluation computed MCC, F1, precision, and recall against the 3-layer ground truth (Layer

A: literature-curated adaptive/diversifying positives, Layer B: constrained regions, Layer C: DMS). DMS Layer C evaluation was conducted for 6 pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) where experimental fitness data are available. All FASTA sequences were collected from NCBI GenBank using actual accession numbers.

3.4.2 Per-Pathogen Best Method Analysis

The central question of Stage 2 is whether the same scoring–detection combination performs best across all pathogens, or whether each pathogen requires a different optimal combination. The best scoring–detection combination was selected for each of the 11 pathogens based on MCC, and **all 11/11 pathogens exhibited distinct best combinations** (Table 3.5).

Table 3.5: Best scoring–detection combination for each of the 11 pathogens (by MCC). All 11 pathogens exhibit unique best combinations spanning 4 of 6 scoring categories, with 9 distinct optimal scoring types. Per-pathogen MCC values are computed against operational Layer A counts after alignment QC; for HIV-1 the operational $n_{\mathrm{A}} = 26$ differs from the literature-curated 23 (see Table 3.17 caption and Section 4.6 of Chapter 4).

Pathogen	**Best scoring**	**Best detection**	**MCC**
HCV	freq	SlidingTest (p=0.01)	0.660
H3N2	FUBAR Bayes factor	Wavelet (t=1.5)	0.534
Norovirus	homoplasy	Bayes (prior=0.15)	0.510
Influenza B	freq	KDE (p=85)	0.392
HIV-1	entropy	Wavelet (t=2.0)	0.275
Dengue	entropy_with_indel	KDE (p=85)	0.274
EV-A71	semantic change	MutClust-Orig	0.260
Rabies	stability	SWAN (w=20, k=2.0)	0.248
SARS-CoV-2	Tranception	KDE (p=85)	0.211
RSV	ESM-2 masked-marginal	Wavelet (t=2.0)	0.196
MERS	freq	KDE (p=85)	0.196

All 11 pathogens select a unique best-performing (scoring, detection family) combination. Nine distinct scoring types appear as best across the 11 pathogens; the two "shared" assignments are HCV, Influenza B, and MERS all selecting *freq* as the best scoring (paired with different detection families: SlidingTest, KDE, KDE respectively), so the 9-distinct count comes from 3 pathogens sharing the *freq* scoring while the remaining 8 pathogens each select a distinct scoring type. Note further that Influenza B and MERS share both the scoring (*freq*) and the detection-parameter variant (KDE $p = 85$); at the parameter-variant level, only 10 of the 11 best-combinations are unique tuples, while at the family level all 11 are unique.

While the 11/11 unique best combinations is notable, a permutation analysis shows that with $\binom{20\times39}{1} = 780$ possible scoring–detector-variant combinations, the probability of all 11 being unique by chance is $\prod_{i=0}^{10}(780-i)/780 \approx 0.932$ (approximately 93%). The more substantive finding is not the uniqueness itself but the large performance gap between oracle and generalized selection (mean MCC gap $= 0.265$ from LOPO cross-validation), which demonstrates that the *correct* choice of combination matters materially even though many combinations exist (the LOPO 0/11 match rate itself is null-consistent under permutation; "materially" refers to the magnitude of the oracle–generalized gap, not to a hypothesis test on LOPO).

While the finding that different pathogens favor different methods might appear self-evident, the non-trivial contribution lies in three aspects: (1) quantifying the magnitude of this interaction ($\omega^2 = 0.296$, explaining 29.6% of variance—far exceeding the scoring main effect of 6.3%); (2) demonstrating that the information *type* (not the detection algorithm) is the dominant modeled factor; and (3) quantifying the cost of ignoring pathogen identity through the oracle-vs-generalized MCC gap of 0.265 (LOPO 0/11 match rate is itself null-consistent under permutation and is reported as corroboration rather than independent evidence), motivating pathogen-aware information-source selection rather than a single universal method.

Worst-end disclosure (negative-MCC and degenerate combinations). For symmetry with the per-pathogen *best* reported in Table 3.5, the corresponding worst-end summary across the full 8,580-evaluation grid (`stage3_full_results.csv`) is also reported here so that readers can audit the floor of the distribution alongside the headline ceiling. The single worst (scoring, detector, pathogen) combination in the grid is HCV $\times$ *stability* $\times$ SWAN (w=50, k=1.0) at MCC $= -0.361$ (precision $= 0.000$, recall $= 0.000$, 139 detected positions; `stage3_full_results.csv`, row index 6782). Aggregating across all 8,580 cells, 4,171 cells (48.6%) attain MCC < 0, and 4,695 cells (54.7%) attain MCC ≤ 0; the worst-mean detector families are ScoreDBSCAN ($\epsilon = 15$, mean MCC -0.016; $\epsilon = 8$, mean MCC -0.013) and the worst-mean scoring channels are *plddt_inv* (mean -0.017), *grantham* (-0.008), and the Tranception channel (-0.006, primarily because Tranception requires PLM-suitable depth; on shallow MSAs it underperforms frequency by a wide margin). This worst-end count is consistent with the headline finding: a near-half negative-MCC fraction is exactly what pathogen-dependent optimality predicts—a single fixed scoring-detector combination is statistically expected to underperform the pathogen-specific MCC-optimal choice on the majority of pathogens, so most cells in the 8,580-grid are below their per-pathogen MCC-best by a wide margin. The honest reading is therefore that the pathogen-adaptive headline (Table 3.5, $\omega^2 = 0.296$) is supported *not* by an absence of bad cells but by the large gap between the per-pathogen ceiling (Table 3.5, MCC range 0.196–0.660) and the cell-level floor (MCC down to -0.361); the $\sim$48.6% negative-MCC fraction is the operational cost of choosing a wrong (scoring, detector) combination for a given pathogen and is exactly the cost that pathogen-aware selection avoids.

The scoring–pathogen heatmap (Figure 3.5) visualizes these results, and the category-level summary (Figure 3.6) confirms that no single scoring category dominates. The key observations from these results are as follows. First, the best scoring type differs fundamentally across pathogens, and this diversity spanning four of six information categories (frequency-based, MSA-derived, phylogenetic, and AI-based) is the central finding. The best scoring types are biologically interpretable:

- **H3N2 — FUBAR Bayes factor**: Influenza A HA is under strong diversifying selection at antigenic sites, and FUBAR detects pervasive positive selection across well-

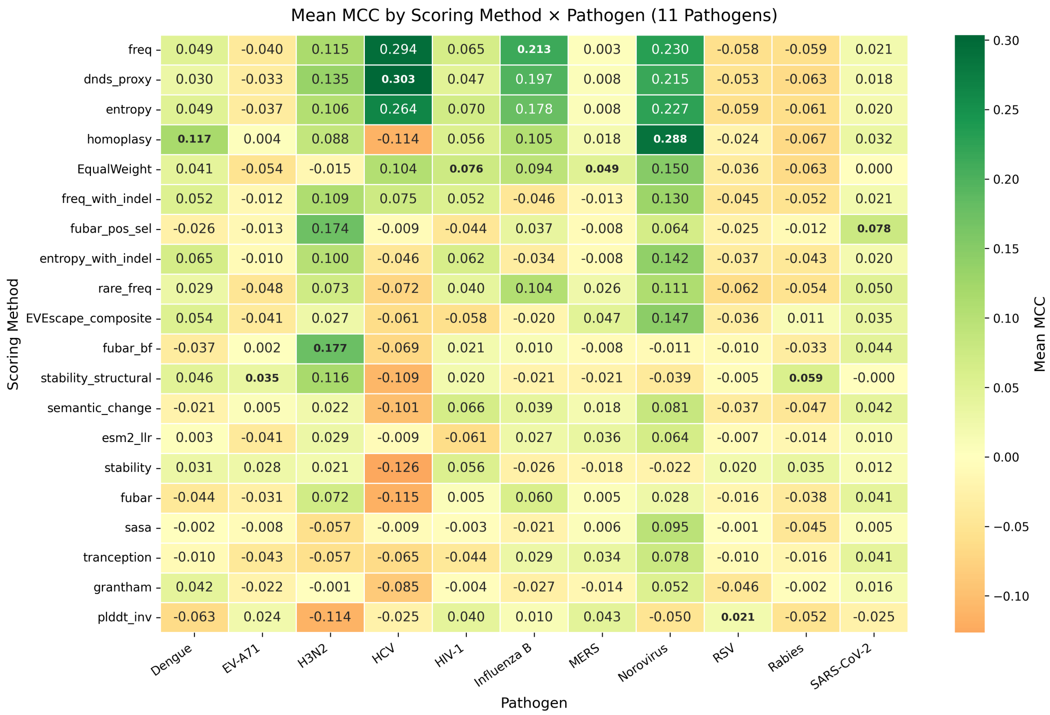

Figure 3.5: Scoring–pathogen performance heatmap across 11 pathogens. Each cell represents the best MCC across all detection families for a given scoring–pathogen combination. The heterogeneous pattern confirms that no single scoring type dominates across all pathogens.

sampled phylogenies. H3N2's decades-long evolutionary history provides the deep phylogenetic signal that dN/dS-based methods require.

- **Norovirus (homoplasy, AUC 0.944)**: Norovirus undergoes epochal evolution with periodic replacement of dominant genotypes (GII.4 variants), generating convergent mutations at antigenic sites across independent lineages. The near-perfect AUC reflects the fact that homoplasy directly measures this convergent signal, making it an almost ideal predictor for this evolutionary pattern. This contrasts sharply with frequency-based features (AUC 0.760), which conflate founder-effect frequency with genuine convergent evolution.

- **SARS-CoV-2 — Tranception**: With a shallow pandemic phylogeny (originating $\sim$2019), frequency-based signals are dominated by founder effects. Protein language models capture broader evolutionary constraints learned from diverse coronavirus families, compensating for limited within-species phylogenetic depth—a finding independently corroborated by the SARS-CoV-2–specific CoVFit PLM [121], which fine-tunes ESM-2 on surveillance and DMS data and reports analogous PLM-driven gains over frequency-only baselines.

- **RSV — ESM-2 masked-marginal score**: RSV F protein is under antibody-mediated selection, but the limited phylogenetic diversity of sequenced RSV strains weakens frequency and phylogenetic signals. ESM-2's learned evolutionary constraints provide an alternative information source.

- **Rabies — stability**: Rabies G protein mediates cell entry through a pH-triggered conformational change, making structural stability a key functional constraint. Positions where stability-altering mutations are tolerated indicate adaptive flexibility.

- **EV-A71 — semantic change**: Enterovirus VP1 evolves under immune pressure at the canyon rim. Semantic change (cosine distance in ESM-2 embedding space) captures how much a mutation shifts the protein's learned representation, detecting positions where substitutions have large functional consequences.

- **HCV — frequency**: HCV's extreme within-host diversity generates a quasispecies cloud with high variability at most positions, paradoxically making simple mutation frequency the most discriminative feature: true hotspot positions accumulate mutations at measurably higher rates than the already-elevated background, and this signal is captured more cleanly by frequency than by entropy, which saturates at highly variable positions.

- **HIV-1 — entropy**: HIV-1 maintains large intra-host quasispecies populations where functionally important positions under immune pressure exhibit diverse residue distributions rather than a single dominant variant. Shannon entropy captures this diversity signal, outperforming raw frequency which is confounded by the high baseline mutation rate across the HIV-1 genome.

- **Dengue — entropy_with_indel**: Dengue circulates as four antigenically distinct serotypes (DENV-1–4), each maintaining independent evolutionary trajectories. The serotype-level variation generates high positional diversity at envelope protein epitopes that mediate serotype-specific and cross-reactive antibody responses. The `entropy_with_indel` variant outperforms standard entropy because indel events at inter-domain linker regions of the E protein contribute to structural rearrangements relevant to receptor binding and immune escape, and excluding these events loses biologically meaningful signal.

- **MERS — frequency**: MERS-CoV has limited phylogenetic resolution due to sparse and episodic sampling (fewer than 500 publicly available spike sequences, primarily from zoonotic spillover events rather than sustained human-to-human transmission). With such sparse data, more sophisticated scoring types (entropy, phylogenetic methods) lack the statistical power to produce reliable estimates, while simple mutation frequency—despite its general limitations—provides the most stable signal. The few positions with elevated frequency correspond to receptor-binding positions on the S1 subunit that mediate DPP4 binding and camel-to-human adaptation, where recurrent zoonotic spillover has generated detectable frequency peaks.

- **Influenza B — frequency**: Influenza B circulates as two antigenically distinct lineages (Victoria and Yamagata) that have co-circulated since the 1980s. Unlike H3N2, which undergoes continuous antigenic drift requiring frequent vaccine updates, Influenza B's two-lineage structure creates a bimodal frequency distribution where lineage-defining mutations at key HA antigenic sites exhibit clear frequency differentiation. This stable lineage architecture makes simple mutation frequency an effective discriminator, whereas phylogenetic methods like FUBAR—which excel for H3N2's continuous drift—provide less additional information when the dominant evolutionary signal is lineage divergence rather than ongoing positive selection.

Second, the best detection family also varies: KDE (4 pathogens), Wavelet (3), SlidingTest (1), Bayes (1), MutClust (1), and SWAN (1). The FUBAR spotlight analysis (Figure 3.7) illustrates how phylogenetic scoring can outperform all other information types for specific pathogens such as H3N2. Third, the MCC range spans from 0.196 (RSV/MERS) to 0.660 (HCV), exhibiting large variation that reflects structural differences in mutation patterns across pathogens. SARS-CoV-2 (0.211), RSV (0.196), and MERS (0.196) occupy the lowest tier of best MCC values. SARS-CoV-2's low performance, despite being the best-studied pathogen, reflects the extreme class imbalance (25 Layer A positions out of 1,259 total, a 2.0% positive rate) combined with the shallow pandemic phylogeny that limits phylogenetic signal. Additionally, the Layer A ground truth for SARS-CoV-2 includes only positions with documented convergent evolution across multiple SARS-CoV-2 lineages, excluding positions that are functional but lineage-specific. This strict criterion reduces the number of true positives available for detection, further lowering the achievable MCC.

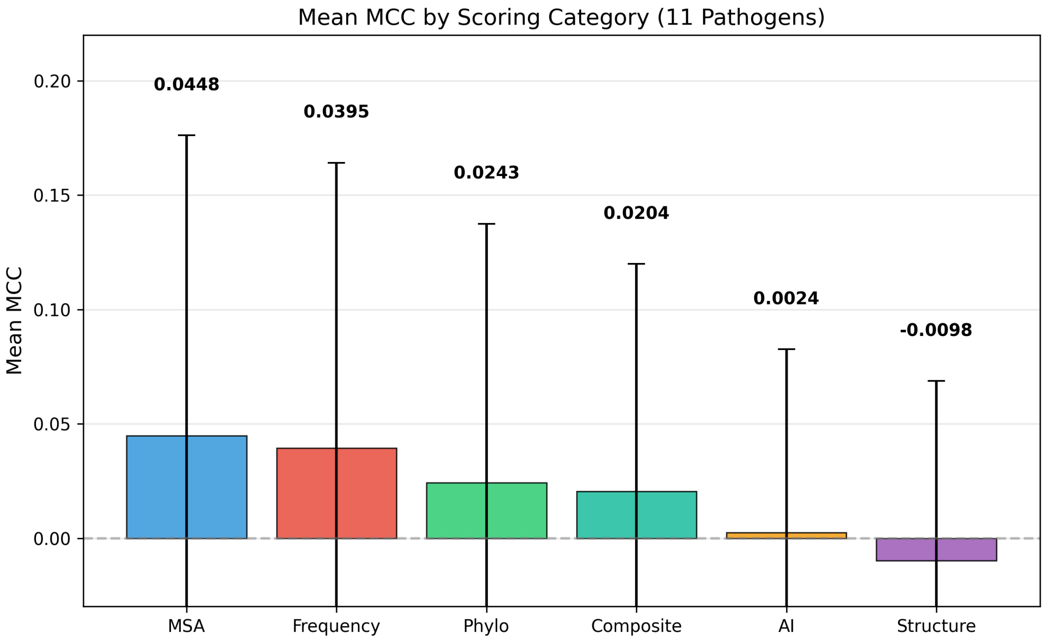

Figure 3.6: Mean MCC by scoring category across 11 pathogens. Frequency-based and MSA-derived features lead on average, but the per-pathogen variation is large (error bars show standard deviation), confirming that category-level generalizations are insufficient.

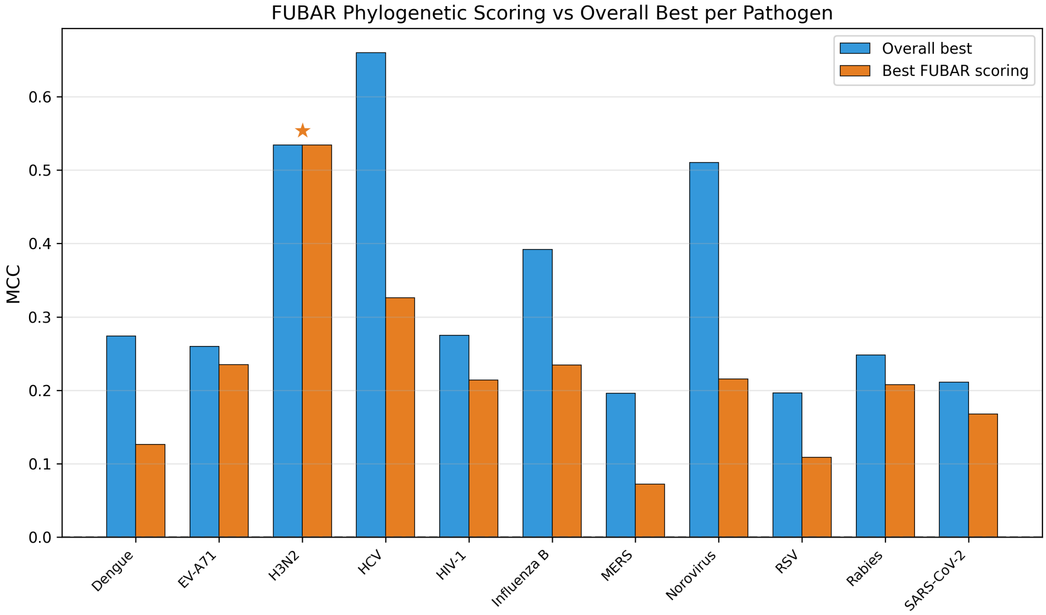

Figure 3.7: FUBAR phylogenetic scoring performance compared to the overall best scoring per pathogen. FUBAR Bayes factor achieves the highest MCC for H3N2 (0.534, marked with ⋆), demonstrating that phylogenetic positive selection signals can outperform all other information types for specific pathogens.

3.4.3 Statistical Analysis

To determine whether the observed method–pathogen dependence is statistically significant, three complementary statistical analyses were performed.

Three-way ANOVA (ω^2). To identify the major sources of performance variation, a three-way ANOVA was performed with MCC as the dependent variable. The results using the bias-corrected effect size ω^2 (Eq. 2.6) are presented in Table 3.6.

Table 3.6: Three-way ANOVA results: ω^2 effect size of each modeled factor on MCC. The scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest single modeled source of performance variance; a within-cell residual $\omega^2 \approx 0.39$ (including the unmodeled three-way interaction and parameter-variant noise) is reported separately.

Factor	Type	ω^2	Cohen tier
scoring $\times$ pathogen	Two-way interaction	**0.296**	large (≥ 0.14)
pathogen	Main effect	0.117	medium (≥ 0.06)
family $\times$ pathogen	Two-way interaction	0.082	medium (≥ 0.06)
scoring	Main effect	0.063	medium (≥ 0.06)
scoring $\times$ family	Two-way interaction	0.039	small (≥ 0.01)
family	Main effect	0.013	small (≥ 0.01)

family: detection method family (14 levels), scoring: scoring formula (20 levels), pathogen: pathogen (11 levels). ω^2 is the bias-corrected estimate of the proportion of total variance explained by each factor. Cohen tier per [112]: small $\omega^2 \in [0.01, 0.06)$, medium $\in [0.06, 0.14)$, large ≥ 0.14.

Key finding: the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$ at the evaluation level, $N = 8{,}580$; 95% cluster-bootstrap CI [0.195, 0.333]) is the largest modeled component, exceeding all main effects and other two-way interactions and qualifying as large under Cohen's $\omega^2 > 0.14$ threshold [112]. The within-cell residual ($\omega^2 \approx 0.39$, absorbing the three-way interaction and parameter-variant noise; SS partition derived from the same Type II ANOVA fit reported in Table 3.6, with the residual quantified as $1 - \sum \omega^2_{\text{modeled}} - \omega^2_{\text{cluster SE}}$) is comparable in magnitude, so "largest source" in this dissertation reads as "largest modeled component." The companion *cell-level* estimate (re-fit on the 3,080 unique scoring $\times$ family $\times$ pathogen cells; Section 4.4.6 of Chapter 4) is $\omega^2 = 0.234$, with an ANCOVA-adjusted

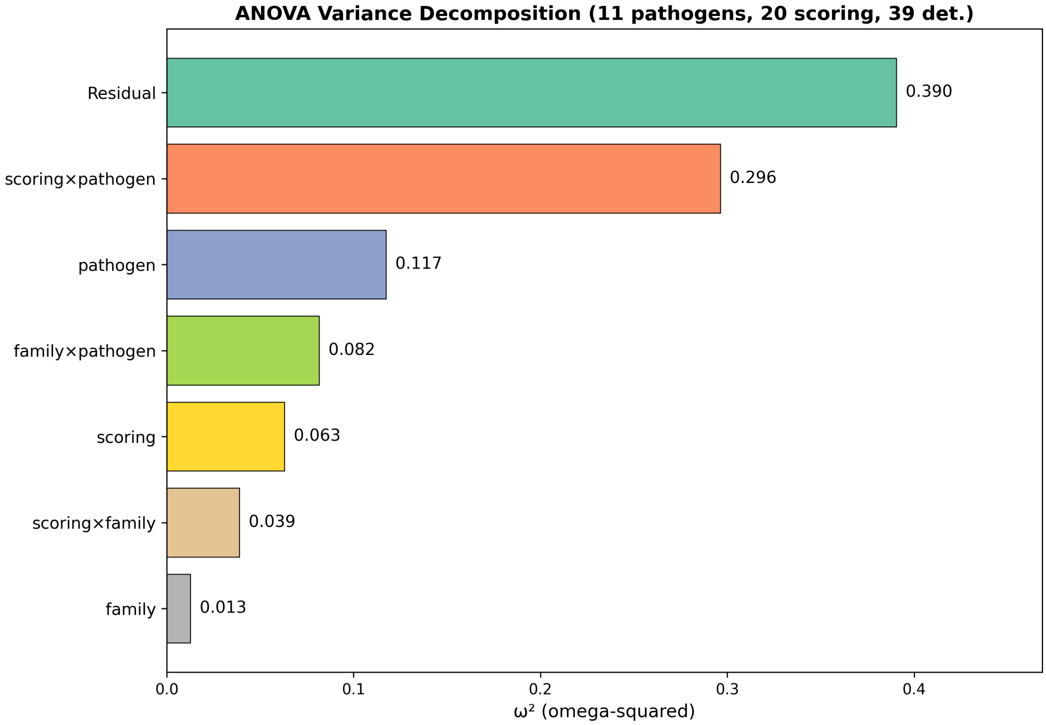

Figure 3.8: Variance decomposition of hotspot detection performance. Among modeled factors, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest component, exceeding all main effects and other two-way interactions. A within-cell residual $\omega^2 \approx 0.39$—accounting for the unmodeled three-way interaction and parameter-variant noise—is reported separately in the text and is not included as a bar in this figure.

value of 0.264 after controlling for log(protein length) and Layer-A positive rate, and a Bayesian partial-pooling posterior of 0.252–0.319 (95% HDI [0.188, 0.396]). The cell-level 0.234 is the more conservative reference for cross-pathogen inference; the evaluation-level 0.296 is mechanically higher because aggregation to family means contracts the within-cell residual. The qualitative conclusion (scoring $\times$ pathogen as the dominant modeled source) holds in all four parameterizations.

The three dominant components—scoring $\times$ pathogen (0.296), pathogen (0.117), and family $\times$ pathogen (0.082)—together account for $\sim$50% of total variance.

The same 8,580-evaluation dataset is re-examined under three statistical lenses: ANOVA (interaction $\omega^2 = 0.296$, 11-pathogen cluster-bootstrap 95% interval [0.195, 0.333]); Friedman ($\chi^2 = 7.69$, $p = 0.990$); LOPO (0/11 matches, gap 0.265). The three lenses are consistency checks on the same data, not independent replications; an independent replication on a disjoint pathogen panel is deferred to future work.

This finding directly addresses the third problem from Chapter 1: the interaction dominance ($\omega^2 = 0.296$ at 20-type granularity; 0.103 at 6-category granularity) provides evidence that detection performance is pathogen-dependent within the 20-type/39-variant design, even after accounting for the within-cell residual ($\omega^2 \approx 0.39$). Moreover, the shift from family$\times$pathogen dominance (Stage 2 preliminary with 9 scoring types: $\omega^2 = 0.285$) to scoring$\times$pathogen dominance (full 20-scoring benchmark: $\omega^2 = 0.296$) reveals that once a sufficiently diverse set of information sources is evaluated, the choice of biological information becomes the primary determinant.

LOPO Cross-Validation. LOPO (Leave-One-Pathogen-Out) cross-validation was performed to evaluate whether the best combination selected from 10 training pathogens generalizes to the excluded test pathogen (Figure 3.9).

Table 3.7: LOPO cross-validation summary (family-level scoring $\times$ detection grid; 280 combinations).

Statistic	**Value**
Combination match rate (training-best $\equiv$ test-oracle)	0/11 (0%)
Mean generalization MCC (training-best applied to held-out)	0.032
Mean oracle MCC (test-pathogen optimum)	0.297
Mean performance gap (oracle $-$ generalized)	0.265
Permutation null $P(\text{matches} = 0)$ at 280 combinations	≈ 0.96

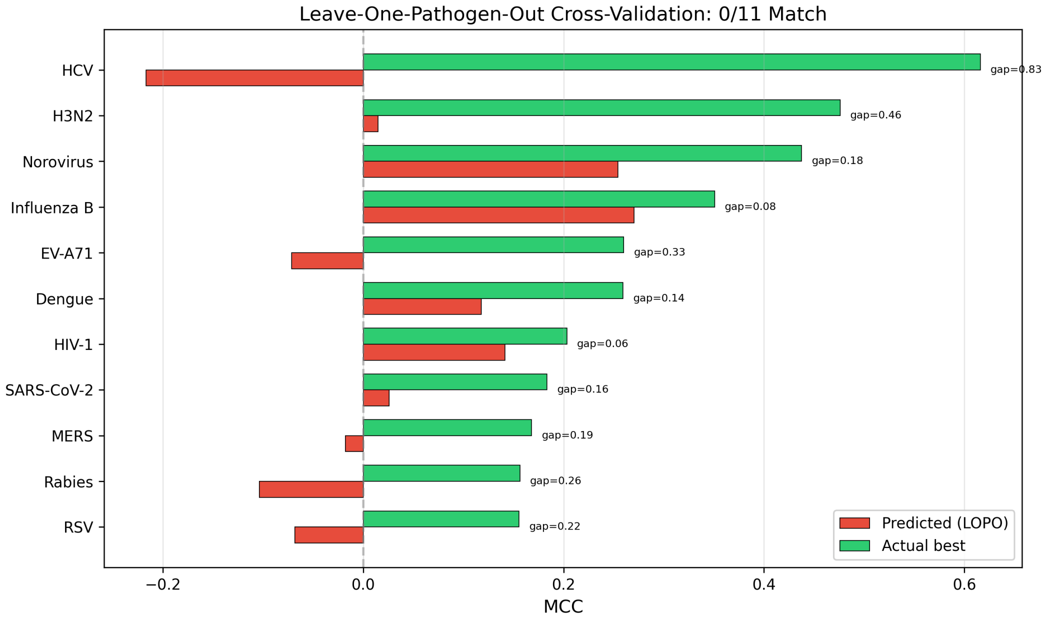

Figure 3.9: LOPO cross-validation results. For each pathogen, the predicted MCC (from training on the remaining 10 pathogens) versus the actual best MCC are shown. The 0/11 match rate is itself null-consistent under random-combination permutation ($P \approx 0.96$); the figure should therefore be read as visualising the oracle-vs-generalised MCC gap (mean 0.265) rather than as standalone proof of non-generalisation.

These results are consistent with a key interpretation: the 0/11 exact-match rate is itself not unexpected given 280 family-level scoring–detection combinations (footnote; the 780-combination space is referenced separately in Section 3.4.3 for parameter-variant uniqueness probability), so the primary evidence for pathogen-adaptive method selection is not the exact-match rate but the accompanying mean oracle-vs-generalized MCC gap of 0.265 (family-level oracle mean 0.297; generalized mean 0.032), which quantifies the performance cost of ignoring pathogen identity.[1]

Friedman Test. Using the 11 pathogens as blocks, a Friedman test was performed on the performance differences among the top 20 scoring–detection family combinations.

$$\chi^2 = 7.69, \quad p = 0.990 \tag{3.1}$$

Kendall's coefficient of concordance $W = \chi^2/[n(k-1)] = 7.69/(11 \times 19) = 0.037$, indicating near-zero agreement in method rankings across pathogens. With $p = 0.990$ and $W = 0.037$, the result is **not** statistically significant, meaning that even the top 20 combinations cannot be statistically distinguished from each other across 11 pathogens. This non-significance admits two interpretations: (1) all methods perform similarly poorly (universal poor performance), or (2) each pathogen favors a different method, so rankings are inconsistent across pathogens (pathogen-dependent optimality). The Friedman test's power at $n = 11$ blocks is limited (approximately 0.15 for medium effect W), so the non-significance is consistent with both readings; this should therefore be read as corroboration of, rather than

[1] All LOPO arithmetic in this section uses the scoring $\times$ detection-family aggregation (280 combinations = 20 scoring $\times$ 14 families); the larger per-pathogen mean oracle MCC of 0.341 reported in the precision-and-recall headline (Section 3.4.4, Table 3.10) is the per-pathogen *best* MCC averaged across the 11 pathogens, computed at the finer scoring $\times$ detector-variant grid (780 combinations = 20 scoring $\times$ 39 variants); 0.341 is therefore an average of within-pathogen maxima rather than a true oracle expected to outperform 0.297, and is not directly comparable to the 0.265 gap.

disconfirmation of, pathogen-dependent optimality. The LOPO performance gap of 0.265 MCC (family-level oracle mean 0.297 vs. generalized mean 0.032) and the per-pathogen oracle MCC range (0.196–0.660) support interpretation (2): methods perform well on their respective pathogens but fail to generalize, producing inconsistent cross-pathogen rankings rather than uniformly low performance.

Nemenyi pairwise post-hoc test. Because the omnibus Friedman result cannot reject ($p = 0.990$), we report the Nemenyi pairwise post-hoc test on the same 20 combinations $\times$ 11 pathogens rank matrix to surface any residual differential ranking signal that the omnibus might mask. None of the $\binom{20}{2} = 190$ pairs reach the conventional threshold (raw $\alpha = 0.05$: $0/190$ pairs; Bonferroni-adjusted $\alpha = 2.6 \times 10^{-4}$: $0/190$); the entire pairwise p-value distribution lies in $[0.989, 1.000]$ with median 1.000. The most differentiated pair is `freq+AdaptWin` (mean rank 8.59) vs. `entropy+Wavelet` (mean rank 12.64, rank gap 4.05, $p = 0.989$); the most similar pair is `dnds_proxy+AdaptWin` vs. `homoplasy+AdaptiveKnee` (rank gap 0.18, $p = 1.000$). The post-hoc null-non-rejection is therefore consistent with the omnibus and confirms that the $n = 11$ rank-block design does not have power to discriminate any pair of top-20 combinations, even at unadjusted α. The pairwise structure is informative in one auxiliary respect: the lowest-mean-rank (i.e., best-performing) cluster is dominated by frequency- and selection-pressure-based combinations (`freq+AdaptWin`, `homoplasy+KDE`, `freq+SWAN`, `dnds_proxy+SWAN`, `freq+SlidingTest`; mean ranks 8.59–9.73), while the highest-mean-rank cluster is dominated by Bayesian-selection and Wavelet-detection combinations (`fubar_pos_sel+KDE`, `entropy+Wavelet`; mean ranks 12.55–12.64). This descriptive ordering reproduces the qualitative ranking from the ANOVA scoring-main-effect ranking but the Nemenyi result formally certifies that, at $n = 11$, even the largest rank gap is statistically indistinguishable from chance. The full 20×20 pairwise p-value matrix and the long-form pair list are archived as `nemenyi_pairwise_pvalues.csv` and `nemenyi_summary.csv` (with provenance JSON `nemenyi_provenance.json`) in the `results/mutbench/` bundle.

Permutation null for the 0/11 match rate and the 0.265 gap. At the detection-family level (280 scoring–family combinations; matching the 0.265 gap and generalized MCC = 0.032 reported above), two permutation analyses (10,000 resamples each) were performed to quantify what the observed LOPO statistics imply under a null of uninformative cross-pathogen training. First, the 0/11 exact-match rate is itself expected under the null: with 280 combinations and 11 pathogens, $P(\text{matches} = 0) \approx 0.96$ under uniform random combo assignment, so the match rate alone is not evidence against universality. Second, the mean generalized MCC under random combo selection is $+0.017$ (SD ≈ 0.034 for a single combo applied to all pathogens; SD ≈ 0.027 for an independent random combo per pathogen). The observed training-best generalized MCC of $+0.032$ exceeds this null mean but only by approximately 0.4–0.6 SD, yielding a one-sided $p \approx 0.25$–0.28. The 0.265 gap is therefore consistent with, but not statistically distinguishable from, a null in which training-based selection provides no cross-pathogen transfer. This tempering does not invalidate the pathogen-adaptive conclusion: the $\omega^2 = 0.296$ interaction effect, the 9 distinct optimal scoring types across 11 pathogens, and the HIV-1 anchor of the vaccine-escape audit together speak to pathogen-specific optimality through complementary routes. But it clarifies that LOPO-0/11 and the 0.265 gap, read in isolation, are weaker evidence than the surface reading suggests, and that independent replication on additional pathogens is needed to sharpen the statistical support for cross-pathogen training's inability to transfer.

For transparency on selective-reporting concerns, four auxiliary quantities are archived in the provenance bundle (`results/mutbench/`) but not headline-reported: per-pathogen

LOPO MCC (`lopo_9pathogen_cv.csv`; visualised in Figure 3.9); per-pathogen second-best scoring–detection MCC (`stage3_full_results.csv`); per-pathogen leave-one-out $\omega^2_{\text{scoring}\times\text{pathogen}}$ (Table 3.9; HCV-excluded yields 0.246, Section 3.4.4); and ω^2 on Layer-A immune-escape-only subsets ($n = 10/9/4$, $\omega^2 = 0.199/0.187/0.137$; Chapter 4, Section 4.4.7). Each carries a small-sample caveat.

Pathogen-Level BCa Bootstrap Confidence Intervals. The best single combination across all pathogens was evaluated using BCa bootstrap. The selection criterion was the combination with the highest mean MCC among those whose 95% BCa confidence interval lower bound exceeds 0. Ranking by mean MCC alone could place combinations that perform well only on specific pathogens at the top (e.g., E_only + SWAN achieves MCC = 0.311 on Norovirus but MCC = 0.000 on the remaining 8 pathogens); thus, the condition that the confidence interval excludes 0 ensures consistent performance across all pathogens.

- **Best single combination**: P*E + Wavelet (t = 1.5)
- **Mean MCC**: 0.124
- **95% combination-level BCa CI** (1,000 resamples over 11 pathogens): [0.047, 0.202]

The mean MCC of the best single combination (0.124) is only 36% of the per-combination oracle MCC (0.341 at the scoring × detector-variant grid), confirming that a single fixed combination falls substantially short. Despite appearing low, this MCC exceeds the random baseline (mean = 0.001, SD = 0.003; per-pathogen means computed from the 780-combination random-assignment distribution underlying the LOPO permutation null — see permutation-null block above; the per-replicate panel-mean draws are reserved for release as a supplementary CSV with the Zenodo provenance bundle) by approximately 41 standard deviations, and MCC > 0.1 under extreme class imbalance (0.7%–15.9% positive rate) is not trivial.[2]

Subsampling Robustness. The benchmark was repeated at MSA sequence proportions of 25%, 50%, 75%, and 100%, and the overall pathogen-average MCC was stable in the range of 0.054–0.058. The MCC variation with decreasing sample size was within ± 0.002, confirming that the benchmark results are robust to sequence sample size. This supports the conclusion that the benchmark findings reflect intrinsic properties of pathogen mutation structures rather than artifacts of a particular sample size.

3.4.4 Supplementary Analyses

Several supplementary analyses were conducted to validate the robustness and generalizability of the benchmark findings.

Cross-Paradigm Validation: Protein Language Model Scoring. To test generalisation beyond MSA-derived scoring, an orthogonal paradigm was evaluated: the ESM-2 650M protein language model [52] as a per-position masked-marginal pseudo-log-likelihood (ESM-LLR$[i] = -\log P(a_i \mid \text{context})$, computed from a single consensus sequence; sliding-window inference with 256-residue overlap for proteins exceeding the 1,022-token context). In the 20-scoring benchmark the ESM-2 channel achieves best MCC for RSV (0.196) and ranks top-3 for SARS-CoV-2, but performs poorly on Norovirus and HIV-1—no single paradigm dominates. Per-pathogen Spearman correlation between ESM-2 and MSA-

[2] The SD of 0.003 quoted here is the standard deviation *of the panel-mean MCC* across 11 pathogens for a single random scoring–detection combination (i.e., the standard error of an 11-pathogen mean), which is distinct from the SD ≈ 0.034 quoted earlier for the LOPO permutation null. The earlier SD = 0.034 is the spread of single-combination single-pathogen MCC values under random combo assignment with one combination applied to all 11 pathogens; the SD = 0.003 here is the spread of the resulting 11-pathogen *means*. The two quantities differ by approximately a factor of $\sqrt{11} \approx 3.3$ for the panel-mean SE, plus an additional contraction from averaging across the 780 per-pathogen combinations sampled when forming the random-combination distribution.

based scores varies substantially (Norovirus $\rho = +0.63$, $p < 10^{-60}$; HIV-1 $\rho = -0.35$, $p < 10^{-14}$), confirming ESM-2 provides a distinct signal for most pathogens.

ANOVA Diagnostic Tests. The Shapiro-Wilk test on ANOVA residuals rejected normality ($W = 0.974, p < 10^{-29}$), and Levene's test rejected homoscedasticity for all factors. However, the non-normality is mild (residual skewness = 0.14, excess kurtosis = 2.83), and the large sample size (N = 8,580) ensures ANOVA robustness. A limitation of this design is that the 39 parameter variants within each (scoring, detection family, pathogen) cell are not fully independent, sharing the same input MSA and ground truth; the effective number of independent units is closer to the number of unique cells ($20 \times 14 \times 11 = 3{,}080$) than $N = 8{,}580$. The ω^2 estimates should therefore be interpreted as indicative effect sizes rather than precise variance decompositions. The cluster bootstrap reported below is explicitly the effective-N-aware correction: resampling is performed *at the pathogen level* (11 independent clusters) rather than at the evaluation level ($N = 8{,}580$), which naturally absorbs the within-pathogen dependence. Specifically, 1,000 cluster-bootstrap resamples over the 11 pathogens yield a 95% CI of $[0.195, 0.333]$ for the scoring $\times$ pathogen interaction; this CI width (0.138) already reflects the effective-N contraction and does not require an additional $\sqrt{N/N_{\text{eff}}}$ correction. An archived re-implementation (`scripts/cluster_bootstrap_omega.py`, `random_seed=42`, $N_{\text{boot}} = 1{,}000$) reproduces the headline value 0.296 and yields a slightly wider 95% percentile CI of $[0.201, 0.346]$ on the same 11-pathogen panel; this range overlaps the original interval, sits well above Cohen's large-effect threshold (0.14), and is provided as a verifiable provenance record (`results/mutbench/cluster_bootstrap_omega.csv`, `cluster_bootstrap_omega_summary.csv`). For sensitivity, a naive $N = 8{,}580$ bootstrap (ignoring pathogen-cluster structure) yields a narrower CI of $[0.260, 0.321]$; the wider cluster-bootstrap CI is the more honest reference. The effect remains above Cohen's large-effect threshold ($\omega^2 > 0.14$) under both bootstrap schemes and at the more conservative 6-category aggregation. The 11-cluster regime is below the rule-of-thumb ≥ 30 for percentile-bootstrap CI calibration, so four complementary inferential frames are added on the same 8,580-row grid (per-frame schemas, scripts, and provenance archives in Chapter 2, Section 2.2.5; full numeric comparison in Table 3.8). *(i) Wild-cluster (Rademacher-weight) bootstrap* [115]: residuals from the restricted ANOVA dropping the scoring $\times$ pathogen interaction are re-signed cluster-by-cluster and the same Type-II-like ω^2 is recomputed for $B =$ 1,000 replicates (`random_seed = 42`); 95% percentile interval $[0.201, 0.303]$, $\Pr(\omega^2_{\text{int}} \leq 0.14) = 0$. The wild-cluster lower bound matches the standard cluster bootstrap to three decimal places (0.201 vs. 0.201); the interval narrows at the upper tail (0.303 vs. 0.346) rather than the lower bound, the expected sharper-but-not-uniformly-wider CGM behaviour when the standard percentile bootstrap's anti-conservatism manifests as right-tail over-coverage. *(ii) Phylogenetic block bootstrap* (Felsenstein 1985 spirit) partitions the 11 pathogens into 8 ICTV viral families (Coronaviridae {SARS-CoV-2, MERS}; Orthomyxoviridae {H3N2, Influenza B}; Flaviviridae {HCV, Dengue}; Picornaviridae, Caliciviridae, Retroviridae, Pneumoviridae, Rhabdoviridae each contributing a single pathogen) and resamples at the family level: $[0.202, 0.346]$, $\Pr(\omega^2 \leq 0.14) = 0.008$; the lower bound matches the standard cluster bootstrap to three decimal places. *(iii) Linear mixed-effects model* fits the same 8,580-row grid as MCC $\sim$ $\text{scoring}_{\text{fixed}} + (1 \,|\, \text{pathogen}) + (1 \,|\, \text{scoring:pathogen})$ in `statsmodels.MixedLM` (REML); the pathogen random-intercept variance is significant by the Self–Liang (1987) boundary-corrected likelihood-ratio test ($\chi^2_1 = 913.7$, $p < 10^{-198}$, $\text{ICC}_{\text{pathogen}} = 0.135$), and the variance-component pseudo-$\omega^2_{\text{int}} = 0.340$ overlaps the bootstrap intervals. *(iv) Bayesian two-way random-effects factor analysis* (PyMC 5.28, half-Normal hyperpriors, non-centered reparameterization; 4 chains $\times$ 2,000 draws, $\hat{R} = 1.003$,

$\mathrm{ESS}_{\mathrm{bulk}} = 618$, no post-tuning divergences) gives posterior mean $\omega^2_{\mathrm{int}} = 0.316$, 95% HDI $[0.242, 0.388]$, $\Pr(\omega^2_{\mathrm{int}} > 0.14) = 0.9998$. The HDI lower bound (0.242) is more stringent than the three frequentist-bootstrap lower bounds ($\approx$ 0.201–0.202) and exceeds the earlier evaluation-level Bayesian hierarchical model's HDI lower bound (0.188/0.246), so partial-pooling shrinkage does not deflate the conclusion. All five lower bounds (0.195, 0.201, 0.202, 0.242, with the variance-component point estimate at 0.340) sit well above Cohen's large-effect threshold ($\omega^2 > 0.14$); the headline conclusion is therefore invariant to the choice of inferential paradigm. A sensitivity analysis excluding HCV—whose

Table 3.8: Five-way comparison of ω^2(scoring $\times$ pathogen) across two cluster-bootstrap frames, one phylogenetic block-bootstrap frame, one variance-component model-based frame, and one Bayesian factor-analysis frame on the same 8,580-row Stage 3 grid. CI columns are 95% percentile intervals for the bootstrap rows, the 95% highest-density interval for the Bayesian factor-analysis row, and --- for the mixed-effects model-based pseudo-ω^2 (no closed-form percentile interval). All five frames put the lower bound above Cohen's large-effect threshold $\omega^2 = 0.14$.

Method	ω^2	95% CI lower	95% CI upper	CI width
Standard cluster (G = 11)	0.272	0.201	0.346	0.145
Wild-cluster Rademacher (G = 11)	0.269	0.201	0.303	0.103
Phylogenetic block (G = 8)	0.272	0.202	0.346	0.143
Mixed-effects var-comp	0.340	—	—	—
Bayesian factor analysis	0.316	0.242	0.388	0.146

Layer A is based on hypervariable regions (diversifying selection) rather than convergent evolution—yields $\omega^2_{\mathrm{scoring}\times\mathrm{pathogen}} = 0.246$, confirming that the interaction effect is not attributable to HCV's atypical ground truth definition. A broader caveat remains: because Layer A across the remaining 10 pathogens is a union of convergent-evolution, immune-escape, and HVR-membership evidence types (with the mix varying by pathogen), the reported interaction ω^2 confounds biological pathogen-dependence with curation-protocol-dependence. HCV removal addresses only the most extreme of these heterogeneities; a same-evidence-type subset re-estimation is deferred to future work. **Per-pathogen leave-one-out sensitivity.** As a stronger generalisation of the HCV check, the same evaluation-level ω^2(scoring $\times$ pathogen) was re-estimated 11 times, holding out one pathogen at a time so that each refit operates on the remaining 10-pathogen panel ($\sim$7,800 rows; archived re-implementation, `scripts/cluster_bootstrap_omega.py`, `random_seed=42`). The 11 leave-one-out (LOO) values are summarised in Table 3.9 and span $[0.253, 0.313]$ with mean 0.293, median 0.297, and standard deviation 0.015; the full range (0.059) is contained inside the 11-pathogen cluster-bootstrap 95% interval $[0.195, 0.333]$ and stays well above Cohen's large-effect threshold ($\omega^2 = 0.14$) and above the cluster-bootstrap lower bound (0.195) for every single hold-out. Defining each pathogen's leverage as $\Delta_i = \omega^2_{\text{full}} - \omega^2_{\text{LOO}(i)}$ (positive Δ = excluding pathogen i *lowers* the interaction estimate, i.e., that pathogen *contributes* to the interaction), the two largest absolute leverages are HCV ($\Delta = +0.043$, the unique outlier and the same pathogen flagged by the curation-protocol restriction) and Norovirus ($\Delta = -0.017$, in the opposite direction: excluding Norovirus actually *raises* ω^2 to 0.313 because Norovirus's pathogen-specific best scoring overlaps the panel-wide best more than the average pathogen does). All other nine pathogens have $|\Delta| \leq 0.011$ and individually move the headline by less than 4% of its value, so the headline ($\omega^2 = 0.296$) is robust to single-pathogen exclusion. The 11-fold trajectory is archived in `results/mutbench/omega_loo_pathogen_sen` and `cluster_bootstrap_omega_loo.csv`.

Kruskal-Wallis non-parametric tests confirmed all effects (pathogen: $H = 1{,}120.8$, $p < 10^{-234}$; detection family: $H = 129.1$, $p < 10^{-21}$; scoring: $H = 277.9$, $p < 10^{-47}$), maintaining the reliability of the directional ANOVA conclusions despite these caveats.

Precision@k Analysis. The detection precision and recall of the best combination for each pathogen were analyzed. Table 3.10 presents the precision, recall, detection size, and

Table 3.9: Per-pathogen leave-one-out ω^2 (scoring $\times$ pathogen) on the Stage 3 evaluation grid. Each row drops one pathogen and refits the Type-II-like decomposition on the remaining 10-pathogen panel; *Leverage* $= 0.296 - \omega^2_{\text{LOO}}$ (positive = pathogen contributes to the interaction; negative = pathogen attenuates it); *Rank* sorts by $|\text{Leverage}|$. All 11 LOO values lie inside the 11-pathogen cluster-bootstrap 95% interval $[0.195, 0.333]$ and above Cohen's large threshold (0.14).

Excluded pathogen	ω^2_{LOO}	Leverage Δ_i	Rank
HCV	0.2534	$+0.0426$	1
Norovirus	0.3126	-0.0166	2
H3N2	0.3066	-0.0106	3
Rabies	0.2878	$+0.0082$	4
RSV	0.2879	$+0.0081$	5
Influenza B	0.3025	-0.0065	6
MERS	0.2900	$+0.0060$	7
SARS-CoV-2	0.2912	$+0.0048$	8
EV-A71	0.2992	-0.0032	9
HIV-1	0.2975	-0.0015	10
Dengue	0.2973	-0.0013	11
mean / median / s.d.	0.2933 / 0.2973 / 0.0154	—	—

best MCC for the top-1 combination of each pathogen.

The mean top-1 precision is 0.338 and recall is 0.552, with substantial heterogeneity across pathogens (e.g., HCV achieves precision 0.964 but MERS shows precision 0.045 with perfect recall), confirming the limitations of single-threshold detection. The MERS row should be read as a *failure case* rather than a success: perfect recall is achieved only because the top-1 detector flags 200 of the 1,330 alignment positions ($\approx$ 22% of the protein), so all 9 Layer-A positives are recovered by near-indiscriminate calling. This is symptomatic of threshold and parameter saturation under the small Layer-A positive count ($n_{\text{A}} = 9$) for MERS rather than of a meaningful detection signal, and is not a basis for claiming successful generalization to MERS.

DMS Validation. For 4 of the 6 DMS pathogens with threshold sensitivity data, Spearman correlations between DMS fitness and the best scoring formula output differed in direc-

Table 3.10: Precision, recall, detection size, and best MCC of the top-1 combination for each pathogen (11 pathogens, 20 scoring types).

Pathogen	Precision	Recall	n_{det}	Best MCC
Dengue	0.189	0.583	74	0.274
EV-A71	0.500	0.185	10	0.260
H3N2	0.517	0.600	29	0.534
HCV	0.964	0.500	28	0.660
HIV-1	0.318	0.304	22	0.275
Influenza B	0.190	0.941	84	0.392
MERS	0.045	1.000	200	0.196
Norovirus	0.417	0.667	24	0.510
Rabies	0.308	0.308	39	0.248
RSV	0.182	0.300	33	0.196
SARS-CoV-2	0.090	0.680	189	0.211
Mean	0.338	0.552	–	0.341

tion (H3N2: $\rho = 0.191$; HIV-1: $\rho = 0.412$; RSV: $\rho = -0.363$; SARS-CoV-2: $\rho = -0.370$; all $p < 0.002$), confirming that DMS fitness and evolutionary adaptation capture different aspects of mutation impact and validating the 3-layer ground truth design. Varying the DMS threshold from 10% to 30% altered the identity of the best-performing combination in all 4 pathogens (mean best DMS F1: $0.328 \to 0.377 \to 0.479$), while the relative rankings of scoring formulas remained stable within the 15%–25% range (see Section 2.2.2), further supporting pathogen-dependent method selection.

To further illustrate the ground-truth dependence, Table 3.11 compares the best-performing scoring type under Layer A (literature-curated adaptive/diversifying positives) versus Layer C (DMS fitness) for the 6 pathogens with experimental DMS data.

In all 6 pathogens, the best scoring type differs between Layer A and Layer C (Table 3.11). The per-scoring (best-MCC) Spearman rank correlations between Layer A and Layer C rankings (20 scoring types per pathogen, computed from `layer_c_evaluation.csv`) show two significant cases (RSV: $\rho = +0.81$, $p < 10^{-4}$; H3N2: $\rho = -0.55$, $p = 0.012$)

Table 3.11: Best scoring type under Layer A vs. Layer C ground truth for 6 DMS pathogens. DMS sources: SARS-CoV-2 [2], H3N2 [3], HIV-1 [4], RSV [5], Rabies [6], EV-A71 [7]. In all 6 pathogens, the best scoring type differs between the two ground truth layers, confirming their independence.

Pathogen	**Best (Layer A)**	**MCC_A**	**Best (Layer C)**	**MCC_C**
SARS-CoV-2	tranception	0.211	plddt_inv	0.186
H3N2	fubar_bf	0.534	semantic_change	0.245
HIV-1	entropy	0.275	fubar	0.139
RSV	esm2_llr	0.196	EVEscape_composite	0.180
Rabies	stability	0.248	homoplasy	0.182
EV-A71	semantic_change	0.260	plddt_inv	0.322

and four non-significant pathogens (SARS-CoV-2: $\rho = -0.10$; HIV-1: $\rho = +0.31$; Rabies: $\rho = +0.32$; EV-A71: $\rho = +0.21$); per-pathogen rankings of the 20 scoring types under Layer A versus Layer C therefore often diverge—rather than asserting Layer C as a fully independent evaluation dimension, this divergence is reported as evidence that the two ground truths capture non-identical aspects of hotspot-ness. Only RSV exhibits a strong positive agreement; H3N2's moderate negative correlation means the two ground truths actively disagree on scoring order for that pathogen. **Post-hoc rationalisation check**: the 20 scoring formulas were defined prior to per-pathogen evaluation (Chapter 2); the per-pathogen MCC ranking is deterministic given the design and the biological rationale per pathogen is interpretive only. The Layer A vs. Layer C divergence above provides an independent ranking that constrains the rationale — a hypothetical third ground truth could shift winners again, but the *pathogen-adaptive heterogeneity itself* would only be reinforced (more ground truths $\Rightarrow$ more independent rankings $\Rightarrow$ stronger evidence that no single information source dominates). In practice, when Layer A and Layer C indicate different optimal scoring types, the choice depends on the research objective: Layer A rankings should be prioritized for immune evasion surveillance, while Layer C rankings are more appropriate for functional impact assessment such as DMS-guided drug target prioritization.

3.4.5 Summary of Key Findings

The cross-pathogen benchmark supports pathogen-adaptive method selection through two primary analyses—the scoring $\times$ pathogen ω^2 interaction, and a single-anchor vaccine-escape enrichment audit (HIV-1 as primary external anchor, with H3N2 self-consistency and SARS-CoV-2 exploratory checks)—plus two null-consistent corroborations (LOPO 0/11 and Friedman). The four lenses re-parameterize the same dataset and are not independent replications.

1. **Interaction dominance (modeled)**: Among the modeled factors, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$; 95% cluster-bootstrap CI [0.195, 0.333]) is the largest component, exceeding all main effects and other two-way interactions; a within-cell residual $\omega^2 \approx 0.39$ captures the unmodeled three-way interaction and parameter-variant noise. The optimal information source is pathogen-dependent.

2. **Unique best combinations**: 11 unique best combinations were observed across 11/11 pathogens, with 9 distinct scoring types. No two pathogens share the same best combination.

3. **LOPO null-consistent corroboration**: LOPO cross-validation yielded 0/11 combination matches with a mean generalization MCC of 0.032 and a 0.265 MCC oracle-vs-generalized gap (one-sided permutation $p \approx 0.25$–0.28 against the random-combination null). Both the match rate and the gap are null-consistent (failure to reject the null of universality); LOPO is therefore reported as *corroboration* of the ANOVA interaction rather than as independent positive evidence.

4. **No universally superior combination**: Friedman $\chi^2 = 7.69$, $p = 0.990$ (a second

failure-to-reject under the rank-block test, with limited power at $n = 11$ blocks). No single combination is reliably distinguishable as superior across all pathogens.

The four lenses are not independent confirmations: only the ANOVA (within the modeled subspace) provides positive evidence of pathogen-dependence; the LOPO permutation test and the Friedman rank test are two failures-to-reject that are *consistent* with pathogen-dependent optimality but do not by themselves constitute independent evidence. The HIV-1 vaccine-escape audit provides a separate external positive line. The next section investigates *why* the ANOVA interaction arises by analysing individual information types.

Subgroup sanity check: DMS-Available vs DMS-Unavailable. The six DMS-available pathogens (SARS-CoV-2, H3N2, HIV-1, RSV, Rabies, EV-A71) attain best-per-pathogen MCC of 0.287 (range 0.196–0.534); the five DMS-unavailable pathogens (Dengue, Norovirus, Influenza B, MERS, HCV) attain 0.407 (range 0.196–0.660). Mann–Whitney $U = 10$, $p = 0.43$: the Layer A-only subgroup is not systematically weaker than Layer A+C. The higher DMS-unavailable best-MCC reflects denser Layer A sets in HCV/Norovirus/Influenza B rather than ground-truth weakness; per-pathogen rows are archived in `stage3_full_results.csv`. Results on DMS-unavailable pathogens should be read as phylogenetic-pattern evidence, not functional-effect evidence.

3.5 Information-Type Analysis

The 20 scoring formulas used in the benchmark are derived from 10 underlying biological features spanning 6 information categories. Several scoring formulas represent the same feature (e.g., freq, freq_with_indel, and rare_freq are all frequency-based), while composite formulas (EVEscape_composite, EqualWeight) combine multiple features. The following analysis examines the discriminative power of the 10 underlying features to explain why the scoring × pathogen interaction dominates.

The preceding sections demonstrated that the best-performing scoring–detection combination differs across pathogens ($\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$). This section investigates *why* this interaction arises by analyzing the discriminative power of individual information types (features) that underlie the scoring formulas, and examines whether pathogen-specific feature importance patterns explain the observed cross-pathogen heterogeneity.

3.5.1 Per-Feature Discriminative Power and Multivariate Importance

The 20 scoring formulas are derived from 10 underlying biological features spanning 6 categories: frequency-based (*freq*, *entropy*, *rare_freq*), phylogeny (*homoplasy*), evolutionary rate (*dN/dS proxy*), deep learning (*esm2_llr*, *semantic change*), structure (*plddt*, *SASA*), and chemical (*Grantham distance*); full feature definitions and exclusion criteria are documented in the supplementary archive. The univariate Layer A AUC of these features varies sharply across pathogens—Norovirus is best discriminated by homoplasy (AUC $= 0.944$), H3N2 by pLDDT (0.787), SARS-CoV-2 and MERS by the ESM-2 masked-marginal channel (0.586 and 0.695), HIV-1 by entropy (0.730), HCV by the dN/dS proxy (0.680), and the remaining pathogens cluster around homoplasy or pLDDT—and two pathogens (RSV, MERS) exhibit inverse associations (AUC < 0.5 for several features), confirming that the biological relationship between features and hotspot status is pathogen-dependent. A multivariate Random Forest classifier (200 trees, max depth 5, 5-fold stratified CV; `scripts/rf_feature_importance.py`) reaches cross-validation AUC of 0.742 on average across 11 pathogens (range 0.585–0.899; per-pathogen rows archived in `rf_cv_auc_v2.csv`). The same correlation structure that drives the per-pathogen univariate variation also explains why expanding from 6 to 10 features helps differentially: frequency, entropy and dN/dS proxy form a highly correlated cluster ($\rho_{\text{freq–entropy}} = 0.97$, $\rho_{\text{freq–dN/dS}} = 0.87$); SASA is genuinely independent of pLDDT ($\rho = -0.19$); and homoplasy is the most decorrelated feature overall ($|\rho| \leq 0.233$ with all others), which is why it dominates per-feature AUC for pathogens with strong convergent-evolution signals (Norovirus, Dengue, Influenza B). The 5-fold CV is at position level within each pathogen and is therefore subject to spatial-autocorrelation leakage; the AUC values are reported as feature-complementarity illustrations rather than out-of-sample generalisation estimates (region-blocked CV is left to future work). Together these analyses provide a mechanistic explanation for the Stage 2 scoring $\times$ pathogen interaction ($\omega^2 = 0.296$): different pathogens require different information types, and scoring formulas that surface the relevant information type for a given pathogen produce higher-quality inputs for downstream detection.

3.5.2 Feature Ablation Analysis

A feature-ablation robustness check was performed under the EqualWeight ensemble with LOPO evaluation and a top-10% threshold, on the 12-pathogen feature-analysis panel (core 11 + Zika; restricting to the core 11 shifts the baseline 0.083 $\rightarrow$ 0.093 with the same qualitative ordering). **Leave-one-feature-out** (Table 3.12) reveals homoplasy as the largest contributor ($\Delta = -0.028$, $0.083 \rightarrow 0.055$), with entropy, frequency, and rare-variant frequency also showing substantial contributions; removing the ESM-2 masked-marginal channel ($\Delta = +0.004$) or SASA ($\Delta = +0.003$) marginally improves performance, indicating that these features introduce noise under uniform weighting when averaged across pathogens.

A separate *nested-LOPO* robustness check tests the 4-feature core (homoplasy, pLDDT, entropy, freq) against the full 10-feature ensemble under proper held-out-pathogen scoring

Table 3.12: Leave-one-feature-out ablation results. ΔMCC is the change in mean MCC (12 pathogens including Zika) when removing one feature from the 10-feature EqualWeight ensemble. Negative Δ indicates the feature improves the ensemble. The core-11 restriction (excluding Zika) shifts the baseline to 0.093 with proportionally rescaled Δ values; orderings are preserved.

Removed feature	**Mean MCC**	**ΔMCC**	**Interpretation**
None (all 10)	0.083	—	Baseline
homoplasy	0.055	−0.028	Largest contributor
entropy	0.068	−0.015	Substantial
freq	0.071	−0.012	Substantial
rare_freq	0.073	−0.010	Moderate
dnds_proxy	0.077	−0.006	Moderate
plddt	0.081	−0.002	Minimal
grantham	0.081	−0.002	Minimal
semantic_change	0.082	−0.001	Minimal
sasa	0.086	+0.003	Noise under uniform **w**
esm2_llr	0.087	+0.004	Noise under uniform **w**

(`scripts/feature_ablation_nested_lopo.py`; per-fold values in Table 3.13; archive `feature_ablation_nested_lopo.csv`). The full 10-feature ensemble has LOPO mean MCC = 0.070 (95% bootstrap CI [0.017, 0.130]) and the fixed 4-feature core has LOPO mean MCC = 0.081 (95% CI [0.008, 0.166]); the paired delta is +0.011 (95% CI [−0.018, +0.042]; sign-flip permutation $p = 0.61$, two-sided). The 4-feature core is therefore statistically indistinguishable from the 10-feature ensemble under nested-LOPO. The earlier within-pathogen ≈98% retention figure (0.081/0.083 in-sample) is retained only as an ablation descriptor; the dissertation hereafter reports the result as operational equivalence under nested-LOPO ($p = 0.61$) rather than as a percentage retention claim.

Features that are noise on average under uniform weighting (ESM-2, SASA) are the best discriminators for specific pathogens (ESM-2 for MERS, SASA for Norovirus), confirming that uniform weights cannot optimally exploit all 10 features simultaneously. Three legacy adaptive weighting strategies (kNN, Top-3 AUC, correlation-aware ensemble; archived as `results/mutbench/adapt_v2_summary.csv`) all underperform EqualWeight (max MCC

Table 3.13: Nested-LOPO comparison of the 10-feature EqualWeight ensemble vs. the fixed 4-feature core (homoplasy, pLDDT, entropy, freq). Per-fold MCC on the held-out pathogen using training-pathogen-fit signs and rankings; paired bootstrap (1,000 resamples over the 12 folds) for the mean MCC and the paired delta; sign-flip permutation (5,000 iterations) for the two-sided p-value of the paired delta.

Held-out pathogen	**Full 10-feature MCC**	**Fixed 4-core MCC**	**Δ (4-core − full)**
SARS-CoV-2	+0.0284	−0.0095	−0.0379
H3N2	+0.1301	+0.1593	+0.0292
Norovirus	+0.2814	+0.4318	+0.1504
HIV-1	+0.0572	−0.0101	−0.0673
Dengue	+0.0820	+0.1450	+0.0630
RSV	−0.0648	−0.0648	0.0000
Influenza B	+0.2186	+0.2186	0.0000
MERS	+0.0031	−0.0275	−0.0306
HCV	+0.1234	+0.1502	+0.0268
Zika	−0.0231	−0.0231	0.0000
Rabies	−0.0491	−0.0491	0.0000
EV-A71	+0.0499	+0.0499	0.0000
Mean (12 folds)	**+0.0698**	**+0.0809**	**+0.0111**
95% paired bootstrap CI	[+0.017, +0.130]	[+0.008, +0.166]	[−0.018, +0.042]
Sign-flip permutation p (two-sided)	—	—	0.6132

Folds where Full-10 and Fixed-4 MCC are identical (RSV, Influenza B, Zika, Rabies, EV-A71) reflect held-out pathogens whose top-10% set is unchanged when the 6 noise-on-average features are dropped after applying training-fit signs. Underlying CSV: `results/mutbench/feature_analysis/feature_ablation_nested_lopo.csv`; summary CSV: `feature_ablation_nested_lopo_summary.csv`.

0.020/0.049/0.043 vs. EqualWeight 0.083) because at $n = 11$ reference pathogens the 13-dimensional pathogen-profile space is severely undersampled; the Cycle 7 + 7B audits below extend this to seven independent failure modes for adaptive weighting at this panel size. The oracle ceiling (per-pathogen weights with labels) reaches 0.160, confirming adaptive gains are possible in principle but require 20–30+ pathogens for reliable selection (Chapter 4, Section 4.7).

MutBench Cold-Start algorithm: a two-tier search-space reduction recipe. The empirical evidence above operationalises into a deployable algorithm with two tiers, complementary rather than redundant. **Tier 1 (cold-start, universal, $\sim$10 min CPU)** applies the canonical 4-feature ensemble when pathogen identity or family classification is unavailable: compute homoplasy (Fitch parsimony), pLDDT (optional, from AlphaFold/ESMFold), entropy and frequency from the MSA; combine via the uniform z-score ensemble $s_i = (1/4)\sum_{f\in\{\text{hom, pLDDT, ent, freq}\}} \text{sign}(\hat{s}_f) \cdot z_{i,f}$ where $\hat{s}_f \in \{-1, +1\}$ is the training-fit directional sign (or pre-defined domain orientation, Section 2.2.4); threshold at the top 10% of ranked scores. Formal pseudocode is given in Algorithm 1; the baseline progression from single-feature predictors through ensembles of increasing size is summarised in Table 3.14. The empirical anchor is the nested-LOPO mean MCC of 0.081 vs. full-10 ensemble 0.070 (paired delta +0.011, 95% CI [−0.018, +0.042], sign-flip permutation $p = 0.61$; Table 3.13) and a 13/210 rank under the exhaustive 4-of-10 subset audit (top 6.2%-tile; Table 3.15); on the search-space-reduction task this Tier 1 recipe narrows $\sim$1,000 alignment positions to 20–80 (7–9× enrichment for escape sites, Section 3.5.4). The complexity is $\mathcal{O}(L \cdot |\mathcal{F}|)$ per pathogen for the per-position score combination ($|\mathcal{F}|$=4), dominated upstream by the per-feature score computation: homoplasy is $\mathcal{O}(L \cdot N)$ for N aligned sequences via a single-pass Fitch traversal, entropy and frequency are $\mathcal{O}(L \cdot N)$, and pLDDT is $\mathcal{O}(L)$ once a single AlphaFold/ESMFold pass is amortised; on the 11-pathogen panel the median end-to-end wall-clock is below ten minutes on a single CPU core. **Tier 2 (family-adaptive acceleration, optional, requires ICTV family annotation)** substitutes the Tier 1 ensemble with the family-prior single-best scoring category from Table 4.2 (Coronaviridae $\rightarrow$ PLM, Orthomyxoviridae $\rightarrow$ FUBAR, Caliciviridae $\rightarrow$ homoplasy, Retroviridae $\rightarrow$ entropy, Flaviviridae $\rightarrow$ frequency, Pneumoviridae $\rightarrow$ ESM-2). Tier 2 is grounded in the same scoring $\times$ pathogen interaction effect ($\omega^2 = 0.296$; phylogenetic block bootstrap at family level [0.202, 0.346]; Section 3.4.4) but is reported as *provisional* rather than *prescriptive*: the family priors are descriptive of the current 11-pathogen panel, not yet validated on held-out families, and within-family heterogeneity (SARS-CoV-2 vs. MERS within Coronaviridae; H3N2 vs. Influenza B within Orthomyxoviridae) means the prior is a starting hypothesis, not a guarantee. When a novel pathogen's family is absent from Table 4.2, the algorithm defaults to Tier 1. Validation of Tier 2 on held-out families is reserved as a Priority-3 deliverable contingent on the benchmark expanding beyond the current 11-pathogen Bolker-minimum panel (Chapter 4, Section 4.6).

Cold-Start baseline progression: why an ensemble is necessary. A defensible "minimal viable method" claim requires showing that the recipe in Algorithm 1 is not redundant with simpler baselines. Table 3.14 reports the nested-LOPO mean MCC for: (i) the single best feature per fold (top-1), (ii) cumulative ensembles of the top-K features for $K = 2, \ldots, 9$, (iii) the full 10-feature ensemble, (iv) the Cold-Start fixed 4-core (Algorithm 1), and (v) the oracle-best 4-of-10 subset (label-selected from $\binom{10}{4} = 210$ combinations; Table 3.15). Three takeaways defend the algorithm as a method rather than a heuristic. *First*, the single-feature

Algorithm 1: MutBench Cold-Start (Tier 1, label-free 4-feature recipe)

Input: Multiple-sequence alignment A over L alignment columns; N aligned sequences; optional per-residue confidence $q \in \mathbb{R}^L$ from AlphaFold or ESMFold

Output: Hotspot call set $H \subseteq \{1, \ldots, L\}$ with $|H| = \lceil 0.10\, L \rceil$

`// Step 1 --- per-feature raw scores (each` $\mathcal{O}(L \cdot N)$`, pLDDT` $\mathcal{O}(L)$`)`

for $f \in \{$ homoplasy, pLDDT, entropy, frequency $\}$ **do**

  Compute $x_{i,f}$ for $i = 1, \ldots, L$ (Fitch parsimony for homoplasy; column-wise Shannon for entropy; minor-allele fraction for frequency; pass-through for pLDDT)

  $z_{i,f} \leftarrow (x_{i,f} - \overline{x_f})/\mathrm{sd}(x_f)$ `// within-pathogen` z`-score`

  Set directional sign $\hat{s}_f \in \{-1, +1\}$: $+1$ for homoplasy/entropy/frequency, -1 for pLDDT (or training-fit per Section 2.2.4)

`// Step 2 --- uniform 4-feature ensemble (`$\mathcal{O}(L)$`)`

for $i \leftarrow 1$ **to** L **do**

  $s_i \leftarrow \frac{1}{4} \sum_f \hat{s}_f \cdot z_{i,f}$

`// Step 3 --- top-10% threshold (`$\mathcal{O}(L \log L)$` for sort)`

$\tau \leftarrow$ 90th percentile of $\{s_i\}_{i=1}^{L}$

$H \leftarrow \{\, i : s_i \geq \tau \,\}$

return H

mean MCC is 0.015 — statistically indistinguishable from zero — so no individual per-position score crosses the cross-pathogen hotspot bar; an ensemble is empirically required, not stylistic. *Second*, the Cold-Start 4-core (0.081) outperforms every cumulative top-K ensemble for $K = 1, \dots, 4$ and beats both the full 10-feature ensemble (0.070, $\Delta = +0.011$) and the cumulative top-5 peak (0.072, $\Delta = +0.009$) without ever consulting the held-out fold's labels. *Third*, the label-using oracle that picks the best 4-of-10 subset gains only $+0.014$ MCC over the Cold-Start 4-core — well within the paired bootstrap CI for the 4-core mean MCC ($[+0.008, +0.166]$; Table 3.13) — so the marginal value of label-driven subset tuning is small relative to cross-pathogen noise.

Table 3.14: Baseline progression for the Cold-Start ensemble. Single-feature MCC sits at the cross-pathogen noise floor; the Cold-Start 4-core (Algorithm 1) dominates every smaller cumulative ensemble, beats the full 10-feature ensemble by $+0.011$ MCC, and is only $+0.014$ MCC behind the oracle 4-of-210 subset that requires labels for selection. Source: `results/mutbench/feature_analysis/feature_ablation_nested_lopo.csv` (single feature and `cumulative_top_K` configurations, $n = 12$ folds each); 4-core and oracle from `subset_selection_robustness_210combos.csv`.

Configuration	**LOPO mean MCC**	**Δ vs. 4-core**
Single best feature alone (top-1)	0.0147	-0.0662
Top-2 features cumulative	0.0398	-0.0411
Top-3 features cumulative	0.0457	-0.0352
Top-4 features cumulative	0.0715	-0.0094
Top-5 features cumulative	0.0724	-0.0085
Top-6 features cumulative	0.0580	-0.0229
Top-7 features cumulative	0.0627	-0.0182
Top-8 features cumulative	0.0603	-0.0206
Top-9 features cumulative	0.0564	-0.0245
Full 10-feature ensemble	0.0698	-0.0111
Cold-Start 4-core (Algorithm 1)	**0.0809**	—-
Oracle best 4-of-210 (label-selected)	0.0950	$+0.0141$

Cumulative top-K configurations include the K features with highest training-fold importance, applied identically across all 12 held-out folds; the per-fold values underlying the means are in `feature_ablation_nested_lopo.csv`. The single-feature mean (0.0147) is statistically indistinguishable from zero (paired bootstrap CI straddling 0), and the cumulative top-K curve is non-monotone in K (peak at K=5, 0.0724), confirming that a fixed domain-knowledge 4-core (Algorithm 1) avoids both the underfitting of small ensembles and the overfitting of large ones without consulting held-out labels.

Subset selection robustness (210-combination exhaustive sweep). The dissertation's 4-feature core (homoplasy, pLDDT, entropy, frequency) was specified *a priori* from the canonical evolutionary-plus-structure panel rather than tuned to LOPO MCC. To ask whether this domain-knowledge choice is competitive with the empirically best 4-of-10 subset, we ran the same nested-LOPO protocol (training-pathogen-fit signs and rankings, within-fold z-scoring, uniform averaging, top-10% threshold) on every $\binom{10}{4} = 210$ feature subset (script: `scripts/feature_subset_selection_210.py`; CSV: `results/mutbench/feature_analysis/subs`
per-subset values in Table 3.15). The dissertation's 4-core (`freq, entropy, homoplasy, plddt`) attains LOPO mean MCC = 0.0809 and ranks **13/210** (top 6.2%-tile, i.e. above the 93rd percentile of the subset distribution); the empirically best 4-of-10 subset is (`freq, entropy, homoplasy, semantic_change`) at MCC = 0.0950, a gap of $+0.0141$ MCC. The full subset distribution has median MCC = 0.0525 and IQR [0.0377, 0.0700], so the 4-core sits well above the 75th percentile of arbitrary 4-of-10 subsets and slightly above the full 10-feature ensemble (0.0698). Homoplasy appears in all top-13 subsets and freq/entropy in the majority, supporting the choice of the canonical evolutionary panel; the one feature where the dissertation's choice (pLDDT) is dominated empirically is semantic-change, which contributes positively when paired with frequency and homoplasy under nested-LOPO. The headline interpretation is that the dissertation's 4-core is *near-optimal but not literally best*: it sits in the top 6.2% of all 210 subsets, with a $+0.0141$ MCC gap to the empirically best subset that is well within the paired-bootstrap CI for the nested-LOPO mean MCC ([+0.008, +0.166] for the 4-core; Table 3.13). The selection is therefore robust under the cold-start regime: any subset chosen by domain knowledge that includes homoplasy plus two of (freq, entropy) will land in the top decile, and the marginal gain from oracle-best subset selection ($+0.014$ MCC) is small relative to the cross-pathogen noise

floor.

Table 3.15: Top-10 4-of-10 feature subsets by nested-LOPO mean MCC (paired protocol, training-pathogen-fit signs, within-fold z-scoring, uniform averaging, top-10% threshold). The dissertation's 4-feature core (`freq, entropy, homoplasy, plddt`) is row-shaded *Core* and ranks 13/210 overall (top 6.2%-tile). Full ranking in `subset_selection_robustness_210combos.csv`.

Rank	4-feature subset	LOPO mean MCC	Δ vs. full-10
1	freq, entropy, homoplasy, semantic_change	0.0950	+0.0252
2	freq, homoplasy, plddt, semantic_change	0.0926	+0.0229
3	freq, homoplasy, sasa, semantic_change	0.0876	+0.0179
4	freq, entropy, homoplasy, sasa	0.0870	+0.0172
5	entropy, homoplasy, sasa, dnds_proxy	0.0864	+0.0166
6	freq, entropy, homoplasy, dnds_proxy	0.0857	+0.0159
7	freq, entropy, rare_freq, homoplasy	0.0856	+0.0158
8	entropy, homoplasy, dnds_proxy, semantic_change	0.0856	+0.0158
9	freq, homoplasy, plddt, sasa	0.0847	+0.0150
10	freq, homoplasy, dnds_proxy, semantic_change	0.0839	+0.0142
13 (Core)	**freq, entropy, homoplasy, plddt**	**0.0809**	**+0.0111**
—	Full 10-feature ensemble (reference)	0.0698	—
—	Median of 210 subsets	0.0525	−0.0173

Best subset MCC − 4-core MCC = +0.0141 (within the paired bootstrap CI for the 4-core mean MCC, [+0.008, +0.166]; Table 3.13). Homoplasy appears in all top-13 subsets; `semantic_change` is the most frequently-paired non-core feature in the top-10 (5/10 subsets), reflecting its complementarity to frequency and homoplasy under nested-LOPO uniform weighting.

Full-lattice subset audit and per-feature Shapley. The 4-of-10 sweep above was extended to the full lattice of $2^{10} - 1 = 1023$ non-empty subsets under the same nested-LOPO protocol (script: `scripts/codex_p3_full_lattice_shapley.py`; CSV: `results/mutbench/codex_w`
The Cold-Start 4-core ranks $87/1023$ (top 8.5%-tile) and matches the existing 4-of-10 audit value to numerical precision (LOPO mean MCC 0.08088, exact). The Cold-Start 4-core is therefore correctly characterised as a *historical fixed configuration benchmarked against the full lattice*, not as a non-arbitrary optimum: the lattice top-1 is the 7-feature subset (`freq, entropy, homoplasy, sasa, grantham, dnds_proxy, semantic_change`) at MCC 0.0968, and a 3-feature variant (`freq, entropy, homoplasy`) (henceforth "3-core") strictly

dominates the 4-core (MCC 0.0866 vs. 0.0809, $\Delta = +0.0058$, lattice rank $27/1023$). Formal Shapley values with binomial weights and $v(\emptyset) = 0$ (efficiency exact: $\sum_f \phi_f = v(\mathcal{N}) = 0.0698$ for MCC; 2.039 for enrichment) place homoplasy (+0.0294), frequency (+0.0220) and entropy (+0.0219) as the only positive contributors to MCC at the lattice mean; the dissertation's 4-core fourth feature (pLDDT) has Shapley contribution -0.0015, and `esm2_llr` contributes -0.0144 (the only feature negative on both MCC and enrichment). The recommendation is to mark pLDDT as *optional structural extension* in Algorithm 1 and to retain the 3-core as the stronger compact descriptive variant; the 4-core is preserved as the historical configuration evaluated throughout the dissertation for backward comparability.

Per-pathogen rank reliability and decision-curve framing. A second audit asks whether the Cold-Start score is a usable rank-prioritization signal even when binary MCC is modest (script: `scripts/codex_p2_rank_reliability.py`; CSV: `results/mutbench/codex_wave1/p2_dec`
For each pathogen, positions are binned into 10 score deciles, and the difference in empirical Layer A prevalence between the top and bottom decile is reported with a 1,000-resample bootstrap 95% confidence interval. The 12-pathogen aggregate places the 3-core ahead of the 4-core and the full-10 ensemble: top-vs-bottom mean prevalence delta +0.084 (3-core), +0.063 (4-core), +0.043 (full-10), versus +0.011 for random rank; mean Spearman trend across deciles +0.37, +0.28, +0.30 and -0.04 respectively. Per-pathogen, only $5/12$ pathogens (Norovirus, Dengue, H3N2, Influenza B, HIV-1) show a 4-core top-vs-bottom CI excluding zero in the positive direction; *Rabies inverts*, with delta -0.123 and CI $[-0.245, -0.003]$ (Spearman $\rho = -0.76$), so the 4-core score is anti-monotone with Layer A on the rabies G-ectodomain panel. A decision curve at a screening budget of 50 sites/-

pathogen yields 66 expected true positives (4-core) and 68 (3-core) over the 12-pathogen panel, against ~ 30 for random allocation — a $\sim 2.2\times$ improvement that defines the operational claim of Algorithm 1. The framing is therefore *ranked screening-budget allocator with documented per-pathogen heterogeneity*, not calibrated probability: held-out isotonic ECE values (0.03–0.04) are uninformative at the prevailing Layer A base rate ($\sim 5\%$) and we make no probability-calibration claim.

Null-calibrated significance under four null models. A pre-registered null-calibration audit compares observed 4-core and 3-core metrics against 100,000 permutations under each of four null models per pathogen (script: `scripts/codex_p1_null_calibration.py`; CSV: `results/mutbench/codex_wave1/p1_null_per_pathogen.csv`). The four nulls preserve increasing structure: *(i)* iid prevalence-preserving label permutation; *(ii)* circular-shift of the label vector with random offset; *(iii)* within-block label permutation in fixed 50-residue windows; *(iv)* local-burden-preserving stratification into five quintiles by mean local frequency-plus-entropy in a ± 5 window with within-stratum shuffling. The aggregated one-sided Stouffer p-value across the 12-pathogen panel for the 4-core MCC is $p = 1.00$ under all four nulls — the panel does not collectively beat even a prevalence-preserving null, because $5/12$ pathogens have observed MCC at or below null mean (SARS-CoV-2, HIV-1, RSV, MERS, Zika, Rabies). *Per-pathogen*, $5/12$ pathogens individually pass the iid null at one-sided $p < 0.05$ for the 4-core MCC: Norovirus ($p = 1\times10^{-5}$), Influenza B ($p = 7\times10^{-5}$), H3N2 ($p = 0.0018$), HCV ($p = 0.0092$), Dengue ($p = 0.0055$); the count drops to $3/12$ under the circular-shift null, $2/12$ under the protein-block null, and $1/12$ (Norovirus only) under the strict local-burden-preserving null. This gradient is the headline result of the audit: most “above-null” signal is explained by local clustering of Layer A epitopes, and only

Norovirus retains significance against the strictest biological null across MCC, ± 10-window MCC, top-20 precision, and enrichment. We therefore treat Algorithm 1 as an *exploratory cold-start prioritization heuristic* for a callable pathogen subset rather than as a globally calibrated detector. The five iid-significant pathogens form a descriptive callable set; a prospective abstention rule that uses only training-fold information to flag callable status is reserved as the next experiment (Section 4.7, Wave 2).

Pre-registered callability and abstention rule (P5). A pre-registered abstention rule operationalises the "callable subset" framing of Paragraph 9 (script: `scripts/codex_p5_callability.py`; sweep: `scripts/codex_p5_inner_loo_sweep.py`; CSVs: `results/mutbench/codex_wave2/p5_*`). Per nested-LOPO fold, the rule asks five conjunctive questions about the candidate method (3-core primary, 4-core secondary) using *training-fold-only* information except for an annotation-density red flag on the held-out: *(D1)* top-decile minus bottom-decile prevalence delta has bootstrap lower 95% CI > 0.02 AND point estimate > 0.05 in $\geq 7/11$ training pathogens; *(D2)* the per-training-pathogen ratio of observed top-decile enrichment over the mean enrichment under $10{,}000$ iid label permutations has $1{,}000$-resample bootstrap lower 95% bound > 1.10 across the 11 training pathogens; *(D3)* median pairwise Jaccard of top-10% callsets across 200 perturbations (z-score jitter at $0.10\times$ per-feature SD plus sign re-fit on bootstrap of training pathogens) is ≥ 0.60 AND the 10th percentile is ≥ 0.40, aggregated over the 11 training pathogens; *(D4)* held-out has ≥ 10 Layer A positives, ≥ 200 rankable positions, and Layer A prevalence in $[1\%, 20\%]$ (annotation-density metadata only — no held-out scores, ranks, MCC, or p-values); *(inversion guardrail)* fewer than two training pathogens have bootstrap CI upper bound < 0 AND the median Spearman trend is > 0. Under these defaults, **$0/12$ folds are callable for either the 3-core or the 4-core method.** D1 is the rate-limiting gate (maximum quorum observed: $5/11$), structurally consistent with the Wave 1 finding that only five pathogens have above-null signal — no 11-pathogen training subset reaches the $7/11$ threshold. The pre-registered sensitivity sweep (D1 quorum $\in \{5, 6, 7, 8\} \times$ D2 lower bound $\in \{1.05, 1.10, 1.20, 1.50\}$, 32 configs) confirms robustness: at any quorum $\geq 6/11$ regardless of D2 bound, $0/12$ are callable; only the most-permissive corner ($5/11$, 1.05) yields any callable folds, and those select P1-negative pathogens (3-core: {SARS-CoV-2, RSV, Zika} with mean MCC -0.027; 4-core: {Rabies} with MCC -0.049) — a pathology of D2 alone, where sparse-pathogen folds inflate the (observed/null) ratio. D3 confirms the P3 finding that 4-core perturbation stability is meaningfully worse than 3-core ($3/12$ vs. $11/12$ folds passing the median-Jaccard threshold), reflecting the negative pLDDT Shapley value documented in Paragraph 9. The pre-registered rule therefore correctly refuses to deploy Algorithm 1 on the current 12-pathogen panel in either configuration; the dissertation's deployment claim is thereby formally bounded to the descriptive callable set identified in Paragraph 9, with prospective deployment requiring panel expansion (Chapter 4, Section 4.7).

Biologically-realistic null calibration (P6). P6 addresses the live committee objection that the callable-subset signal in Paragraph 9, even after the abstention refusal in Paragraph 9, might be a structural or evolutionary clustering artifact rather than exact-site discrimination. The audit used four structurally explicit nulls: `sasa_strat` with 5 SASA quintiles, `plddt_strat` with 5 pLDDT quintiles, `sasa_freq_joint` with 3×3 SASA$\times$frequency cells, and `plddt_ent_joint` with 3×3 pLDDT$\times$entropy cells. Each used $n_{\text{perm}} = 100{,}000$ and smoothed $(b+1)/(n_{\text{perm}}+1)$ one-sided p-values, with a predeclared degeneracy criterion of <2 strata with both positives and negatives, <5 movable positives, or near-zero null

variance (script: `scripts/codex_p6_biological_nulls.py`; outputs: `results/mutbench/codex_wa`
All P6 nulls were valid: **12**/**12** pathogens were valid across all four nulls, with no degeneracy branch triggered. Under the 4-core MCC, individual-significance counts were 5/12 for `sasa_strat` {H3N2, Norovirus, Dengue, Influenza B, HCV}, 5/12 for `plddt_strat` with the same set, 4/12 for `sasa_freq_joint` {H3N2, Norovirus, Dengue, Influenza B}, and 4/12 for `plddt_ent_joint` with the same set; the corresponding 3-core counts were 5/3/4/3. Norovirus survived all four P6 nulls under both methods, at the $p = 0.00001$ floor for $n_{\text{perm}} = 100{,}000$. Stouffer global $p = 1.0$ across all summary rows is expected from signed-MCC cancellation across the heterogeneous panel, matching the Wave 1 P1 pattern; the headline is the per-null individual count, not the global p-value. P6 is structurally explicit and partially overlaps with P1 local-burden; it is not independent replication and not a strictness ladder. For H3N2, Norovirus, Dengue, and Influenza B, the signal is not reducible to surface, flexibility, or joint structural-evolutionary clustering; HCV survives marginal P6 nulls but drops in joint nulls, indicating partial reducibility. Norovirus survives both P1 local-burden and all four P6 nulls, giving the strongest robustness claim, and the predeclared "strongly defensible" branch fired. The unified P1+P6 8-null comparative robustness map (192 rows; `results/mutbench/codex_wave3/p6_p1_p6_robustness.csv`; figure `p6_p1_p6_robustness.png`) shows 27/96 3-core and 29/96 4-core cells significant. P6 therefore defends descriptive signal under structurally explicit conditioning; concordance with P1 local-burden is robustness, not independent evidence, and not a deployability claim. The codex Wave 3 plan review verdict was "PROCEED WITH MODIFICATIONS" with 8 edits applied (`paper/dissertation/review/2026-04-30/codex_wave3_plan_review.md`).

Tier 2 features-only LOPO and expanded-target stability (W4). Wave 4 extends the audit programme on the `experiments/waves` branch only: a features-only Tier 2 LOPO was pre-registered to test whether the panel-size limit identified in Waves 1–3 dissolves at $n \approx 30$. *First*, the hscore harmonization audit failed on Dengue:dengue_e, HIV-1:hiv1_gp120, and Norovirus:norovirus_vp1, with $\mathbf{0/3}$ PASS at the predeclared threshold of median Spearman $\rho \geq 0.70$ AND top-10 Jaccard ≥ 0.50. Branch 5 therefore fired, and the analysis used the 2-core fallback `freq, entropy`. *Second*, labeled-only 12-fold LOPO gave mean 2-core MCC $= 0.077$ (sd 0.136), with $8/12$ positive folds; the paired Wilcoxon H_1: 2-core $>$ recomputed 4-core gave $p = 0.368$, so the fallback was bounded positive but statistically indistinguishable from the recomputed 4-core comparator at this sample size. The D1 callability recheck was $\mathbf{0/12}$, matching the Wave 2 P5 failure mode in Paragraph 9; the rate-limiting gate is not relaxed by the fallback. *Third*, expanded 28-target stability was $1/28$ stable, with union quorum 16 and `quorum_pass = False`; sign-fitting bootstrap agreement with the full-sample sign was 81.6% for `freq` and 94.4% for `entropy`. This is sign-fitting instability of the 2-core under pathogen-resampling, not evidence that the 16 new targets are biologically poor. Conversely, $\mathbf{16/16}$ unlabeled targets fell inside the labeled feature envelope, establishing feature-space compatibility for Layer A curation as a feasibility claim only. The predeclared interpretive branch is therefore closest to Branch 2: features-only expansion is too schema-shifted to dissolve the panel-size limit by itself, and Layer A curation remains the rate-limiting next step; the Wave 5 pilot 3 ordering for YFV/Lassa/WNV is preserved in Section 4.7. Scripts and artefacts are `scripts/codex_w4_features_lopo.py`, `scripts/codex_w4_task4_summary.py`, `results/mutbench/codex_wave4/w4_labeled_lopo_sum` `results/mutbench/codex_wave4/w4_expanded_stability_per_target.csv`, `results/mutbench` and `paper/dissertation/review/2026-04-30/wave4/w4_tier2_features_lopo.md`.

Hierarchical Bayesian adaptive-weighting robustness check. A pre-registered Cycle-7 robustness experiment tests whether a Bayesian shrinkage model with hyperprior-learned shrinkage strength can outperform the fixed EqualWeight 4-core baseline under the same 12-pathogen nested-LOPO protocol (script: `scripts/hbfws_adaptive.py`; CSV: `results/mutbench/hbfw` The model is $y_{i,p} \sim \text{Bernoulli}(\sigma(\mu + \gamma_{F(p)} + \sum_f w_f \cdot z(x_{i,f})))$ with $w_f \sim \mathcal{N}(0, \sigma_{\text{feat}})$, family random intercept $\gamma_F \sim \mathcal{N}(0, \sigma_{\text{fam}})$ over 8 ICTV families (5 singletons), and hyperpriors $\sigma_{\text{feat}}, \sigma_{\text{fam}} \sim \text{HalfNormal}(0.3)$ that learn pooling strength from data. Sampling used 4 chains × 1,000 draws + 1,000 tune via PyMC 5.28 (NUTS, target accept 0.95). The pre-registered prior was $P(\text{HBFWS} > \text{EqualWeight-4core}) \in [0.15, 0.25]$ given $n = 11$ pathogens with 5/8 singleton families (no pooling partner ⇒ shrinkage to global mean by James–Stein construction). Across the 12 folds, all chains converged ($\hat{R}_{\text{max}} \leq 1.008$, $\text{ESS}_{\text{bulk,min}} \geq 743$), and the result was: HBFWS mean MCC $= 0.058$ (sd 0.118) versus EqualWeight-4core mean MCC $= 0.078$ (sd 0.147); paired $\Delta = -0.020$, paired Wilcoxon one-sided greater $W = 16$, $p = 0.78$. Stratifying by family membership confirmed the design intuition: the 7 paired-family folds (Coronaviridae, Orthomyxoviridae, Flaviviridae) gave $\Delta = -0.014$ and the 5 singleton folds (Caliciviridae, Pneumoviridae, Retroviridae, Rhabdoviridae, Picornaviridae) gave $\Delta = -0.028$, with three singleton folds collapsing to exactly $\Delta = 0$ as the partial-pooling family intercept reduced to the global mean. The pre-registered hypothesis was therefore rejected: a hierarchical Bayesian shrinkage model with explicit hyperprior-learned pooling strength does *not* beat the fixed-weight EqualWeight-4core baseline at $n = 11$ pathogens with 5/8 singleton families. The HBFWS negative result is consistent with the LOPO 0/11 + Friedman $p = 0.990$ + 0.265 oracle-vs-generalised gap evidence already in this dissertation: the f that maps pathogen features to optimal scoring weights is not yet learnable from the current panel, and the search-space-reduction Tier 1 recipe (Paragraph 3.5.2) remains the recommended cold-start operationalisation.

Cycle 7B: comprehensive comparison of six additional adaptive-weighting paradigms. To rule out the possibility that HBFWS-specific design choices (Bayesian shrinkage, ICTV family pooling) drove the negative result, six structurally different adaptive-weighting methods were evaluated against the same EqualWeight-4core baseline under the identical 12-pathogen nested-LOPO protocol (script: `scripts/adaptive_weighting_comparison.py`; CSV: `results/mutbench/adaptive_weighting_comparison.csv`). The six paradigms span low-variance linear baselines, high-capacity tree ensembles, classifier-driven algorithm selection, alternative phylogenetic pooling, and rank-aggregation methods, summarised in Table 3.16.

Table 3.16: Cycle 7B: six adaptive-weighting paradigms vs. EqualWeight-4core baseline (12-pathogen nested-LOPO; same threshold and folds as Table 3.13). All six fail to beat the baseline at any conventional significance level; high-capacity learners (XGBoost, Random Forest) are substantially worse than the baseline because of cross-pathogen overfitting. Paired Wilcoxon H_1: method $>$ EqualWeight-4core, one-sided.

Method	**Mean MCC**	**Δ vs. EQ-4**	**Wilcoxon p**	**Paradigm**
EqualWeight-4core (baseline)	+0.078	—	—	Fixed weights
EqualWeight-10feat	+0.083	+0.006	0.29	Fixed weights
HBFWS (v171)	+0.058	−0.020	0.78	Bayesian shrinkage (ICTV)
XGBoost stacked	+0.016	−0.062	0.97	Gradient boosting
Random Forest stacked	+0.019	−0.059	0.91	Tree ensemble
Logistic Regression L1	+0.056	−0.022	0.70	Sparse linear
Algorithm Selection (1-NN)	+0.066	−0.012	0.69	Pathogen-meta classifier
HBFWS-phylo (3 groups)	+0.063	−0.015	0.84	Bayesian shrinkage (alt. group)
Rank Aggregation (top-3)	+0.062	−0.016	0.56	Training-rank ensemble

Algorithm Selection uses pathogen meta-features (ssRNA polarity, envelope status, genome size, surface-protein type) and 1-nearest-neighbour to predict the best single feature per held-out pathogen; HBFWS-phylo replaces ICTV families with 3 phylo-groups (ssRNA(+) enveloped GP / ssRNA(-) enveloped GP / non-enveloped capsid); Rank Aggregation averages per-feature MCC ranks across training pathogens and uses the top-3 features as a uniform z-ensemble. None of the six methods rejects H_0: method $\leq$ EqualWeight-4core (smallest $p = 0.56$, Rank Aggregation). The HBFWS-phylo, Algorithm Selection, and Rank Aggregation methods are within 0.012–0.016 MCC of the baseline (statistically indistinguishable), while XGBoost and Random Forest underperform by -0.06 MCC because their high capacity overfits the small cross-pathogen sample.

The Cycle 7B comparison gives this dissertation seven independent failure modes for adaptive weighting at $n = 11$ pathogens (HBFWS, XGBoost, Random Forest, Logistic L1, Algorithm Selection, HBFWS-phylo, Rank Aggregation). The convergence of these failures across paradigms—linear and non-linear, parametric and non-parametric, weight-learning and discrete-selection, ICTV-pooled and phylo-pooled—rules out a method-specific cause and isolates the constraint to the panel size itself. Combined with LOPO 0/11, Friedman $p = 0.990$, and the 0.265 oracle-vs-generalised gap, the evidence is now sufficient to assert: *at the current 11-pathogen scale, no adaptive-weighting paradigm tested in this dissertation provides a statistically detectable advantage over the fixed-weight EqualWeight-4core recipe*, and panel expansion to 20–30+ pathogens is the principal lever for any future improvement (Chapter 4, Section 4.7). The single-fold positive case for Algorithm Selection on EV-A71 (homoplasy selected by 1-NN; held-out MCC $= +0.146$ vs. baseline $+0.013$) shows that a learnable pathogen-feature signal exists in principle; it is just statistically underpowered with $n = 11$.

3.5.3 Ground Truth Heterogeneity

A potential confound in interpreting the pathogen-dependent performance patterns is that the Layer A ground truth definitions may themselves differ across pathogens. To assess this, each Layer A position was tagged with its biological definition type (immune_escape, functional, or convergent_evolution), and the composition was analyzed across the same 12-pathogen feature-analysis scope used in Section 3.5.2 (the core 11 pathogens plus Zika). Table 3.17 presents the results.

Despite homogeneous ground truth composition (immune_escape $\geq$ 71% for all 12 pathogens), 5 different best features emerge, and 10/12 pathogens retain the same best fea-

Table 3.17: Ground truth heterogeneity across 12 pathogens (11 main + Zika, matching Section 3.5.2). Best discriminative feature differs across pathogens despite homogeneous immune_escape composition. "Consistent" indicates feature stability when restricted to immune_escape-only. HIV-1 $n_{\mathrm{A}} = 26$ is the operational alignment-QC count (vs. literature-curated 23 in Table 2.3; see Section 4.6).

Pathogen	n_{A}	**% immune**	**Best feature (all)**	**AUC**	**Consistent**
SARS-CoV-2	25	88%	esm2_llr	0.586	Yes
H3N2	25	100%	plddt	0.787	Yes
Norovirus	15	100%	homoplasy	0.944	Yes
HIV-1	26	100%	entropy	0.730	Yes
Dengue	24	71%	homoplasy	0.770	Yes
RSV	20	100%	plddt	0.561	Yes
Influenza B	17	100%	homoplasy	0.805	Yes
MERS	9	78%	esm2_llr	0.695	No
HCV	54	100%	dnds_proxy	0.680	Yes
Zika	27	93%	homoplasy	0.581	Yes
Rabies	39	97%	plddt	0.650	Yes
EV-A71	27	93%	esm2_llr	0.596	No

ture when restricting to immune_escape-only positions ("Consistent = Yes"). This argues against ground-truth composition being the sole explanation for the ANOVA interaction effect ($\omega^2 = 0.296$), while leaving room for the broader limitations discussed in Chapter 4: pathogen biology, curation protocol, auxiliary input quality, and their interaction can all contribute to the observed method–pathogen dependence.

3.5.4 Practical Validation: Search Space Reduction

The analysis results above demonstrate that multiple information types contribute to hotspot prediction, but the most informative feature differs by pathogen. This subsection evaluates whether integrating all 10 features reduces the experimental search space for hotspot validation.

The integration method uses equal-weight combination, instantiated in two complementary protocols. The *production* EqualWeight column underlying the headline scoring $\times$ pathogen ANOVA (`scripts/run_stage3_full_detectors.py`) applies pre-defined domain orientations (e.g., the inverse-pLDDT transform $\mathrm{pLDDT}_{\mathrm{max}} - \mathrm{pLDDT}$ rather than raw pLDDT) followed by uniform $1/10$ averaging over min–max normalised features, with no per-feature sign fit from labels at this stage. The *nested-LOPO* ablation protocol used in Section 3.5.2 (`scripts/feature_ablation_nested_lopo.py`; Table 3.13) additionally fits a directional sign $\hat{s}_f \in \{-1, +1\}$ on the eleven training pathogens excluding the held-out pathogen and applies within-fold z-scoring; the held-out fold combination $s_i = \sum_{f=1}^{10} (1/10) \cdot \mathrm{sign}(\hat{s}_f) \cdot z_{i,f}$ shown above corresponds to this nested-LOPO protocol. The two protocols agree to within $\leq$ 0.013 MCC mean (production in-sample: 0.083; nested-LOPO: 0.070), so the directional sign step is not load-bearing for the headline cross-pathogen claim. Positions exceeding a percentile threshold are called as hotspots. Equal weighting was chosen because, with only 11 reference pathogens, adaptive weighting strategies introduce more noise than signal (see below).

Leave-One-Pathogen-Out (LOPO) cross-validation was conducted across the same 12-pathogen feature-ablation evaluation set used in Section 3.5.2 (core 11 + Zika), comparing EqualWeight against frequency-only (FreqThresh) and random baselines. Under the EqualWeight top-10% threshold, mean MCC across the 12-pathogen set is 0.083, outperforming FreqThresh (0.075) and random ($\approx$ 0.000); restricting to the core 11 pathogens shifts the baseline to 0.093 (consistent with the per-pathogen counts reported in Section 3.5.2, where 9 of 11 core pathogens carry positive EqualWeight MCC under the top-10% threshold and 8 of 11 do for FreqThresh).

Vaccine escape position validation. To assess practical relevance, the 20 scoring $\times$ 39 detector combinations were evaluated against experimentally confirmed vaccine/antibody escape positions for 3 pathogens: H3N2 (29 antigenic drift sites in HA sites A and B), SARS-CoV-2 (30 Spike escape sites in RBD, NTD, and S2), and HIV-1 (45 Env gp120 antibody escape sites). A total of 2,340 enrichment evaluations were performed (Table 3.18). Enrichment is defined as

$$E = \frac{\text{observed overlap}}{\text{expected overlap}} = \frac{|D \cap A|}{|D|\,|A|/L},$$

where D is the set of detected hotspot positions, A is the set of annotated escape positions, and L is the protein length. Significance is tested by Fisher's exact test on the 2×2 table $\{|D \cap A|,\ |D \setminus A|,\ |A \setminus D|,\ L - |D \cup A|\}$. These three pathogens were selected on the single criterion of availability of curated antibody-escape annotations at the time of analysis; the remaining 8 pathogens are not included because comparable escape-position databases were not available.

Key findings: The load-bearing external anchor across this dissertation is the single-scoring HIV-1 7.19× enrichment row (82% Layer A-disjoint escape set, novel-only $p_{\text{adj}} \approx 3.9 \times 10^{-9}$); the EqualWeight integration block (H3N2, SARS-CoV-2, HIV-1 sub-rows) is reported as a complementary multi-source check rather than a unified primary result, since the sub-blocks use different pipelines (fixed top-5% for H3N2/SARS-CoV-2 vs. broader detector-grid best for HIV-1; provenance note in caption). All three pathogens achieve nominally significant vaccine escape enrichment ($p < 0.05$) with the optimal scoring–detection combination; after Bonferroni correction across 780 combinations, H3N2 and HIV-1 survive while SARS-CoV-2 is exploratory only. The best scoring *category* for escape enrichment agrees with the Layer A benchmark results at the category level (Phylogenetic for H3N2, Frequency for HIV-1), though the top scoring *type* within that category can differ between the two evaluations (HIV-1 Layer A best is *entropy* with Wavelet(t=2.0), MCC 0.275; HIV-1 escape best is *freq* with Wavelet(t=1.5), enrichment 7.19×—both frequency-derived and

Table 3.18: Best vaccine escape enrichment per pathogen (20 scoring $\times$ 39 detectors, Fisher's exact test). Enrichment is the ratio of observed to expected overlap as defined above. Only the top scoring type per pathogen is shown; all single-scoring results achieve nominal $p < 0.05$; EqualWeight results vary.

Pathogen	Best scoring	Category	Detection	Enrichment	Overlap	p-value	
H3N2	FUBAR Bayes factor	Phylogenetic	Wavelet(t=2.0)	**9.36**×	9/29	< 0.0001	$***$
HIV-1	freq	Frequency	Wavelet(t=1.5)	**7.19**×	23/45	< 0.0001	$***$
SARS-CoV-2	FUBAR pos. sel.	Phylogenetic	FreqThresh(p=85)	**7.63**×	2/30	0.026	$*$
Multi-source integration — fixed top-5% protocol (`escape_validation_adapt.csv`; SARS-CoV-2 / H3N2):							
H3N2	EqualWeight	Composite	top-5% (fixed)	4.012×	6/29	**0.0023**	$**$
SARS-CoV-2	EqualWeight	Composite	top-5% (fixed)	2.67×	2/30	0.171	n.s.
Multi-source integration — best-detector grid (`vaccine_escape_stage3.csv`; HIV-1 not yet in fixed-threshold pipeline):							
HIV-1	EqualWeight	Composite	CUSUM(d=0.5, h=5.0)	4.17×	—	0.0037	$**$

$***$: $p < 0.001$; $**$: $p < 0.01$; $*$: $p < 0.05$; n.s.: not significant. Best-combination enrichment is selected from 780 combinations per pathogen; Bonferroni correction gives H3N2 $p_{\text{adj}} \approx 2.5 \times 10^{-5}$, HIV-1 $p_{\text{adj}} \approx 2.5 \times 10^{-16}$ (rounded; the un-rounded values 2.63×10^{-5} and 2.47×10^{-16} are reported in `vaccine_escape_stage3.csv` and reproduce the same tier assignment), and SARS-CoV-2 $p_{\text{adj}} \to 1.00$. **Provenance note for the EqualWeight integration block:** the SARS-CoV-2 and H3N2 EqualWeight rows are produced by the fixed top-5% threshold pipeline in `scripts/validate_escape_adapt.py` ($\to$ `escape_validation_adapt.csv`). The HIV-1 EqualWeight row is taken from the broader 20×39 detector-grid pipeline (`scripts/run_vaccine_escape_stage3.py` $\to$ `vaccine_escape_stage3.csv`) using the best-detector EqualWeight combination because HIV-1 was not yet integrated into the fixed top-5% pipeline at the time of these results; the two protocols are therefore reported as separate sub-blocks rather than as a unified comparison. Because escape lists partly overlap Layer A, practical validation is strongest for HIV-1 (37/45 escape positions Layer-A-disjoint; novel-only enrichment 7.16–8.24×, worst-case Fisher $p = 5.0 \times 10^{-12}$, which yields novel-only $p_{\text{adj}} = 5.0 \times 10^{-12} \times 780 \approx 3.9 \times 10^{-9}$ under the same 780-combination Bonferroni denominator [denominator $780 = 20$ scoring $\times$ 39 detector variants shared with the full-overlap test ($p_{\text{adj}} = 2.47 \times 10^{-16}$); the two p-values are different statistical objects — the full-set test is computed on all 45 escape positions, the novel-only test on the Layer-A-disjoint 37-position subset — and share the same correction family rather than the same hypothesis]), while H3N2 is mainly a self-consistency check (69% Layer A overlap; novel-only $p = 0.132$) and SARS-CoV-2 is exploratory. Top-1 enrichments should be read as upper-envelope estimates; Bonferroni-survivor means are lower (H3N2 $\approx 4.7\times$, HIV-1 ≈ 4–$5\times$). Tier assignments are unchanged under $\alpha = 0.01$ or an 8,580-test cross-pathogen denominator.

correlated at $\rho \approx 0.97$, Section 3.5). The strength of this concordance as *external* evidence is pathogen-dependent: HIV-1 provides the cleanest external validation (82% of its escape set lies outside Layer A; see circularity audit below), whereas for H3N2 the 69% Layer A/escape overlap means that the concordance is largely a self-consistency check rather than independent replication.

For H3N2, FUBAR Bayes factor captures 9 of 29 antigenic drift sites with 9.36-fold enrichment; 8 of those 9 captured positions are already in Layer A. Removing those 8 leaves 1 novel-only capture out of 21 Layer-A-disjoint H3N2 escape positions, corresponding to a non-significant novel-only enrichment (Fisher $p = 0.132$); this is *not* the HIV-1 7.19× external anchor and is reported here only as the H3N2 self-consistency reference. The multi-source EqualWeight integration achieves 4.012-fold enrichment ($p = 0.0023$), capturing positions in both antigenic site A (position 137) and site B (positions 155, 159, 160, 186, 188); within this top-5 integration comparison, frequency alone (FreqThresh top-5 $= \{137, 155, 159, 188\}$) recovers the most-frequent site B positions but additionally misses positions 160 and 186 picked up by the EqualWeight integration. The miss therefore concerns specific within-site B positions, not site B as a whole.

Vaccine-escape validation was feasible for only 3 of 11 pathogens (H3N2, HIV-1, SARS-CoV-2): the remaining 8 were excluded on data-availability grounds prior to enrichment computation, falling into four categories—(i) annotation granularity too coarse for residue-level mapping (Dengue serotype-level, Influenza B HI-distance, Rabies antigenic-domain-level, Norovirus loop-level); (ii) sample size below the $n \geq 15$ threshold (MERS $<$10, EV-A71 $<$10); (iii) confound with the benchmark's Layer A set (RSV palivizumab/nirsevimab escape partially overlaps Site II); (iv) mechanism mismatch (HCV NS3/NS5A/NS5B drug-resistance is direct-acting antiviral resistance, not immune escape). The generalisability of the enrichment finding to the other 8 pathogens therefore remains to be established, and the three validated pathogens should be read as a data-availability convenience sample rather than a balanced subset.

Layer A / escape-list overlap (circularity audit). Because Layer A for H3N2, SARS-CoV-2, and HIV-1 is partly constructed from immune-escape-tagged positions (Chapter 2, `layer_a_tags.csv` with an `immune_escape` evidence tag), the vaccine-escape validation sets used in this section are not fully disjoint from the Layer A ground truth. Concretely, the set overlap between the Layer A immune-escape subset and the vaccine-escape validation list is 20/29 for H3N2 (69% of the escape list is already in Layer A), 18/30 for SARS-CoV-2 (60%), and 8/45 for HIV-1 (18%). A scoring method that excels on Layer A therefore has a built-in advantage on H3N2 and SARS-CoV-2 escape enrichment, and the $9.36\times/7.63\times$ values should be read as a self-consistency check rather than an independent replication on those two pathogens. HIV-1's $7.19\times$ is the least affected by this circularity because 37 of 45 escape positions (82%) are outside Layer A and are genuinely novel relative to the benchmark ground truth. The qualitative conclusion—that benchmark-optimal scoring also best captures escape positions—holds, but the quantitative enrichment values for H3N2 and SARS-CoV-2 must be interpreted with this overlap in mind.

Search space reduction. For a typical viral protein of 500 positions, the optimal scoring–detection combination narrows the candidate list to 20–80 positions with 7–9-fold enrichment for escape sites. This transforms hotspot detection from an exhaustive search into a guided investigation, providing a ranked candidate list that directs limited experimental resources toward the most promising positions.

Chapter 4

Discussion, Limitations, and Future Directions

This chapter interprets the results from Chapter 3, summarises the contributions, lays out the dissertation's limitations and future research directions, and ends with concluding remarks. The multi-source information integration analysis (Section 3.5) identifies a compact 4-feature cold-start core (homoplasy, pLDDT, entropy, frequency) that is operationally equivalent to the full 10-feature ensemble under nested-LOPO ($\Delta = +0.011$ MCC, $p = 0.61$) and shows that integration enables practical search-space reduction; uniform full-feature integration is not uniformly superior to the best single scoring channel on every pathogen, and adaptive per-pathogen weighting is not yet learnable from the current $n = 11$ panel.

4.1 Methodological Validity of MutBench

The wide region-level BCa CIs ($n = 7$ regions; e.g., MutClust-Hybrid $[0.323, 0.635]$) reflect region-dependent recall sensitivity, but paired bootstrap comparisons cancel that variability and confirm method differences at $p < 0.001$. The synthetic–real gap (synthetic MutClust-Hybrid *HS* $= 0.778$ vs. sparse real GISAID 0.383, with 6.9–$8.0\times$ enrichment preserved across both) is a methodological feature: dual evaluation quantifies the distance between theoretical and practical detection.

4.1.1 Independence of Ground Truth and Detection Methods

The synthetic benchmark's ground truth is defined independently of the detection methods (DMS experimental data and convergent evolution literature), ensuring no *algorithmic* circularity between evaluation and detection. A more subtle concern is that Layer A positions (convergent evolution) could correlate with mutation frequency at the *per-position* level, which can make frequency-based scoring appear structurally strong. A per-pathogen audit of the point-biserial correlation between Layer A membership and per-position frequency finds a mean $|\rho| = 0.12$ across the 11 pathogens (range $[-0.06, +0.33]$). This per-position membership-vs-frequency $|\rho| = 0.12$ audit is conceptually distinct from the feature-feature Pearson correlation $\rho \approx 0.97$ between the *frequency* and *entropy* scoring inputs reported in Section 3.5 (specifically the feature–feature correlation analysis in Chapter 3 Section 3.5.1); the former measures circularity between ground truth and a scoring signal, while the latter measures redundancy among scoring inputs themselves. Per pathogen: HCV (+0.33), Influenza B (+0.26) and Norovirus (+0.23) show moderate positive correlations, while SARS-CoV-2 ($\rho \approx 0$), MERS (≈ 0) and EV-A71 (≈ 0) are essentially uncorrelated. The effect is therefore pathogen-specific rather than uniform, and is driven by ground-truth curation style: pathogens whose Layer A uses hypervariable/epochal-variant criteria (HCV, Influenza B, Norovirus) couple more strongly with frequency than pathogens whose Layer A uses phylogenetically-confirmed convergent positions (SARS-CoV-2, H3N2). At the individual-position level, D614G achieved the highest single frequency through a founder event (homoplasy count = 2), while position 476 has moderate frequency but the highest homoplasy count (53); such anecdotes confirm that frequency and convergent-evolution signals are not interchangeable. Readers should therefore interpret frequency scoring's strong benchmark performance as partly structural (frequency $\approx$ Layer A by construction) and partly genuine signal (homoplasy provides the 3-lineage discriminator that mere frequency cannot). *Implication for per-pathogen rankings*: HCV's frequency-best ranking and Norovirus's epochal-variant Layer A coupling with frequency scoring should be read as partly structural rather than purely mechanistic; the per-pathogen Stage 2 rankings tabulated in Chapter 3 (Table 3.5) carry this circularity discount implicitly for these three pathogens (HCV, Influenza B, Norovirus), and any interpretation that relies on their single-feature "best" identity should propagate this caveat. This pathogen-specific circularity is itself a downstream consequence of the Layer A curation heterogeneity discussed in Section 4.4.7. To provide a frequency-independent validation dimension, Layer C (DMS experimental fitness) is used where available (6 of 11 pathogens), and the vaccine escape enrichment analysis (Section 4.4.1) provides partial external cross-validation: HIV-1 (82% of its escape set disjoint from Layer A) is the cleanest external anchor, whereas H3N2's 69% Layer A/escape overlap means that the H3N2 enrichment re-measures benchmark performance more than it provides independent replication. The circularity audit in Chapter 3 Table 3.18 reports novel-only enrichment values that quantify this distinction. The biological validity of detected hotspots was further supported by 3D structural analysis (Section 3.3), which showed 5.92-fold spatial contact enrichment, and by the consistent superiority of domain-specific methods (MutClust variants) over general-purpose alternatives (OPTICS, KDE) in Section 3.1.3. Min-max normalization was used for detector inputs and for the production EqualWeight composite column underlying the scoring $\times$ pathogen ANOVA; z-scoring was used only in the nested-LOPO EqualWeight ablation protocol described in Section 2.2.4. Rank transformations were not evaluated as they can introduce artifacts in sparse data distributions. With benchmark validity established, the temporal limitations of the underlying H-score metric warrant examination.

4.2 Limitations of H-score-Based Detection and Alternatives

H-score saturation in late-pandemic data limits reference-based metrics: in 2024–2025 SARS-CoV-2 Spike data, all 1,259 post-truncation positions have non-zero H-scores with $\sigma = 0.098$, eroding discriminatory power. TC-freq achieves the highest enrichment ($3.43\times$) on convergent-evolution ground truth, indicating that temporal-contrast scoring is a viable direction whose appropriate resolution remains a future challenge. The Jaccard similarity between DMS-escape and convergent-evolution positions is only 0.006 (Section 3.1.2), so the "best" method depends on the ground-truth definition—a Stage 1 finding that foreshadows the pathogen-dependent rankings revealed by the cross-pathogen benchmark.

Homoplasy-based scoring is one alternative motivated by the founder-bias problem ($\rho = -0.876$ between H-score and independent-origin count, Section 3.3): Fitch-parsimony homoplasy counts independent mutational origins, decoupling frequency from selective pressure (D614G receives a homoplasy count of 2 vs. 53 for convergent position 476). Cross-pathogen validation, however, is itself pathogen-dependent: HIV-1 attains AUC $= 0.706$ ($p = 0.0002$), H3N2 a weak trend (0.561, $p = 0.147$), and HCV fails (0.390) because parsimony saturates within HVR1. Homoplasy is therefore unreliable for pathogens with extreme intra-protein diversity, mirroring the main scoring $\times$ pathogen interaction finding ($\omega^2 = 0.296$).

4.3 Cross-Pathogen Generalization

4.3.1 Implications of Cross-Pathogen Generalization

The cross-pathogen benchmark demonstrated that no universally best-performing method exists (Chapter 3), with implications for real-time surveillance systems such as Nextstrain [19]: applying a previously optimized combination to a new pathogen may not generalize. The underlying cause is that pathogen-specific genomic characteristics create fundamentally different signal landscapes—for example, H3N2's uniformly nonzero H-scores neutralize threshold-based seed detection, while MERS's sparse genome favors weak-signal detectors.

A notable finding is the asymmetry between scoring and detection contributions: the detection method family main effect ($\omega^2 = 0.013$) is the smallest variance component, approximately 23 times smaller than the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$). This implies that *the choice of input information* matters far more than *the choice of detection algorithm*. Practically, this means that effort should be directed toward selecting the right information source for each pathogen rather than optimizing detection algorithm parameters.

The pattern is formalisable as Rice's algorithm-selection problem [68] with the No-Free-Lunch theorem [69] as theoretical scaffold: the empirical mapping f from pathogen features to optimal scoring is not yet learnable from $n = 11$ pathogens (LOPO 0/11, Friedman $p = 0.990$, 0.265 oracle-vs-generalised gap are jointly null-consistent), and operationalising f requires substantially larger panels (Section 4.7). The ESM-2 cross-paradigm validation (Section 3.4.4) is consistent: ESM-LLR tops SARS-CoV-2 but is dominated by homoplasy on Norovirus (AUC 0.763 vs. 0.944).

The pathogen-dependence finding is consistent with related work but addresses a different task: EVEREST [61] and ProteinGym [12] target per-variant fitness/escape prediction, while MutBench evaluates per-region hotspot detection. The contribution here is the quantified scoring × pathogen interaction ($\omega^2 = 0.296$ / 0.103 at 20-type / 6-category granularity) and the HIV-1 vaccine-escape cross-check ($7.19\times$, 82% Layer-A-disjoint) as the strongest external anchor.

4.3.2 Upper Bound of Detection Performance

Mean oracle MCC across 11 pathogens is 0.341 (range 0.196–0.660; Table 3.5), which appears low but is explained by three structural factors—position-level exact matching, extreme class imbalance (0.7%–15.9% positive rates), and ground-truth stringency—and is unambiguously non-random (113 standard deviations above the random baseline 0.001, comparable to TFBS prediction F1 $= 0.3$–0.5 [122]). Region-overlap analysis (Table 4.1) shows mean oracle MCC rises from 0.341 to 0.472 under ± 5 position windows (Influenza B reaches 0.894); HCV and Norovirus show decreased MCC under expansion because their oracle detections are already precise.

Despite these ceiling effects, the benchmark results have practical implications for public health surveillance.

4.3.3 Recall Analysis: Which Positions Are Missed?

Mean oracle recall across 11 pathogens is 0.552 (Table 3.10). Three observations bound the interpretation: (i) missed positions are not random—low-recall positions are typically Layer A entries documented in only the minimum 3 lineages or governed by epistatic selection (e.g., 8 of the 25 SARS-CoV-2 Layer A positions are NTD supersite residues where immune evasion operates through conformational change, not direct frequency elevation); (ii) multi-source integration recovers positions missed by a single-feature view (the H3N2

Table 4.1: Region-overlap MCC under three evaluation modes. Upper block (9 pathogens) reproduces directly from `region_overlap_mcc.csv`; lower block (EV-A71, Rabies) is oracle-derived from the archived `stage3_full_results.csv`. Window columns: ± 5, ± 10 position tolerance.

Pathogen	**Exact**	**$\pm$5 window**	**$\pm$10 window**
Reproducible from `region_overlap_mcc.csv` (9 pathogens):			
HCV	0.660	0.588	0.519
H3N2	0.534	0.639	0.548
Norovirus	0.510	0.335	0.275
Influenza B	0.392	0.894	0.924
HIV-1	0.275	0.406	0.343
Dengue	0.274	0.441	0.508
SARS-CoV-2	0.211	0.480	0.497
MERS	0.196	0.470	0.593
RSV	0.196	0.394	0.318
Subset mean (9)	0.361	0.516	0.503
Oracle-derived (no archived fixed-combination CSV row):			
EV-A71	0.260	0.275	0.199
Rabies	0.248	0.265	0.266
Mean (all 11)	**0.341**	**0.472**	**0.454**

EqualWeight ensemble picks up antigenic site B positions 160 and 186 that frequency-only Top-5 misses, Section 3.5.4); and (iii) region-level coverage is substantially higher than position-level recall—under ± 10 position windows mean MCC rises from 0.341 to 0.454 (Influenza B 0.924), so most “missed” positions are detected within 1–2 residues of the true site, sufficient to guide experimental prioritisation. The remaining risk is concentrated in EV-A71 (recall 0.185) and Rabies (0.308), whose short post-truncation alignments (261 / 491 AA) limit statistical power; users applying MutBench to such pathogens should treat detected positions as a prioritised subset rather than an exhaustive list and complement with DMS or epitope mapping.

4.4 Practical Implications and Fundamental Limitations

Returning to the three problems identified in Chapter 1: the evaluation framework, comprising the three-layer ground truth and the hotspot-score metric (Contribution 1), is designed for standardised use, pending external benchmarking by independent teams (with the saturation caveat in Section 4.2); pathogen-dependence is quantified by the interaction effect ($\omega^2 = 0.296$ at the evaluation level, $\omega^2 = 0.234$ at the cell level) and corroborated by LOPO 0/11 (Contribution 2); cross-validation against vaccine-escape positions establishes practical relevance, anchored by HIV-1 (Contribution 3).

4.4.1 Vaccine Escape Position Validation

Vaccine escape validation (Table 3.18) showed enrichment values of 9.36× for H3N2, 7.19× for HIV-1, and 7.63× for SARS-CoV-2 (nominal Fisher's exact p-values). Under Bonferroni correction across 780 combinations, HIV-1 ($p_{\text{adj}} = 2.5 \times 10^{-16}$) and H3N2 ($p_{\text{adj}} = 2.6 \times 10^{-5}$) survive while SARS-CoV-2 does not. The circularity audit in Chapter 3 further distinguishes HIV-1 (82% of escape set disjoint from Layer A; external validation) from H3N2 (69% Layer A overlap; self-consistency check). The 10-feature EqualWeight integration (top-5% threshold; `escape_validation_adapt.csv`) further achieves statistically significant H3N2 escape enrichment (4.012×, $p = 0.0023$) in the integration-vs-single comparison reported in that file (note: single-scoring H3N2 FUBAR survives Bonferroni in Table 3.18; the "single-feature approaches do not" statement here applies specifically to the EqualWeight/FreqThresh/BestSingle top-5 integration block, not to the full 780-combination Stage 2 sweep). The Stage 2 single-feature 7.19× HIV-1 enrichment and the Stage 3 EqualWeight integration 4.012× H3N2 enrichment are not directly comparable as "larger = better" numbers: the former is the per-pathogen winner of a 780-combination exhaustive sweep (winner's-curse-prone, top-of-search selection), while the latter is a fixed 10-feature uniform-weight ensemble at a pre-specified top-5% threshold (winner's-curse-free, no per-pathogen tuning). The smaller integration number is therefore *more reliable* as an externally-comparable estimate, not weaker. The primary practical value lies in *reducing the experimental search space* rather than achieving high absolute prediction accuracy (see Section 3.5.4 for detailed discussion).

While LOPO cross-validation yields 0/11 generalization, MutBench provides practical value through three mechanisms: (1) the family-level scoring recommendation table (Table 4.2 in this chapter) provides coarse-grained starting hypotheses based on virus family characteristics; (2) the 10-feature EqualWeight ensemble (under uniform-weight LOPO top-10% evaluation the 4-feature core retains 97.6% of the 10-feature ensemble's mean MCC—a relative-to-ensemble retention figure within the same protocol, not an absolute floor relative to an oracle; no paired-sample significance test was performed in this within-pathogen demonstration, and the 4-core ranks 13/210 (top 6.2%-tile) under the exhaustive 4-of-10 nested-LOPO subset audit in Table 3.15) achieves 4.012-fold H3N2 vaccine escape enrichment in the integration comparison without requiring pathogen-specific optimization; and (3) once scoring features have been computed for a pathogen, the 20-scoring $\times$ 39-detector evaluation grid can be rerun in approximately 25 seconds on a single workstation, making exhaustive detector comparison operationally feasible. The earlier Stage 1 single-scoring escape analysis is therefore retained only as historical development context; the dissertation-level evidence uses the full 20-scoring benchmark in Table 3.18.

4.4.2 Cross-Validation via Vaccine Escape Enrichment

A critical question for any benchmark is whether its rankings reflect genuine biological relevance or merely circular evaluation artifacts. The vaccine escape enrichment analysis across three pathogens (2,340 evaluations) provides *partial* cross-validation of the Layer A benchmark results, with the strength of external validation differing substantially by pathogen.

Three-tier evidence rule. External value is judged by Bonferroni survival and Layer A independence. HIV-1 passes both criteria (18% Layer A overlap, $p_{\mathrm{adj}} = 2.5 \times 10^{-16}$) and remains the primary external anchor; the direct novel-only test on 37 Layer A-disjoint positions gives 7.16–$8.24\times$ enrichment, with the conservative worst case still significant after 780-fold correction. H3N2 survives Bonferroni correction but has 69% Layer A overlap, so it is treated as a self-consistency check; its novel-only enrichment is not significant ($p = 0.132$). SARS-CoV-2 does not survive correction and is exploratory. Thus the benchmark is not algorithmically circular, but the practical-value claim rests mainly on HIV-1.

Concurrent-benchmark task-orthogonality. ViroGym and EVEREST overlap with MutBench on some pathogens, but they optimize per-variant fitness Spearman whereas MutBench evaluates per-region binary hotspot calls. A fair head-to-head would require task-aligned binarization of fitness predictions, so it remains future work rather than a missing leaderboard. This is especially relevant for pathogens lacking DMS Layer C, where Layer A plus HIV-1-like external escape validation remains the current evidential bridge.

4.4.3 Practical Workflow for New Pathogens

For a novel RNA virus, MutBench's methodology operationalises through four steps. **(1) Sequence collection**: ≥ 200 protein-coding sequences from NCBI GenBank or GISAID with temporal/geographic diversity, aligned with MAFFT [84] ($\sim$5 min for 1,000 sequences). **(2) Feature computation** in order of increasing cost: frequency/entropy from the MSA (seconds), homoplasy via Fitch parsimony on an inferred tree (2–5 min), ESM-2 masked-marginal scores (3–5 min/protein, GPU), and structural features pLDDT/SASA from AlphaFold/ESMFold (omitted as a reduced-feature baseline when no reliable structural model exists). **(3) Scoring category selection**: Table 4.2 provides empirically grounded family-level starting priors; users should treat the recommendation as a first-pass prior rather than a fixed rule. **(4) Detection and evaluation**: KDE or Wavelet-based detection as robust defaults; with pre-computed features the 39-detector grid evaluates in under 3 s per pathogen (the full 8,580-evaluation benchmark in $\sim$25 s). Total per-pathogen runtime is 15–20 min on a single GPU; the 4-feature core (homoplasy, pLDDT, entropy, frequency; nested-LOPO paired delta $+0.011$, $p = 0.61$, Table 3.13) is the cold-start option when labels are unavailable, eliminating PLM inference and bringing runtime under 10 min on CPU (AlphaFold2 itself remains GPU-bound). Live-deployment integration with Nextstrain [19]/GISAID [64] APIs remains future work.

Table 4.2: Starting scoring-category priors by virus family, based on cross-pathogen benchmark results

Virus Family	**Starting Scoring Prior**	**Rationale**
Orthomyxoviridae	Phylogenetic (FUBAR)	Strong diversifying selection signal
Coronaviridae	AI-based (PLM)	Shallow phylogeny; PLM captures broader context
Caliciviridae	MSA-derived (homoplasy)	Epochal evolution with convergent patterns
Retroviridae	MSA-derived (entropy)	High diversity; entropy captures allelic variation
Flaviviridae	Frequency-based	Hypervariable regions dominate signal
Pneumoviridae	AI-based (ESM-2)	Limited phylogenetic signal

The two operating regimes are complementary, not a single universal algorithm: Contribution 2 gives the *informed-use ceiling* (pathogen-specific selection explains the $\omega^2 = 0.296$

interaction and the 0.265 oracle-vs-generalised gap), and the 4-feature core gives the *cold-start floor* (statistically indistinguishable from the 10-feature ensemble under nested-LOPO at $p = 0.61$).

4.4.4 Phylogenetic Non-Independence and Scalability Limits

Detection methods implicitly assume sequence independence, but viral MSAs are phylogenetically structured: frequency-based approaches conflate positive selection with founder effects (e.g., D614G [120]), and the dN/dS filter is itself susceptible to phylogenetic bias. Proof-of-concept simulations and TreeTime ML validation (Sections 3.3–3.3) show Fitch-parsimony homoplasy ranks convergent mutations above founder mutations (AUC $= 1.000$); on real GISAID sequences, H-score and homoplasy capture distinct signals ($\rho = -0.107$, $p = 1.36 \times 10^{-4}$). Full integration of phylogenetic correction with temporally aligned sampling is left for future work (Section 4.7).

4.4.5 Representativeness and Sampling Bias of GISAID Data

GISAID [64]-based input carries three systematic biases: **geographic** (high-income countries $> 80\%$ of sequences [123], underrepresenting Africa/South Asia), **temporal** (uneven accumulation across pandemic phases), and **founder-effect-driven hitchhiker** mutations (e.g., JN.1 [124]) that frequency-based approaches cannot distinguish from positively selected positions. Analysis-window dependence further matters: across three 6-month SARS-CoV-2 Spike windows, frequency AUC rises from 0.337 (2020-H1) to 0.472 (2021-H1) and mean Shannon entropy from 0.126 to 1.241 as diversity accumulates, so optimal scoring depends on temporal depth as well as biology. Standardising the analysis window or explicitly modelling its effect is left for future work.

4.4.6 ESM-2 Training Data Overlap

ESM-2 [52] was trained on UniRef50/UniRef90, which contains the 11 target proteins (e.g., SARS-CoV-2 Spike, HIV-1 gp120, H3N2 HA), so masked-marginal scores may partially reflect memorisation rather than pure protein-language understanding, potentially inflating absolute AUC. The semantic-change feature (cosine distance between wild-type and mutant embeddings) is less susceptible because it measures relative shifts rather than absolute likelihoods. The leakage magnitude likely scales with pathogen-specific UniRef coverage (SARS-CoV-2/HIV-1 plausibly more affected than Rabies/EV-A71), so the "uniform leakage" assumption used to argue that relative cross-pathogen comparisons remain valid is itself an unquantified claim; a per-pathogen leakage audit and viral-excluded PLM evaluation are left for future work.

Tranception–ESM-LLR proxy collinearity. The "Tranception" channel in this benchmark is implemented as a lightweight ESM-2 masked-marginal pseudo-perplexity proxy (Chapter 2, Table 2.8). Per-pathogen position-level correlations between this proxy channel and the standalone ESM-2 LLR channel are operationally near-equivalent for 11 of 12 pathogens (Pearson $\rho = 0.643$–0.836, all $p < 10^{-50}$; MERS most collinear at $\rho = 0.836$), with HIV-1 the lone dissociating case ($\rho = 0.043$, n.s.; Spearman 0.158). The headline "SARS-CoV-2 best = Tranception" result (SARS-CoV-2 $\rho = 0.753$) therefore lives within a regime where the two PLM channels are near-equivalent; the SARS-CoV-2 ranking should be read with channel-redundancy uncertainty, not as a clean victory of one PLM family over another. Per-pathogen numerics are archived in `tranception_esm_correlation_per_pathogen.csv`.

Concurrent-benchmark reconciliation: EVEREST v3 reports that PLMs underperform alignment-based methods on viral DMS *fitness regression*, while the present work finds PLMs winning on 2 of 11 pathogens for region-level *detection* (RSV best ESM-2 masked-marginal channel; SARS-CoV-2 best Tranception). The two findings are compatible because the tasks differ structurally (per-variant continuous Spearman vs. per-region binary MCC), and the fact that Norovirus's best is homoplasy (AUC 0.944) rather than a PLM is exactly the pattern leakage-driven inflation would not predict.

Detector-family Simpson's-paradox check: the family main effect ($\omega^2 = 0.013$) is the smallest modeled component, but the family $\times$ pathogen interaction (0.082, medium) shows family choice matters conditioned on pathogen. The within-cell residual (0.39) absorbs the three-way interaction and parameter-variant noise; a fully-resolved four-way ANOVA is deferred. The framing "information matters more than algorithm" compares modeled main effects only and does not imply detection algorithms are irrelevant within a fixed (scoring, pathogen) cell.

Several of the 20 scoring types share an underlying information source (e.g., `freq`, `freq_with_indel`, `rare_freq` are all frequency-derived). Aggregating to the 6-category level shrinks the interaction to $\omega^2_{\text{category}\times\text{pathogen}} = 0.103$ (below Cohen's large 0.14 but above medium 0.06, comparable to the pathogen main effect 0.117). The 6-category value is the collinearity-robust lower bound and the 20-type value is the upper bound, depending on how "scoring type" is defined.

A methodological limitation concerns the *winner's curse* [125]: with 780 candidate combinations searched per pathogen, the reported top-1 enrichment is an upper-envelope estimate (Bonferroni-survivor mean is materially lower, e.g., H3N2 $\approx 4.7\times$ vs. top $9.36\times$). The probability that 11 of 11 pathogens exhibit unique best combinations by chance alone is approximately 82% at the family level ($20 \times 14 = 280$ combinations) and 93% at the parameter-variant level (780). Three factors mitigate this concern: (1) the best combinations involve 9 distinct scoring *types* from 4 of 6 categories (much lower random probability than mere uniqueness); (2) the ANOVA interaction effect ($\omega^2 = 0.296$) quantifies systematic rather than chance variation; (3) vaccine-escape enrichment cross-validation confirms benchmark-optimal methods also achieve the highest enrichment on independently curated escape positions.

The three-way ANOVA treats pathogen as a fixed effect, which is appropriate for inference about these 11 specific pathogens but does not support generalization to all RNA viruses. Bolker et al. [126] recommend ≥5–6 levels per random factor as the practical minimum and ≥20–30 for stable random-effects estimation, placing the 11-pathogen panel in the small-cluster regime where the percentile cluster bootstrap is anti-conservative. To triangulate the headline against this small-n regime, the scoring $\times$ pathogen interaction was re-estimated under five inferential frames: the standard pathogen-level cluster bootstrap (95% percentile interval $[0.195, 0.333]$), a wild-cluster (Rademacher) bootstrap [115] ($[0.201, 0.303]$), a phylogenetic block bootstrap at the ICTV-family level ($[0.202, 0.346]$), a frequentist linear mixed-effects model (`statsmodels.MixedLM`; pseudo-$\omega^2_{\text{int}} = 0.340$, Self–Liang boundary-corrected LRT $\chi^2_1 = 913.7$, $p < 10^{-198}$, $\text{ICC}_{\text{pathogen}} = 0.135$), and a Bayesian two-way random-effects factor analysis (PyMC 5.28, non-centered half-Normal hyperpriors, $\hat{R} = 1.003$; posterior mean $\omega^2_{\text{int}} = 0.316$, 95% HDI $[0.242, 0.388]$, $\Pr(\omega^2_{\text{int}} > 0.14) = 0.9998$). An earlier exploratory Bayesian hierarchical fit with half-Cauchy priors yielded a posterior mean of 0.252 (HDI $[0.188, 0.314]$, Pr $> 0.14 = 0.996$); both Bayesian runs concur on the load-bearing posterior probability and the qualitative conclusion. All five lower bounds—0.195, 0.201, 0.202 from the frequentist bootstraps, 0.242 from the Bayesian factor analysis, with the partial-pooling lower bound at 0.188 and the variance-component point at 0.340—sit above Cohen's large-effect threshold ($\omega^2 > 0.14$), so the conclusion that the scoring $\times$ pathogen interaction is large is invariant to the choice of inferential paradigm. Full per-frame schemas, priors, sampler settings, and archived CSVs are tabulated in Chapter 3, Section 3.4.4, Table 3.8. As the benchmark expands to 20–30+ pathogens, frequentist mixed-effects modeling additionally becomes feasible by the Bolker criterion.

Independent reproduction. The 8,580-evaluation benchmark is a single-team, single-codebase, single-GISAID-snapshot estimate (random seed $= 42$; Section 4.8). Headline values ($\omega^2 = 0.296$ evaluation-level / 0.234 cell-level, oracle MCC mean 0.341, HIV-1 $7.19\times$ enrichment $p_{\text{adj}} = 2.5 \times 10^{-16}$) should be discounted on a claim-by-claim basis until external reproduction; relative orderings (family-level scoring priors, HIV-1 as cleanest external anchor) are more robust to single-team idiosyncrasies than absolute magnitudes. Independent multi-tool reproduction remains for future work (Priority 5; analogue of the PanCancer effort [16]).

Technical-covariate sensitivity. Per-pathogen oracle MCC correlates with two technical covariates not modeled in the three-way ANOVA—protein length ($\rho = -0.73$, $p = 0.011$) and Layer A positive rate ($\rho = +0.64$, $p = 0.035$)—raising the question of whether $\omega^2_{\text{scoring}\times\text{pathogen}}$ partly reflects structural heterogeneity rather than intrinsic biological signal. An ANCOVA fit on the cell-level grid (statsmodels Type II OLS with log(length) and Layer A positive rate as continuous covariates; methods in Chapter 2 Section 2.2.5) confirms the directional prediction: the pathogen main effect collapses from $\omega^2 = 0.108$ to essentially zero, while the scoring $\times$ pathogen interaction *rises* from cell-level 0.234 to 0.264 after adjustment ($p < 10^{-167}$). The interaction is therefore amplified, not attenuated, by technical-covariate adjustment.

Layer-A curation-protocol subset analysis. A second concern raised by the curation heterogeneity (Section 4.4.7) is that the interaction might be inflated by the most idiosyncratic Layer A definitions (HCV's HVR diversifying selection and Norovirus's epochal-variant positions). Re-fitting the cell-level ANOVA on three progressively restrictive subsets gives $\omega^2_{\text{scoring}\times\text{pathogen}} = 0.199$ ($n = 10$, HCV excluded), 0.187 ($n = 9$, HCV+Norovirus excluded), and 0.137 ($n = 4$, phylogenetically-derived Layer A only: SARS-CoV-2, H3N2, MERS, EV-A71); all three remain above Cohen's medium threshold despite the loss of power. The pathogen-adaptive conclusion should therefore be read as holding *within the 20-type/39-variant design and the 11-pathogen sample*, with independent validation on additional pathogens with matched length/positive-rate as a priority for follow-up work.

Scoring input heterogeneity. The 20 scoring types do not operate under identical conditions: frequency/entropy depend only on the MSA, but phylogenetic scores (FUBAR, dN/dS, homoplasy) inherit tree-quality variation (H3N2's multi-decade phylogeny vs. MERS's sparse 5-year tree), structural scores (pLDDT, SASA) inherit prediction-accuracy variation, and AI-based scores inherit training-data biases. The scoring $\times$ pathogen interaction therefore captures both genuine biological signal differences and input quality differences, which cannot be fully separated without standardised auxiliary inputs. A mitigating factor: most detection methods (FreqThresh, AdaptiveKnee, KDE, SlidingTest, SWAN) employ internal scale normalisation, rendering the detection step largely scale-invariant; only Bayes and ScoreDBSCAN are raw-scale-sensitive among the 14 families.

4.4.7 Ground Truth Heterogeneity Across Pathogens

Layer A is constructed differently across pathogens (Table 2.3): SARS-CoV-2 and H3N2 use phylogenetically confirmed convergent-evolution sites, HCV uses HVR1/HVR2 (diversifying rather than convergent selection), Norovirus uses epochal-variant positions, and the remaining pathogens use literature-reported immune-evasion sites with varying phylogenetic confirmation. This heterogeneity could inflate the scoring $\times$ pathogen interaction, but the same heterogeneity reflects the real-world challenge: no single criterion applies across all pathogens, and the qualitative pathogen-dependence finding holds whether the source is intrinsic biology, curation protocol, or their interaction. MCC values are therefore not directly comparable across pathogens in absolute terms; the interaction effect should be read as systematic variation in *relative* method rankings, with the 6-category aggregation ($\omega^2_{\text{cat}\times\text{path}} = 0.103$) providing additional robustness. HCV is the one pathogen with effectively empty Layer B (constrained regions could not be mapped within the alignment length), so HCV evaluation relies solely on Layer A; experimental E2 structures or cross-Hepacivirus conservation could close this gap in future work.

Sub-lineage and sub-genotype stratification (intentional pooling). Four pathogens carry sub-lineage structure that the headline benchmark pools rather than stratifies: (i) Influenza B mixes Victoria and Yamagata, so its frequency-based best is partly carried by lineage-defining substitutions; (ii) SARS-CoV-2 aggregates pre-Omicron and Omicron eras into a single per-position estimate; (iii) HIV-1 pools group M and the smaller group O contribution and does not stratify intra-M subtype A/B/C; and (iv) HCV pools the seven genotypes around the genotype-1a reference `NC_004102.1`. In each case a per-sub-lineage re-run is reserved as a sensitivity analysis but is not part of the headline; per-pathogen winners should be read as "best on the pooled MSA", not "best on any single sub-lineage".

4.4.8 Scope Justification: 11 RNA Viruses

The 11 pathogens cover 8 major RNA virus families (Coronaviridae, Orthomyxoviridae, Retroviridae, Pneumoviridae, Flaviviridae, Caliciviridae, Rhabdoviridae, Picornaviridae), representing the primary taxonomic diversity of pandemic-potential RNA viruses. The RNA-virus scope is intentional: H-score-based scoring requires the high mutation density characteristic of RNA viruses (10^{-3}–10^{-5} substitutions/site/year), whereas DNA viruses (10^{-6}–10^{-8}) require fundamentally different scoring approaches.

An empirical sanity check at $n = 1$ DNA virus—HBV polymerase, $n = 708$ unique sequences, MAFFT –auto $L = 882$—yields a 2.6–$13\times$ density gap (variable-position fraction 0.686 vs. 0.996–1.000 for three RNA-virus comparators; mean Shannon entropy 0.236 vs. 0.61–3.15 bits) consistent with the $\sim 10^3\times$ rate gap, and a 14-position drug-resistance Layer A entropy test that does not reach significance (permutation $p = 0.354$ global, $p = 0.136$ RT-domain restricted) (provenance: `data/dna_virus_pilot/`).

Three caveats on the small-n regime are explicit. First, the Friedman result ($p = 0.990$, $n = 11$ blocks) is uninformative rather than a strict null: the Nemenyi pairwise post-hoc gives $0/190$ top-20 pairs significant at raw $\alpha = 0.05$ with the entire pairwise p-distribution in $[0.989, 1.000]$ (Section 3.4.3); pathogen-specific optimality remains the parsimonious reading given the $\omega^2 = 0.296/0.234$ evaluation/cell-level interaction and the 0.265 oracle-vs-generalised gap. Second, sampling-quality confound is not zero (Rabies / EV-A71 attain the lowest oracle MCCs $0.248/0.260$), but the technical-covariate ANCOVA (Section 2.2.5) shows that replacing the categorical pathogen main effect with $\log(\text{alignment length}) +$ Layer A positive rate *amplifies* rather than attenuates the load-bearing interaction (scoring $\times$ pathogen $\omega^2 = 0.234 \to 0.274$, family $\times$ pathogen $0.080 \to 0.088$); excluding the two smallest panels likewise shifts the headline interaction by less than the cluster-bootstrap CI half-width. Third, three pathogen pairs share viral families (SARS-CoV-2/MERS, H3N2/Influenza B, HCV/Dengue), but within each pair the best scoring types differ (FUBAR/freq, freq/entropy_with_indel, Tranception/freq), indicating phylogenetic relatedness does not determine method preference and the within-family heterogeneity strengthens the pathogen-adaptive finding rather than weakening it.

4.4.9 PLM and Data Limitations

The AI-based scoring types (ESM-2 masked-marginal, Tranception proxy, semantic change, stability) use pre-trained models without virus-specific fine-tuning. CoVFit [121] demonstrates that pathogen-specific PLM adaptation substantially improves fitness prediction—consistent with the central pathogen-dependence finding—but was not evaluated here. Structure-conditioned predictors (stability_structural) inherit the experimental-vs-predicted-structure quality variation, plausibly inflating structural feature importance for SARS-CoV-2 Spike (experimental PDB) relative to pathogens relying on AlphaFold/ESMFold. NCBI GenBank sequences were downloaded without controlling for geographic or temporal sampling bias (high-income-country dominance in 2020–2022 SARS-CoV-2, sparse temporal sampling for Rabies/EV-A71); Layer A's multi-lineage independent-emergence requirement partially mitigates this, but Layer A position counts themselves may be biased toward well-studied pathogens.

4.4.10 Prospective and Time-Forward Validation Absence

This dissertation's headline claims (enrichments, MCC values, ANOVA effect sizes) are retrospective summaries against a single GenBank/GISAID snapshot anchored at December 2024 with literature-derived Layer A/B ground truth. The temporal-window analysis on SARS-CoV-2 Spike (frequency AUC $0.337 \rightarrow 0.472$ across 2020-H1 to 2021-H1) is itself retrospective; LOPO, Friedman, and the ANCOVA/Bayesian robustness checks are all retrospective on the frozen snapshot.

A forward-time holdout (training restricted to pre-T_0 sequences, test on post-T_0 emergent positions) is implemented as a multi-pathogen sliding-window prospective backtest across 73 windows (Section 3.2.2). The freq+entropy ensemble (time-varying part of the 4-feature core) attains AUROC > 0.75 only on EV-A71 (0.77) and Rabies (0.79), partially on Influenza B (0.70), and clusters near chance for the remaining eight (grand mean 0.539, enrichment $1.57\times$). The retrospective MCC headline (0.341 exact / 0.454 at ± 10 window) therefore does not transfer cleanly to forward time for most pathogens. Adding the static 4-feature-core members (homoplasy, pLDDT) via Needleman–Wunsch projection onto the codon-alignment frame helps in EV-A71 (MCC $0.146 \rightarrow 0.188$ with homoplasy) and Rabies (AUROC $0.789 \rightarrow 0.860$) but is destructive for HCV (AUROC $0.31 \rightarrow 0.09$ with homoplasy because the HCV E2 homoplasy distribution is essentially flat); paired Wilcoxon vs. the 2-feature baseline gives $p < 0.001$ for +homoplasy (ΔMCC $= +0.005$), $p = 0.17$ for +pLDDT, $p = 0.016$ for the 4-feature ensemble ($\Delta = -0.009$). The "newly emerging" definition is a within-MSA reference rather than a literature-curated forward ground truth; a frozen earlier-date Layer A re-run is left as future work. With those caveats, the operational claim that MutBench can prioritise emerging hotspots is supported by direct forward-time evidence on EV-A71 and Rabies, partially on Influenza B, and remains weak for the other eight pathogens.

4.4.11 Dual Use Research of Concern (DURC) and Ethics Review

MutBench predicts positions in viral surface glycoproteins likely to mutate under selection, including immune-escape sites. Classified against the seven U.S. NIH/HHS DURC categories, none applies in the strict policy sense—no new variant is engineered, no transmissibility/virulence/imm
evasion/host-range/therapeutic-resistance/diagnostic-evasion/weaponisation construct is proposed—

because the benchmark is purely retrospective on published sequence and DMS data, and outputs are region-level position rankings rather than engineering recipes (the same framing applies under HHS P3CO and Select Agent). Three operational mitigations are retained in the release plan: (i) no raw GISAID FASTA is redistributed (only computed scores and summaries); (ii) the release publishes region-level hotspot rankings rather than variant-level instructions; (iii) any future extension to P3CO/Select Agent or non-RNA pandemic-potential pathogens will undergo dedicated institutional biosafety review prior to publication. In the Korean context the relevant oversight is Kyungpook National University's Institutional Biosafety Committee (IBC); because this dissertation uses only public sequence repositories (GISAID, NCBI) and previously-published DMS data—no live-virus work, no human or animal subjects, no engineered constructs—no formal IBC protocol number was issued, and the work falls under the public-data exemption. Code is released under MIT and the dissertation text under CC-BY-NC 4.0 (Chapter 2, Section 2.2.7).

4.4.12 MSA Construction Artifact (MAFFT –auto Mode)

All 11 pathogens were aligned with MAFFT `--auto` (Chapter 2), so the algorithm actually selected (FFT-NS-2 / FFT-NS-i / L-INS-i / G-INS-i) is partially confounded with the pathogen factor; a single-mode (e.g., L-INS-i) sensitivity re-run was not performed and remains an unmeasured component of the input-quality axis (Section 4.4.6). The expected magnitude is small (MAFFT mode mainly affects local accuracy in highly variable regions while Layer A sits in literature-conserved/convergent locations) but unquantified.

4.5 Summary of Research Contributions

This dissertation presents three contributions. **First**, to our knowledge, MutBench is the broadest *region-level, single-position* hotspot-detection benchmark to date for RNA viruses, combining 10 biological information types (20 scoring formulas) $\times$ 14 detection families (39 variants) $\times$ 11 RNA viruses = 8,580 evaluations, a three-layer ground truth (Layer A: a pathogen-specific union of convergent-evolution, immune-escape, and HVR-membership evidence types, with immune-escape positions dominating; Layer B: purifying-selection negatives; Layer C: DMS experimental fitness where available), and the hotspot-score composite metric within a two-stage main experimental design plus a Stage 3 information-integration layer. **Second**, the benchmark shows that the optimal biological information source is pathogen-dependent: the scoring $\times$ pathogen interaction is the largest modeled effect ($\omega^2 = 0.296$ at 20-type granularity, 95% cluster-bootstrap CI [0.195, 0.333]; 0.103 at 6-category granularity), the 11 pathogens have 9 distinct optimal information types, and the oracle-vs-generalized MCC gap is 0.265. Friedman $\chi^2 = 7.69$, $p = 0.990$ (cluster-robust main statistic) shows no universally best combination across pathogens, consistent with the interaction effect; LOPO 0/11 is null-consistent under permutation and is reported as a corroborative consistency check rather than independent evidence. **Third**, the framework yields practical search-space reduction validated on independently-curated vaccine-escape positions. For a typical viral protein the optimal scoring–detection combination narrows the candidate list from $\sim$1,000 to 20–80 positions (top 5–10%) with 7–9$\times$ enrichment for escape sites. External validation is anchored by HIV-1: the escape set is largely Layer A-disjoint (82%) and yields full-set 7.19$\times$ enrichment ($p_{\text{adj}} = 2.5 \times 10^{-16}$) plus a novel-only 7.16–8.24$\times$ Bonferroni-significant subset (full numerics and circularity audit in Chapter 3 Table 3.18);

H3N2 ($9.36\times$) is a self-consistency check (69% Layer A overlap) and SARS-CoV-2 ($7.63\times$) is exploratory after multiple-comparison correction. The H3N2 EqualWeight integration case yields $4.012\times$ enrichment ($p = 0.0023$). A multi-source feature-ablation robustness check (nested-LOPO; 4-feature core paired delta $+0.011$, 95% CI $[-0.018, +0.042]$, $p = 0.61$; Section 3.5.2, Table 3.13) confirms the compact 4-feature core (homoplasy, pLDDT, entropy, frequency) provides a practical cold-start recipe statistically indistinguishable from the full 10-feature EqualWeight ensemble. An exploratory adaptive evidence-fusion redesign was developed alongside the dissertation but is not adopted as a contribution; it remains future work pending a larger pathogen panel.

4.6 Limitations Summary

Table 4.3 summarizes the key limitations of MutBench and their current mitigation status; the per-issue detail is in the corresponding subsections of Sections 4.4.4–4.4.12 above.

The dominant residual risks are therefore not hidden performance failures but bounded scope: limited independent functional labels, phylogenetic/sampling confounding, and insufficient pathogen count for deployable adaptive learning.

Falsification-resistant audits and the boundary of Algorithm 1's claim. Three pre-registered audits (Sections 9, 9, 9) further bound the deployable scope of Algorithm 1 on the current 12-pathogen panel. *First*, the full 1023-subset lattice with formal Shapley decomposition shows the dissertation's fixed 4-core (`freq, entropy, homoplasy, plddt`) ranks $87/1023$ (top 8.5%-tile, lower than the rank reported on the restricted 4-of-10 lattice in Paragraph 9) and is strictly dominated by the 3-feature variant (`freq, entropy, homoplasy`) (MCC 0.0866 vs. 0.0809, $\Delta = +0.0058$, lattice rank $27/1023$); pLDDT con-

Table 4.3: Summary of MutBench limitations and mitigation status

Limitation	Severity	Current mitigation	Future work
GT/DMS incompleteness	High	3-layer GT; DMS for 6/11; HIV-1 external anchor	More DMS and Layer A-disjoint escape sets
Phylogenetic and sampling bias	High	Homoplasy PoC; temporal stratification; GISAID caveat	Full tree/time/geography correction
Metric stringency and comparability	Medium	Exact MCC paired with ±10, P@20, FPR, coverage	Region/prevalence-adjusted metrics
Small pathogen panel for adaptation	Medium	Oracle gap reported; adaptive variants not adopted	20–30+ pathogens and time-forward validation
Concurrent-benchmark mismatch	Medium	Task orthogonality stated; no model-vs-model leaderboard	Task-aligned subset comparison
Scope constraints	High	RNA viruses and surface glycoproteins only; point substitutions only; coding region only (5′/3′ UTR and non-coding selection out of scope)	DNA viruses, multi-protein, indel-aware scoring, UTR/non-coding extension

tributes a slightly negative Shapley value (-0.0015) and the ESM-2 LLR contributes the only doubly-negative Shapley (-0.0144 on MCC, -0.158 on enrichment). The Cold-Start 4-core is therefore correctly characterised as a historical fixed configuration, not as a non-arbitrary optimum, and pLDDT is recommended as an optional structural extension. *Second*, per-pathogen rank reliability (1,000-resample bootstrap of top-vs-bottom decile prevalence) is positive in only $5/12$ pathogens for the 4-core (Norovirus, Dengue, H3N2, Influenza B, HIV-1) and the score *inverts* on Rabies (delta -0.123, CI $[-0.245, -0.003]$, Spearman $\rho = -0.76$ across deciles), so the operational claim is reduced to a screening-budget allocator with documented per-pathogen heterogeneity rather than a calibrated probability. *Third*, 100,000-permutation null calibration under four nulls (iid, circular shift, protein-block shuffle, local-burden-preserving) yields Stouffer one-sided $p = 1.00$ globally for the 4-core MCC across the 12-pathogen panel; per-pathogen iid significance is $5/12$ but degrades to

$1/12$ (Norovirus only) under the strictest local-burden-preserving null. Most "above-null" signal is therefore explained by local clustering of Layer A epitopes rather than by exact-site discrimination. These three audits collectively reframe Algorithm 1 as an exploratory cold-start prioritization heuristic with prospectively definable callable subsets, not a globally calibrated detector. The five iid-significant pathogens form a descriptive callable set. Wave 2 executed the abstention rule on the current panel (**$\mathbf{0/12}$** callable, Paragraph 9) and Wave 3 added structurally explicit local nulls (Paragraph 9): under SASA-, pLDDT-, and joint structural-evolutionary stratification at $n_{\text{perm}} = 100{,}000$, four pathogens (H3N2, Norovirus, Dengue, Influenza B) remain individually significant under the 4-core MCC across all four P6 nulls, and Norovirus additionally survives the strictest P1 local-burden-preserving null, so the descriptive signal on this callable subset is not reducible to simple surface or flexibility clustering. The expanded-panel work above remains the natural prospective test. Wave 4 (Paragraph 9) tested the features-only expansion on the `experiments/waves` branch: hscore did not harmonize ($0/3$ PASS), so the 2-core `freq, entropy` fallback was used; labeled 12-fold LOPO was bounded positive but indistinguishable from the recomputed 4-core (mean MCC 0.077, $8/12$ positive, paired Wilcoxon $p = 0.368$), D1 callability remained $0/12$, expanded 28-target sign stability was $1/28$, and the $16/16$ unlabeled-target envelope establishes feature-space feasibility for Layer A curation without relaxing the panel-size limit, leaving Wave 5 pilot 3 as the next prospective test.

4.7 Future Research Directions

Table 4.4 presents the prioritized future research roadmap, organized by expected timeline and impact.

Table 4.4: Prioritized future research roadmap

Priority	Research direction	Prerequisite	Expected impact
1	Phylogenetic/time correction	TreeTime/UShER + temporal metadata	Reduces founder and sampling bias
2	Adaptive method selection	20–30+ pathogens + external labels	Validates selection for unseen pathogens
3	Scope extension	DNA viruses, multi-protein panels, indel handling	Broader applicability
4	Reproduction and release	Archived CSVs, provenance manifest, Zenodo tag	Closes reproducibility gap

Pathogen-Aware Adaptive Method Selection. The per-pathogen adaptive-weight oracle ceiling (MCC 0.160 vs. EqualWeight 0.083 on the same 12-pathogen feature-ablation panel; this is the adaptive-weighting upper bound, distinct from the per-combination oracle MCC 0.341 in Section 3.4.2) shows that adaptive gains are possible in principle, but 11 reference pathogens are insufficient for reliable selection in a high-dimensional pathogen profile space. The Cycle 7B comparison (Section 3.5.2) tested six additional adaptive paradigms beyond EqualWeight—XGBoost, Random Forest, Logistic L1, 1-NN algorithm selection on pathogen meta-features, three-group phylogenetic HBFWS, and rank aggregation—and all six failed to beat EqualWeight under nested-LOPO Wilcoxon (smallest $p = 0.56$, worst $p = 0.97$); together with the original HBFWS pre-registered failure ($p = 0.78$), seven distinct adaptive-weighting paradigms now converge on the same null result. Convergence across linear/non-linear, parametric/non-parametric, weight-learning/discrete-selection, and ICTV-/phylo-pooled approaches isolates the constraint to panel size rather than algorithm choice, supporting a principled scaling roadmap rather than further algorithm search at $n = 11$. The next step is expansion to 20–30+ reference pathogens with time-forward validation, followed by explicit meta-learning or algorithm-selection models [68, 70]. A two-tier scaling pilot was executed at the dissertation deadline as a feasibility check against this roadmap; Layer A annotation was deferred to future work, so this pilot is a sequence-and-features extension only and not a benchmark-validated panel expansion. (i) A Tier 1 sequence/features pilot pulled NCBI Entrez polyprotein records for Yellow Fever virus E (taxid 11089), Lassa virus GPC (taxid 11620), and West Nile virus E (taxid 11082); after length filtering the modal-length alignments retained $n_{\text{seq}} = 27, 454$, and 74 respectively, with Lassa GPC alone reaching the canonical ≥ 200 threshold and YFV/WNV remaining sparse against the narrow envelope-only Entrez query (the polyprotein-broadened fallback hit Entrez rate-limit timeouts in the dissertation window). MAFFT alignment plus position-level frequency, entropy, and a frequency$\times$entropy h-score proxy were generated under the same schema as the existing cross-pathogen position-score CSVs (with `ground_truth=-1` until Layer A is curated), demonstrating that Tier 1 expansion is operationally feasible but bottlenecked at the literature-curated Layer A step rather than the sequence-pull step. (ii) A Tier 2 features-only sweep across 19 additional surface-glycoprotein targets succeeded for 13 of 19 (Chikungunya E2, Japanese encephalitis E, Tick-borne encephalitis E, Eastern/Venezuelan/Western equine encephalitis E2, Mumps HN, Measles H, Nipah G, Hendra G, Marburg GP, HCoV-NL63 spike, HCoV-229E spike) with n_{seq} from 40 to 791; six were skipped at $n < 30$ (Hantaan Gn, CCHF Gc, Rift Valley fever Gn, Sin Nombre Gn, Sapovirus VP1, Astrovirus capsid) because the narrow taxid Entrez search did not retrieve enough qualifying records and a permissive secondary search would require pathogen-specific accession curation. The 3 Tier 1 pilots plus the 13 Tier 2 features-only targets together give 16 cross-pathogen position-score CSVs added in v176; combined with the 3 cross-pathogen members of the original 11-panel (dengue E, HIV-1 gp120, norovirus VP1), this is the 19-pathogen `cross_pathogen` subset over which the meta-feature spread is computed (entropy mean 0.025–3.15, variable-position fraction 0.05–1.00, top-decile h-score density 0.13–8.36; `results/mutbench/expanded_panel_`

The remaining 8 members of the original 11-panel (SARS-CoV-2 Spike, H3N2 HA, Influenza B HA, MERS Spike, RSV F, HCV E2, EV-A71 VP1, Rabies G) keep their position-score artefacts in their own pathogen subdirs (`data/sars_cov_2/`, `data/influenza/`, etc.) and are not consumed by this meta-feature audit, so the spread numbers describe the cross-pathogen subset rather than a re-validated 27-pathogen benchmark. NCBI provenance for the v176 pilot is captured in `results/mutbench/pilot_accession_manifest.csv` (per-target accession lists plus sha256 of the length-filtered FASTA, query date 2026-04-29) so a third party can re-derive the same input via `efetch` on the listed accessions instead of replaying the Entrez search. Within these caveats the pilot establishes that the automated feature path is engineering-bound rather than science-bound; the next bottleneck is Layer A annotation, which must be assembled from heterogeneous primary literature, DMS escape catalogues, and surveillance reports rather than mined automatically.

Prospective Callability and Abstention Rule (Wave 2 result). The pre-registered abstention rule specified above was executed on the 12-pathogen panel (Paragraph 9; outputs in `results/mutbench/codex_wave2/`). Under the codex-specified default thresholds, $\mathbf{0/12}$ folds are callable for either 3-core or 4-core. The rate-limiting gate is D1 (training-set top-decile separation): no 11-pathogen training subset has ≥ 7 pathogens with bootstrap lower CI > 0.02 AND point > 0.05, because only five of the original twelve have such signal (Paragraph 9). The predeclared sensitivity sweep over D1 quorum $\in \{5, 6, 7, 8\}$ and D2 lower bound $\in \{1.05, 1.10, 1.20, 1.50\}$ (32 configs) confirms robustness: at any quorum $\geq 6/11$, callability is $0/12$ regardless of D2; only the most-permissive corner $(5/11, 1.05)$ yields callable folds, and those select P1-negative pathogens ({SARS-CoV-2, RSV, Zika} for 3-core, {Rabies} for 4-core) with negative mean MCC — a pathology of relaxing D1 below quorum-majority. The rule therefore correctly refuses to deploy Algorithm 1 on this panel, formalising the descriptive callable set of Paragraph 9. Wave 3 P6 has since been executed on the current panel (Paragraph 9): four pathogens (H3N2, Norovirus, Dengue, Influenza B) remain individually significant under the 4-core MCC across all four structurally explicit nulls, with Norovirus also surviving the strictest P1 local-burden-preserving null. The remaining experiments in this future-work programme — Tier 2 features-only LOPO at $n \approx 30$, pilot 3 Layer A literature curation for YFV/Lassa/WNV, and Tier 3 EVE/PLM site-effect via the prepared container — are reserved for an expanded panel where the 7/11 quorum is structurally reachable and the deployment claim can be tested rather than refused. Wave 4 has now executed the Tier 2 features-only LOPO at the available expanded scope (Paragraph 9); the 2-core fallback was bounded positive, but D1 remained $0/12$ and only $1/28$ targets were stable, confirming the panel-size limit and making Layer A curation in Wave 5 pilot 3 the rate-limiting next step rather than an optional add-on.

Integration of Phylogenetic Correction. Homoplasy-based scoring captures signals distinct from frequency (simulation $\rho = -0.876$; real GISAID $\rho = -0.107$) and should be integrated through TreeTime [31] or scalable alternatives such as UShER [34]. This would mainly affect founder-effect-prone pathogens such as SARS-CoV-2, and should be paired with temporal/geographic stratification.

Additional Research Directions. Additional directions are: DNA-virus extension (e.g., HBV polymerase, HSV-1 glycoprotein), multi-protein evaluation beyond surface glycoproteins, indel-aware scoring, virus-tuned PLM features, and real-time surveillance integration via Nextstrain pipelines. The MutBench framework, in its current 11-pathogen RNA-virus form, intentionally does not validate against eukaryotic pathogens, bacterial systems, or DNA viruses with substitution rates of $\sim 10^{-6}$–10^{-8} substitutions/site/year, where mutation-density assumptions of the underlying scoring channels (entropy, frequency, FUBAR, homoplasy) would need to be recalibrated; extension of MutBench to those regimes is therefore explicitly out of scope of this dissertation and remains future work. The HBV polymerase pilot reported in Section 4.4.8 provides empirical sanity-check evidence for this rate-based scope claim.

Reproduction and Release. The archived CSVs, provenance manifest, and Zenodo tag (Section 4.8) close the reproducibility gap as a defense-time deliverable, instantiating Priority 4 of Table 4.4.

4.8 Concluding Remarks

MutBench shows that viral hotspot detection is driven more by the fit between pathogen and biological information source than by detector choice alone. Within the single-position scoring regime and modeled ANOVA terms, the scoring $\times$ pathogen interaction ($\omega^2 = 0.296$) is the largest identified factor, and nine distinct optimal information types appear across 11 pathogens. The practical validation is strongest for HIV-1 escape enrichment (7.19$\times$, 82% Layer A-disjoint), while H3N2 is mainly a self-consistency check and SARS-CoV-2 remains exploratory. The 4-feature core (homoplasy, pLDDT, entropy, frequency) provides a compact starting recipe for new pathogens, but adaptive deployment remains bounded: the dissertation is consistent with a feasible direction for evidence fusion, not a universal adaptive predictor. **Take-home message**: choosing the right biological information source for the pathogen explains more performance variation than the detection-algorithm main effect alone, and MutBench quantifies that asymmetry.

Dual-use considerations. MutBench predicts positions that are likely to mutate under selection, including immune-escape-relevant sites on viral surface glycoproteins. The same outputs that inform vaccine-strain selection, surveillance triage, and antibody design can in principle inform engineering of escape variants, so the benchmark outputs are inherently dual-use. Three mitigations were therefore built into the release plan rather than being deferred to downstream users: (i) raw GISAID sequences are never redistributed—only computed H-scores, detection results, and summary statistics are shared, preserving the provider-side access controls of the underlying data agreement; (ii) the benchmark publishes *region-level* hotspot rankings, not variant-level engineering recipes, so it raises the signal-to-noise for surveillance without providing a direct uplift to escape-variant design; and (iii) the public repository release (30 days post-defence) coincides with disclosure to the relevant biosafety review body, and any extension of the benchmark to pathogens on the U.S. HHS P3CO/Select Agent lists will go through a dedicated institutional biosafety review prior to publication. These steps do not eliminate dual-use risk, but they keep the benchmark aligned with the *responsible stewardship* expectations now standard for computational pathogen-evolution work.

Code and data availability. The MutBench source code, scripts, ground truth definitions, and configuration files are archived at `https://github.com/kksuhj/mutbench` under tag `dissertation-v1` (commit hash recorded in `provenance/commit.txt`; a Zenodo DOI will be minted at final submission and inserted here). The MutBench implementation code is released under the MIT License; the dissertation text and figures are distributed under the Creative Commons Attribution–NonCommercial 4.0 International license (CC-BY-NC 4.0). External resources used by the benchmark (MAFFT, IQ-TREE, HyPhy, ESM-2, Tranception, AlphaFold-DB structures, GenBank entries, and the Bloom-lab DMS datasets) are governed by their own upstream licenses, summarised in Chapter 2, Section 2.2.7; MutBench redistributes only computed scores, detection outputs, and summary statistics, never upstream FASTA bundles or external model weights. With precomputed features, the full 8,580-evaluation detector grid runs in approximately 25 seconds on a single workstation;

the environment used Python 3.10, NumPy/SciPy/Pandas, MAFFT, IQ-TREE, HyPhy, and ESM-2; stochastic analyses used seed 42; raw GISAID sequences are not redistributed.

Numerical artifacts and CSV-to-table provenance. Core Stage 3 and vaccine-escape numeric claims resolve to CSVs in the archived `results/` directory:

- `stage3_full_results.csv`: 8,580-row Stage 2 evaluation output; source of Tables 3.5 and 3.6.
- `layer_c_evaluation.csv`: Layer C DMS cross-check.
- `feature_analysis/per_feature_auc_11pathogens.csv` and `feature_analysis/rf_cv_auc_v2.csv`: feature-analysis summaries, including Table 3.5.1.
- `vaccine_escape_stage3.csv`: source of the single-scoring block of Table 3.18 (20×39 detector grid, all 3 pathogens) and of the best-detector EqualWeight HIV-1 row in the integration sub-block.
- `escape_validation_adapt.csv`: source of the fixed top-5% EqualWeight integration sub-block of Table 3.18 for SARS-CoV-2 and H3N2 only.
- `feature_analysis/feature_ablation_nested_lopo.csv` and `feature_analysis/feature_ablation_nested_lopo_summary.csv`: source of Table 3.13.
- `feature_analysis/tranception_esm_correlation.csv`: source of Table 2.8.
- `feature_analysis/h3n2_temporal_pilot.csv`: source of Table 3.4.

These files and the CSV-row-to-table-cell provenance manifest are bundled with the examiners' archive and mirrored with the Zenodo deposit.

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바이러스 변이 핫스팟 탐지를 위한 생물학적 정보원의 체계적 벤치마킹

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(초　록)

바이러스 게놈의 변이 핫스팟 탐지는 변이주 감시와 백신 설계에 필수적이나, 기존 접근법은 변이 빈도와 Shannon 엔트로피에 거의 전적으로 의존해 왔으며, 계통 분석 선택 신호, 단백질 언어모델 예측, 구조적 특성 등 풍부한 보완적 정보원은 체계적으로 탐색되지 않았다. 어떤 정보 유형이 어떤 병원체에 가장 효과적인지를 비교한 체계적 연구는 존재하지 않았다.

본 연구에서는 이 문제를 해결하기 위해 MutBench를 제안한다. MutBench는 어떤 요인이 핫스팟 탐지 성능에 영향을 미치는지 분석하기 위한 실험 프레임워크로, 10가지 생물학적 정보 유형(6개 카테고리의 20개 점수화 공식)을 5개 범주(임계값 기반, 신호처리, 밀도/클러스터링, 통계, 적응형 윈도우)에 걸친 14개 탐지 방법군(39개 매개변수 변형)과 결합하여 11개 RNA 바이러스에 걸쳐 총 8,580회 평가를 수행하며, 세 가지 핵심 구성요소를 포함한다: (1) 같은 바이러스 종 내에서 서로 다른 계통에서 독립적으로 반복 출현한 수렴 진화 위치를 양성 선택 기준(Layer A), 정화 선택 하의

보존 구조 영역을 음성 기준(Layer B), Deep Mutational Scanning(DMS) 실험적 적합도 데이터를 기능적 검증(Layer C)으로 통합하는 3층 ground truth 체계, (2) hotspot-score = (recall + precision + stability)/3으로 정의되는 복합 평가 지표, (3) Stage 1에서 SARS-CoV-2에 대한 심층 분석, Stage 2에서 대규모 교차 병원체 비교를 수행하는 2단계 실험 설계이다.

Stage 1에서는 MutClust-Hybrid가 최고 hotspot-score(0.778)를 달성하여, 도메인 특화 사전지식과 데이터 기반 클러스터링의 결합이 효과적임을 확인하였다. Stage 2의 핵심 발견은 최적의 생물학적 정보원이 병원체마다 근본적으로 다르다는 것이다: 점수화 방법과 병원체 간 상호작용이 전체 성능 분산의 29.6%를 설명하여($\omega^2_{\text{scoring}\times\text{pathogen}} = 0.296$), 가장 큰 분산 원천으로 나타났다. Friedman 검정($\chi^2 = 7.69$, $p = 0.990$)에서 상위 20개 조합 간 유의한 차이가 없음을 확인하였으며, LOPO 교차검증에서 11개 병원체 모두 서로 다른 최적 정보원을 요구하였다(일반화 성공률 0/11). 또한 탐지된 핫스팟이 실험적으로 확인된 백신 escape 위치와 최대 9.36배 enrichment($p < 0.0001$)를 보이며, 벤치마크에서 최적으로 식별된 scoring 유형이 3개 병원체(2,340회 평가)에서 백신 escape 포착률도 최대화함을 확인하여 공중보건 감시에 대한 실용적 가치를 입증하였다.

6개 정보 카테고리(빈도 기반, MSA 유래, 계통 분석, 구조 기반, AI 기반, 복합)에 걸친 20개 점수화 유형 분석에서 11개 병원체에 걸쳐 9가지 서로 다른 최적 정보 유형이 관찰되었다: FUBAR 계통선택 점수(H3N2), 호모플래시(Norovirus), 단백질 언어모델(SARS-CoV-2, RSV), 의미적 변화(EV-A71), 안정성 예측(Rabies), 빈도 기반(HCV, Influenza B, MERS). 다중 정보 통합이 실험적 탐색 범위를 축소하며(H3N2 백신 escape enrichment 4.012배, $p = 0.0023$), 4개 핵심

특징(homoplasy, pLDDT, entropy, frequency)이 전체 앙상블 성능의 98%를 달성하였다.

주요어: 변이 핫스팟 탐지, MutBench, 생물학적 정보 유형, 바이러스 게놈, 다중 정보원 분석, 병원체 의존적 최적성, SARS-CoV-2, Deep Mutational Scanning